

FEASIBILITY OF PAYMENTS FOR ENVIRONMENTAL SERVICES IN THE AMAZON  
REGION: TOOLS FOR *EX ANTE* ASSESSMENT

By

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To my beloved Giulia and Manuela Aitana who deserve a better world

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## LIST OF ABBREVIATIONS

|         |                                                          |
|---------|----------------------------------------------------------|
| APA     | Environmental Protected Area                             |
| BAU     | Business as Usual                                        |
| CAR     | Cadastro Ambiental Rural                                 |
| CBD     | Convention on Biological Diversity                       |
| CDM     | Clean Development Mechanism                              |
| CIFOR   | Center for International Forestry Research               |
| CONAB   | Companhia Nacional de Abastecimento                      |
| CI      | Conservation International                               |
| COP     | Conference of Parties                                    |
| CSV     | Cotriguaçu Sempre Verde Project                          |
| CU      | Conservation Units                                       |
| ELP     | Ethnographic Linear Programming                          |
| Embrapa | Empresa Brasileira de Pesquisa Agropecuária              |
| GDP     | Gross Domestic Product                                   |
| HH      | Household                                                |
| IBAMA   | Institute of Environment and Renewable Natural Resources |
| ICV     | Instituto Centro de Vida                                 |
| INCRA   | National Institute for Colonization and Agrarian Reform  |
| INDEA   | Instituto de Defesa Agropecuária do Mato Grosso          |
| IPAM    | Amazon Institute for Environmental Research              |
| ISA     | Instituto Socioambiental                                 |
| IUCN    | International Union for the Conservation of Nature       |
| KFW     | Kreditanstalt für Wiederaufbau                           |
| MA      | Millennium Ecosystem Assessment                          |

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| MT          | Mato Grosso                                                       |
| NKMCAP      | Noel Kempff Mercado Climate Action Project                        |
| NTFP        | Non Timber Forest Products                                        |
| OECD        | Organization for Economic Cooperation and Development             |
| ONF-I       | Office National des Forêts                                        |
| PA          | Rural Settlement Project                                          |
| PAA         | Programa de Aquisição de Alimentos                                |
| PES         | Payments for Environmental Services                               |
| PPCDQ MT    | Action Plan to Control Deforestation and Fire                     |
| ProAmbiente | Socio-Environmental Development Program for Smallholder Producers |
| PROCERA     | Programa de Créditos Especiais para a Reforma Agrária             |
| REDD        | Reducing Emissions from Deforestation and Degradation             |
| R\$         | Brazilian Reais                                                   |
| SEMA        | Secretaria do Meio Ambiente                                       |
| SES         | Social and Ecological Systems                                     |
| SISA        | Sistema de Incentivo a Serviços Ambientais do Acre                |
| SNUC        | National System of Conservation Units                             |
| SLAPR       | Sistema de Licenciamento Ambiental de Propriedades Rural          |
| TNC         | The Nature Conservancy                                            |
| UF          | University of Florida                                             |
| UNFCC       | United Nations Framework Convention on Climate Change             |
| WTP         | Willingness to Pay                                                |

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FEASIBILITY OF PAYMENTS FOR ENVIRONMENTAL SERVICES IN THE AMAZON  
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This dissertation proposes and applies three analytical tools and approaches to *ex ante* assessment of a potential PES project, conducting analyses at regional, project, and individual-level household productive systems levels. The project site analyzed was Cotriguaçu Sempre Verde (CSV), in Mato Grosso state, part of a REDD-like initiative concerned with decreasing deforestation, greenhouse gas emissions and increasing carbon stocks, while also positively influencing incomes and livelihoods of local producer groups. The initiative initially focused on supporting landowners to come into compliance with environmental regulations which encouraged them to adopt more sustainable production practices.

Conclusions were drawn based on the following aspects of PES: i) considering rural settlements in Cotriguaçu as a Social Ecological Systems (SES), what is the contribution of PES to the resilience of this systems; ii) satisfaction of preconditions established by Wunder for a PES initiative in the CSV context, and iii) impacts and feasibility of a hypothetical PES in a household livelihood system, using the Ethnographic Linear Program modeling tool.



Results showed that PES is not the most appropriate tool that should be used to decrease deforestation and forest degradation in this specific region. Three main reasons justifies this position: i) PES is not the right tool to improve rural settlers' system resilience; ii) Cotriguaçu does not fit (at least yet) all the criteria to implement a PES initiative and; iii) If implemented in this municipality, PES will not improve livelihood systems and will not contribute to deforestation control. In order to be effective, improvement of command and control tools, and strengthening of government presence in those areas, should accompany PES.

## CHAPTER 1 BACKGROUND

There are many arguments in favor of preserving the Amazon. One of the most common is that the Amazon Region is home to various traditional communities, and these people depend on this place to survive. Another important argument is because it is the region with the largest amount of biodiversity in the planet, with an area larger than 6 million km<sup>2</sup>, spread in nine countries of South America (da Silva et al. 2005), or simply because the Amazon has a scenic beauty not found in other places in the world. Although there are some myths around the relevance of the Amazon for the world's balance, such as it being the "lung of the planet" (false, since it absorbs all the oxygen it produces) or having a rich soil (widely variable), it is also true that the Amazon is one of the largest and most diverse tropical areas (da Silva 2005), and its hydric balance enables the generation of electric energy and supports the hydrological balance of the whole planet. By 2005, "around 40,000 plant species, 427 mammals, 1294 birds, 378 reptiles, 427 amphibians, and 3,000 fishes" from the Amazon were identified and classified (da Silva 2005, p. 690). Thus, the Amazon Region is undeniably an important biome and it must be preserved for all the above reasons together. It should also be preserved for something that everybody takes advantage of but not all of us realize: the environmental services the Amazon provides to all of us.

Environmental services (ES) come from the operation of natural ecosystems. They include carbon sequestration, watershed protection, climate regulation biodiversity, and food, among others (Daily 1997, Millennium Ecosystem Assessment 2005). Those services play a crucial role for the maintenance of human life on the planet. They are directly related to the provision of human well-being in terms of security

(personal safety, secure resource access, security from disaster), basic material for a good life (adequate livelihoods, sufficient food, shelter, access to goods), health (strength, well-being, access to clean air and water), social relations (social cohesion, mutual respect, ability to help others), and freedoms of choices and action (opportunity to be able to achieve what an individual values doing and being) (Millennium Ecosystem Assessment 2005).

Yet we are not aware of the total array of ES, nor of the threats currently posed to them. In 2005, the Millennium Ecosystem Assessment Board (MA) completed a study on the current status of global biodiversity. Conducted by 1,300 scientists, the study revealed the rapid increase of biodiversity extinction. Further, the study showed that threatening processes such as pollution, exotic species invasion and climate change will be intensified in the world. If academics and decision makers intend to achieve goals for conserving biodiversity and reducing climate change, it is necessary to better understand the factors that drive human behavior that affects ecosystem services, and how these current destructive actions could be changed in order to design appropriate policies to control such behaviors

Almost twenty years ago, around 50% of primary forest on earth was already gone, especially due to anthropogenic activities (Bryant et al. 1997). As a consequence, environmental goods and services are lost, and the potential for economic development is reduced. To consider these services in market transactions could be a solution to permit to value and compensate producers of environmental services.

Payments for Environmental Services (PES) are mechanisms developed for transferring resources to providers of environmental services, as a strategy to reduce

resource degradation, usually of forests or watersheds. The basis of PES systems is the transfer of resources (monetary or other) to land managers in order to promote sustainable methods of production that do not threaten the provision of ES. Rather than productive activities that can potentially change ecological characteristics of their lands, a rural landholder would have the incentive to keep the forest intact or well managed, for example, thereby maintaining the provision of environmental services such as carbon sequestration.

The potential for PES projects depends on complex economic, social, environmental and political variables that interact to determine the characteristics and resilience of the overall social and ecological system. A PES project could potentially provide economic support to communities, improve the social condition of families, and maintain a healthy environment. For that reasons, many different factors need to be assessed in areas with potential for PES, such as: complexity of systems and resilience analysis if a PES is applied; pre-conditions for a PES, and how this could be met; variability of household situations and potential benefits/costs.

### **Reducing Emissions from Deforestation and Payments for Environmental Services**

One form of PES program currently under discussion and experimentation is REDD (Reducing emissions from deforestation and forest degradation), a mechanism that has been discussed since 2005 by the United Nations Framework Convention on Climate Change (UNFCCC). The two main goals of REDD are to reduce greenhouse gas emissions as well as to improve forest management in order to mitigate climate change.

REDD stimulates financial trades in a multilevel system (global-national-local) of financial compensations, i.e., PES, to promote the increase of forest carbon stocks and

reduction of carbon emissions (Angelsen et al. 2009). Originally, the idea of REDD was limited to PES financial transfers. However, due to the fact that REDD is still a concept that is being created, it has the flexibility to have different formats

According to Sven Wunder et al. (2008), the implementation of REDD schemes will be an opportunity to leverage funds for PES in Brazil. Even if the main goal of REDD is to lower carbon emissions and increase carbon storage, this dynamic will subsidize the maintenance and provisioning of other environmental services, including biodiversity conservation and socioeconomic benefits in forested areas.

### **PES and REDD Project in the Amazon Region**

There are few PES projects or PES-like initiatives currently being implemented in the Amazon Region. The first PES initiative designed for the Amazon was the Socio-Environmental Development Program for Smallholder Producers (ProAmbiente), designed in 2000 (Hall 2008, Wunder et al. 2008, Mattos 2010). After ProAmbiente, other PES-like initiatives arose in the region: the Bolsa Floresta Program, implemented in Amazonas State since 2007; the Sistema de Incentivo a Servicos Ambientais do Acre (SISA); the Cotriguaçu Sempre Verde Project (CSV); the Pilot Project in São Félix do Xingu; and the Transamazon REDD project (Wunder et al. 2008, Mattos 2010). Studies conducted to evaluate the impact of these initiatives have produced mixed results. The Bolsa Floresta Program, for example, presented positive impacts related to income increase of beneficiaries, provoking positive livelihood impacts and increasing equality (Bakkegaard 2013) but there was little evidence that the program is able to reduce deforestation in the Conservation Units (CU) where it is being implemented (Viana 2012). In common, all of these initiatives work through contracts, oriented according to

state laws, among providers and buyers of ES. However, there is not yet a formal environmental services market (Terra de Direitos 2011).

Originally, proponents of the CSV project intended to adopt PES as a tool to distribute financial resources from REDD. However, due to insufficient resources raised, the idea of PES had to be temporarily cancelled. Instead of cash payments, proponents found alternative strategies to implement REDD such as providing capacity building activities, tools and equipment, as well as efforts to improve governance among communities included in the area of the project. This study opted to analyze the implementation of a hypothetical PES in the REDD project, using modeling that included PES as a new variable in the system at small (household livelihood system) scales.

Many studies and evaluations of PES results and impacts are being conducted, such as the impacts of PES on deforestation reduction or decrease in social inequality (Wunder et al. 2008). Some authors highlight that in many cases after PES implementation neither monitoring, quantification or certification of service were effective, nor systematic impact evaluation (Hall 2008) was conducted. However, there are not many *ex-ante* studies about the viability of PES, which could provide crucial information to inform the design of new initiatives and determinants for their success. An example of an *ex-ante* study is Wunder et al. (2008). In this study the authors point out what must be analyzed before implementing a PES, and to what extent the Amazon region as a whole has the potential to implement PES initiatives. Aspects such as a careful prior study of the local context and the need to fit preconditions for successful PES were the most important findings that will support this dissertation. Along with these findings, other complementary analysis such as the study of resilience of the

systems as a whole, and the potential local impacts of a PES are also points to be considered in an *ex ante* investigation.

I analyzed Cotriguaçu Sempre Verde (CSV) project, located in Mato Grosso state. CSV is one of several REDD projects being implemented in Brazil and in other regions that retains tropical forest.

### **Environmental Services: Evolution of the Concept over Time**

There are different approaches to define environmental services (ES). The *Dictionary of Environmental Economics* (Markandya et al. 2001) defines ES as “ecological functions currently perceived to support and protect human activities of production and consumption or affect the overall human wellbeing in some way.” (Markandya et al., p. 69). The Millennium Ecosystem Assessment (MA)<sup>1</sup> (2005) relates ES with the capacity they have to provide benefits for human being. For Daily (1997, p. 3), “ecosystem services are the conditions and processes through which natural ecosystems, and the species that make them up, sustain and fulfill human life”.

In contrast, (Boyd & Banzhaf 2007) disagree with these definitions of ES. For them, ES are the “end-products of nature”, i.e. directly consumed, which should be distinguished from intermediate products, important to the production of services but are not products themselves (Boyd & Banzhaf 2007). Thus, they do not consider functions and processes as services because they are intermediate in the production of the

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<sup>1</sup> The MA grouped ES in four categories: 1) supporting other services (nutrient cycling, soil formation, primary production); 2) provisioning (food, freshwater, wood and fiber, fuel, etc.); 3) regulating (climate regulation, flood regulation, disease regulation, water purification, etc.), and 4) Cultural (aesthetic, spiritual, education, recreational, etc.).

services, not end-products. In this proposal I will be using the approach of ES of the Millennium Ecosystem Assessment (2005), because it is the one more commonly used.

In spite of the relevance of ES to human civilization, modern life habits neglect their existence (Daily 1997) and currently, almost two-thirds of the ES on the planet are under threat (Millennium Ecosystem Assessment 2005). The recognition of services provided by ecosystems is not new. Since Plato, and probably much earlier, humanity has been aware of the existence of environmental services (Daily 1997). The modern concerns of ES come from George Perkins Marsh's 1865 book *Man and Nature*, one of the first to describe anthropogenic impacts over nature. He was concerned over the deterioration of forest resources that provoked erosion of soils and reduction of soil productivity. He influenced public policies and decision makers since his book was significantly important for the creation of Adirondack Park, in New York City.

In 1949, Aldo Leopold introduced the notion of impossible replaceability of those services (Daily 1977). In 1956 Paul Sears explicitly pointed out the importance of recycling services of the ecosystems (Daily 1977). Raymond Lindeman was the one who most influenced the modern era of ecosystem ecology, pointing out in 1942 that the "analysis of food-cycle relationships indicates that a biotic community cannot be clearly differentiated from its abiotic environment; the ecosystem is hence regarded as the more fundamental ecological unit" (Lindeman 1942, p. 415). Since then, influential authors such as Howard Odum (1953), Rachel Carson (1962), Frank Golley (1993) and many others developed studies and theories on the relevance of ecological systems and services.



Although there is a consensus that ES are crucial for our survival on the planet, Kanounikoff (2006) criticizes the fact that all ES definitions are utilitarian and anthropocentric. This is because ES are always characterized in a human context, requiring an 'interaction with humans' (Nasi et al. 2002) and neglecting other functions that could be less important to human well-being but crucial for other species. Certainly, there are also other services, of anthropogenic value or not, yet to be described.

### **Forests as Providers of ES**

Considering that forests are the largest providers of ES (Daily 1997), I will now sketch the ecological processes of some of the most common ES provided by tropical forests, which are relevant to the broad topic of this research. Such services are water provision, climate regulation, and carbon storage. Despite their relevance, other environmental services provided by tropical forests such as pollination, biodiversity, natural pest control, scenic beauty and tourism, will not be addressed here.

### **Watershed Services**

According to Shiklomanov (1993), the earth's surface receives around 119,000 cubic kilometers of water from rain per year. In tropical areas, forests exert a 'sponge effect' because they absorb water before releasing it again to aquifers and surface rivers. The deforestation of these areas provokes a muddy effect on the rivers that reduces the infiltration of water, shrinks the river and reduces the quality of water (Daily 1997; Nasi et al. 2002).

Current discussions among scientists on water-forest links question if rain exists due to the forest cover or if it is humidity that creates forests (Nasi et al. 2002). Some renowned researchers have discovered many myths, such as the belief that the lack of canopy cover was the principal determinant of erosion, whereas the actual determinant

was the ground cover (Wiersum 1984). Another belief that has since been disproved is that deforestation is linked to the decrease of runoff, or flooding, when it turns out that the opposite is true (Calder 1998). More recently, there is a new hypothesis that rainfall occurs in the interior of continents because of the forest cover there, which is the case of the Amazon and Congo (Sheil & Murdiyarso 2009).

### **Climate Regulation**

Scientists have always believed that deforestation has always been responsible for compromising climate regulation on Earth because it reduces rainfall and interrupts the water recycling process by taking off evapotranspiration from the forests (Nasi et al. 2002). In fact, considering that evapotranspiration is responsible for 80% of rainfall (Nasi et al. 2002), we can expect an increase in dry climate after deforestation. However, deforestation also affects other aspects such as surface albedo, aerodynamic forces, clouds, and circulation of air. For this reason, the direct relationship between deforestation and reduction of rainfall is not clear. Local climate seems to be affected by deforestation although the extent of its effect is not clear (Nasi et al. 2002).

### **Carbon Storage**

Carbon emissions are one of the factors responsible for global warming. Forests have the important role of carbon sinks, which contribute to the mitigation of global warming (Woodwell & Mackenzie 1995; Apps & Price 1996). Boreal forests are the earth's largest terrestrial biome and they have around half of the forest carbon in the world, followed by tropical forests which have more than one-third of carbon stocks, and temperate forests with one-seventh of carbon (Daily 1997). When a forest is burned, carbon dioxide and methane are both released into the atmosphere, thus contributing to greenhouse-effect processes.

A tropical primary forest can stock about 250 tons of carbon per ha and could release 200 tons of carbon if converted to slash and burn agriculture, and even more if converted to grazing or a permanent agricultural site (Nasi et al. 2002). It is estimated that the Brazilian Amazon biomass contained 80-100 billions of carbon (Nilson 1993) in the 1990s and 47 billion tons more recently (Nepstad et al. 2007). Reforestation of ten million km<sup>2</sup> of deforested lands in the tropical regions could contribute to the capture of 100-150 billion tons of carbon (Nilson 1993). It could also contribute to the preservation of watersheds, and protection of soils, among other benefits.

There is debate about the potential for PES programs to combat deforestation and the most successful approaches to implement them especially because, if badly implemented, they can provoke the opposite effects when people start to threaten the area more because they want the area to be eligible for PES (Wunder et al. 2008). Besides, the implementation of a PES program can provoke a “demographic magnet effect” attracting people interested in the compensation, provoking more degradation to the ecosystems (GEF 2010).

According to Pagiola et al. (2003), in the 1990s about 15 million ha of forest were lost worldwide. This loss was due to the absence of compensation to the users of these lands for the environmental services that they could generate, among many other problems such as lack of good governance, improper policy processes, and lack of alternative livelihood options. For Mattos & Pereira (2003), small rural producers are not only suppliers of raw materials such as lumber, NTFPs, and agricultural products; they also have a multifunctional role when their economic production is associated with conservation of the environment and the supply of environmental services. As a result,

they should be compensated for this, although there are still obstacles to valuing these services and to understanding their effects.

### **The Amazon Forest as an Important Provider of Environmental Services**

The Amazon Rainforest is still the largest tropical forest in the world (Azevedo-Ramos 2008) and one of the most threatened ecosystems. Contextualizing the Amazon forest in order to better understand its problem is a first step to designing and planning actions to manage the region. The Amazon is denominated as either the Amazon Biome or the Legal Amazon. The first term refers to the entire Amazon region, which spans nearly eight million square kilometers across nine countries in South America (Bolivia, Brazil, Colombia, Ecuador, Guiana, French Guiana, Peru, Suriname and Venezuela).

In Brazil, the Amazon Biome covers about 4.2 million km<sup>2</sup>. In contrast, the Legal Amazon is a political concept defined solely in the Brazilian Federal Constitution in 1953 as the region containing the following nine Brazilian states: Acre, Amapá, Amazonas, Mato Grosso, Maranhão, Pará, Roraima, Rondônia and Tocantins. The Brazilian State conceived this term in order to spur socioeconomic development in the region, which would, in turn, bring about financial benefits. The Legal Amazon includes other biomes aside from the Amazon rainforest such as the Savannah (5.1 million km<sup>2</sup>), which covers approximately two thirds of the country, and is home to nineteen million people (ISA 2008).

The Brazilian Amazon forest is also known for having one of the biggest bio-diverse areas in the world (Filho 2004), and sheltering approximately one-third of all species on the planet (ISA 2008). An example is the diverse vegetation, which averages forty to three hundred species of trees per hectare, in comparison with temperate

forests, which average only four to twenty-five tree species per hectare (Filho 2004). Some authors also highlight that many additional species remain undiscovered.

The region also boasts significant socio and cultural diversity, considering the sixty-six different ethnic groups with unique languages (many of whom face the threat of extinction), the around fifty isolated indigenous groups, as well as other groups believed to remain undiscovered (ISA 2008). Moreover, traditional communities, such as rubber tappers, *babaçu* palm nut breakers, Brazil nut collectors, small farmers, and fishermen, have also inhabited the region for centuries.

Considering all of the above, the Amazon forest deserves respect and protection not only because of its large area and geographical diversity but also for its biodiversity and its social, political and cultural richness. The threat of extinction to what remains undiscovered should also be taken into consideration. Due to such richness and abundance, the Amazon is also a stage for land conflicts. In the past and even up until today, the rightful owners of the land in this region are unclear. This, in turn, spurs conflicts among the region's diverse array of stakeholders.

### **The Frontier of Destruction**

The history of the Post-Columbian Amazon region has repeatedly experienced cycles of exploitation of its diverse natural resources, which resulted in inconceivably high rates of deforestation over vast areas of land. In the last thirty years, around seventeen percent of the Brazilian Amazon forest, or sixty million hectares (an area equivalent to France), has been converted to other land uses (INPE 2008).

Before the discovery by the Europeans, traditional communities extracted natural resources from the Amazon forest only for subsistence purposes. European colonization brought about exploitation of natural resources to meet demands outside of

the region (Becker 2005). During the colonial period, the first wave of European exploitation focused on resources such as guaraná, cinnamon, salsaparilla, cacao, among others (Schmink & Wood 1992). Although the exploitation of these resources continues to this day, it reached its height at the end of the nineteenth century (Bessa, et al. 2005). The rubber boom began at the end of the nineteenth century followed by the extraction of other resources such as Brazil nut, minerals, timber, and animals (Pasquis et al. 2001).

These repeated cycles are referred to as *boom-bust* mechanisms in which resources are exploited until their depletion or to the disappearance of the market (Pasquis et al. 2001; Becker 2005; Betts et al. 2008; ISA 2008). Even if this mechanism brings initial positive impacts due to increases in income, after the boom reaches its peak, it results in undesirable consequences for the local population. These include economic crises as a result of the decrease in jobs, environmental impacts, few social benefits and socioeconomic indicators not significantly bigger than before the beginning of the cycle (Pasquis et al. 2001; Betts et al. 2008; Rodrigues et al. 2009).

The current exploitation pattern in the Amazon forest by small producers is based on the logging of primary areas followed by the implementation of slash and burn agriculture and cattle ranching. Considering that the Amazon region still contains a large undisturbed area, after producers exhaust the soil in one region, they look for another region to continue the cycle of depletion (Pasquis et al. 2001). As long as there are unoccupied lands, producers will keep following this pattern. Schmink & Wood (1992) indicate that this pattern of *boom-bust*, followed by other activities such as

deforestation, land use changes, rural settlements and conflicts over land is, in fact, a consequence of the competition over natural resources by different stakeholders.

Local actors' activities are driven by regional and global economic forces. The Brazilian government offers credits or subsidies to encourage the production of specific commodities with high market value, such as cattle and soy. The current concern is that biofuels will invade the Amazon (Nepstad et al. 2007; Sawyer 2008) which would lead to even greater deforestation rates, and associated social consequences such as displacement of people. However, the recent discovery of a large offshore oil field could make Brazil the tenth biggest oil producer in the world which may, in turn, halt the expansion of biofuels in the Amazon.

Since its discovery until the end of the 1940s, the Brazilian Amazon was considered a region of exploitation for exportation purposes. After the Second World War, though, the region garnered national interest (Guerra 2002). Proof of this is the inclusion of Article 199 in the Constitution of 1946 that created the “Plano de Valorização Econômica da Amazônia” which stated that the federal government should invest at least 3% of its income in the development of region for the following twenty years (Guerra 2002). In 1953, the government created the “Superintendência do Plano de Valorização Econômica da Amazônia – SPVEA” to economically improve it; and at the same period the political concept of the Legal Amazon was created (Guerra 2002).

During the Brazilian military regime, which began in 1964 and lasted until 1984, the region experienced a major period of occupation. The military viewed the region as a means to support capitalism in the country. The military regime also recognized that the region was difficult to access and feared that it could be occupied by other

countries. Embedded in this mentality, the Brazilian government began to implement a development policy in the region with the motto ‘integrar para não entregar’<sup>2</sup> (Guerra 2002).

The above-mentioned colonization policies brought negative consequences in different areas of the Amazon region. Shortly after the discovery of Brazil in 1501, it is estimated that Brazil was home to more than a thousand indigenous groups numbering around five million in the Amazon basin (ISA 2008). Up to now scientists have identified human traces back to twelve thousand years (Filho 2004). Today, about 150,000 indigenous people from sixty-six different groups living in the Legal Amazon have survived “colonialism, the nation-building and the globalization” (Heckenberger 2007). Approximately ten to fifteen percent of these indigenous people now inhabit cities (ISA 2008).

The most significant key drivers of deforestation in the Amazon region are agricultural expansion (especially soybeans), cattle ranching, infrastructure extension and selective logging. These drivers are promoted especially by public policies that ignore environmental issues and respond to: the need of increase economic profits and infrastructure (credits, roads, and resettlement); the demand for soybeans at an international level; the demand for beef at national and global levels; and institutional factors (weak law enforcement in frontier zones and weak property rights) (Chomitz and Thomas 2001; Fearnside 2001; Kaimowitz 2002; Kaimowitz et al. 2004; FBOMS 2005; Naughton-Treves 2005; Chomitz et al. 2006; Azevedo-Ramos 2008;). Altogether they

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<sup>2</sup> This motto means that Brazil should integrate the Amazon into the rest of the nation in order to avoid interference from other countries.



form the patchwork quilt of deforestation in the Amazon. Table 1-1 shows the participation in deforestation in 2013 by different types of land title regimes. Rural settlers from the colonization program were responsible for 29%, followed by land for which titling information was lacking (23%), and private properties (20%). Due to the important participation of rural settlers and private properties in the deforestation rates, my study focuses on both stakeholders. It is worthy to stress that rural settlers in Brazil is the tenure category of people that received areas from the National Institute for Colonization and Agrarian Reform (INCRA) and differs from the other small holders that privately bought their lands.

The consequences of deforestation are huge, and especially affect the provision of environmental services. There are four major environmental consequences to deforestation in the Amazon: loss of forest productivity (on timber and NTFP); changes in hydrological regime; biodiversity loss; and net emissions of greenhouse gases (Fearnside 2005). Protecting this region is wise not only for local communities' survival and national sovereignty, but also for the ecological balance of the whole planet. The following section will explore the conservation strategies currently being undertaken in the Brazilian Amazon forest.

### **Changing the Future**

The 5.1 million km<sup>2</sup> of the Legal Amazon frequently have areas of difficult access. Thus, control and monitoring of this region implies high costs. Besides, Brazilian environmental legislation is too rigid to be respected by small producers without reducing even more their life quality (Wunder et al. 2008).

Therefore, according to the geographer Bertha Becker (2005), even if the greatest challenge today is to modify the current development that is mainly focused on

economic growth, and make efforts to keep the forest exactly how it is, it is not simple to promote the development in the region, which should not be based on the pure “replicability of current western development standards” (CGEE 2009). However, in order to promote conservation and development in the Amazon, it is first necessary to pinpoint the social actors and the most important deforestation drivers in the region so that the government can develop and implement appropriate mechanisms.

In the last decade, the Brazilian government implemented an accelerated strategy of conservation in an attempt to avoid, or at least control, deforestation in the Amazon, fundamentally based on the creation of Protected Areas (PA). Despite the deficiencies in personnel, infrastructure, and management procedures, PAs are by far the best way to conserve natural resources and promote their sustainable use (Silva 2005). Thus, in 2005 the Minister of the Environment (Marina Silva, and current candidate for President), followed Brazil’s endorsement of the Convention on Biological Diversity (CBD) and declared that the Ministry of the Environment’s main strategy for the region would be the expansion of the number of PAs. The basis for the Brazilian Protected Area System is the National System of Conservation Units (SNUC), created by the Federal Government through Law 9.985 in 2000; it establishes the procedures for the creation, implementation and management of PAs. The SNUC is divided into two big categories: Direct and Indirect Use. In the first there are people living in the areas (corresponding to the VI category of IUCN), while in the latter, permanence of humans is forbidden (corresponding to I to V IUCN categories).

Up to now about 37% of the Amazon has been legally protected/ conserved through Protected Areas (Azevedo-Ramos et al. 2006). Despite these numbers, some

PAs are still paper parks due to the difficulties related to the management of those areas and high deforestation rates continue. The graphs below (Figures 1-1 and 1-2) show the evolution of deforestation in the Brazilian Amazon from 1988 until 2013. Deforestation rates systematically reduced after 2004, when deforestation rates were 27,000 km<sup>2</sup> per yr, to about 11,000 km<sup>2</sup> per yr in 2007. However, the reduction detected in 2005 and 2006 seems to be associated more to the decrease of prices of international commodities (soya and cattle) (Betts et al. 2008) than to the creation of PAs, especially because when those commodities' prices increased again, after 2007, deforestation rates also got higher (Wunder et al. 2008). In 2013, there was a considerable increase in deforestation rates (28%), most of it illegal in abandoned areas (IPAM et al. 2013). This is considered unacceptable because the government already has the means to monitor these areas (IPAM et al. 2013).

The current global trend is for even more demand for those commodities, which could be reflected in an increase of deforestation rates in the Amazon. In light of this context, PAs alone in the Amazon are not the most efficient tool to protect the biome, and new solutions are an urgent necessity.

Considering that Brazilian command-and-control instruments are not effective in reducing deforestation rates in the Amazon, implementing financial mechanisms such as Payments for Environmental Services (PES) and REDD+ can bring innovations to the current policies to enhance management in the region, especially due to the flexibilities of these tools, such as voluntary access and also the capacity of self-monitoring (Wunder et al. 2008).

Within this financial approach, it is inevitably required to improve the “marketing” of nature in the Amazon, in which natural resources are being transformed into “fictitious commodities” (Becker 2005) such as the carbon market. This is because those resources have not been produced for the market but, instead, the “market created this necessity” with the goal of obtaining financial benefit (Becker 2005). Moreover, keeping the economic value of the standing forest higher than the value of wood, cattle or soy is the best way to compete with the high opportunity costs of protecting the forest.

This trend, even if appears to be risky, is seen by some as a good opportunity. Almost twenty years ago, Fearnside (1997) pointed out the urgency of local populations of the Amazon converting environmental services such as biodiversity maintenance, carbon stocks, and water cycle into monetary flows. Fearnside argued that Amazonian people should sell what they have in order to survive, and environmental services could be a wise strategy for generation of incomes that would not put the area in risk and would be a long-term strategy for the survival of these people.

### **Payments for Environmental Services**

The critical ideas behind ES are related to the need to preserve these services, and the correct approach to value them. Most environmental services (and also goods) are not valued in a correct way by our society, partly because of the lack of tools to do so. The knowledge of ecosystem services and their functions, as well as their values, is crucial not only to the comprehension of nature as a system but also to the development of the best strategies to preserve these systems.

Payments for environmental services (PES) are part of a new paradigm of market-oriented environmental conservation that holds the promise of benefiting both

the ‘sellers’ and the ‘buyers’ of these services, which, at least in theory, in turn contributes to both the conservation of nature and to economic improvement. In theory there are different ways of making the monetary transactions from buyers to sellers of environmental services, shown in Figure 1-3. One is the direct purchase without intermediation, in which buyers, who can be the beneficiaries or the government, directly access and pay the sellers, or producers of ES. Government is necessary when the services are public goods and the provider does not have the conditions to value it (such as carbon provision). The other way is a direct purchase with intermediation, through an institution that could be the local government, for example. The last form is an indirect purchase by a mediator, such as a non-government agency. Direct purchase is still the least common (Wunder et al. 2008). In sum, PES have many different arrangements, which can lead to situations where sometimes the “sellers” are not actually the ES producers, and sometimes the “buyers” are not actually the direct ES consumers.

PES has the advantage to internalize indirect externalities that come from the “indirect use of ecosystems” (Wertz-Kanounnikoff 2006). According to Pagiola et al. (2005), there is a market failure involving ES since they are seen as public goods, and beneficiaries do not pay for the services received. Consequently, PES appears to be a way to deal with these failures. Next, some definitions and concepts will be introduced to better understand these economic tools.

### **Definition of PES**

There is not yet, in the literature, a formal concept of PES. Instead, there are criteria and categories of PES according to the kind of services, the geographic area, the market or even the kind of payment realized. Wunder (2005) has defined PES using a

Coasean<sup>3</sup> approach (1960) which is purely based on the market as well as on the ability to naturally recover from market failures, according to five criteria: “(1) a voluntary transaction in which (2) a well-defined environmental service (or a land use likely to secure that service) (3) is “bought” by a (minimum of one) buyer (4) from a (minimum of one) provider (5) if and only if the provider continuously secures the provision of the service (conditionality).” This approach derives from environmental economics, where all the externalities can be valued, and which requires property rights over natural resources and environmental services (Mattos 2010).

As a matter of fact, in practice, there are not too many projects that totally satisfy those five criteria, although there are many ‘PES-like’ schemes that satisfy some of them (Wunder 2005). From 287 projects identified around the globe, not more than 20% fit all the criteria above. Exactly because of this, a group of researchers proposed a different way to describe PES (Muradian et al. 2010, p. 1205). For them, social and cultural aspects should be included in the PES concept, providing a new vision for decision makers. The alternative conceptualization for them is: “PES is the transfer of resources between social actors, which aims to create incentives to align individual and/or collective land use decisions with the social interest in the management of natural resources.” This definition was promptly rejected by Wunder due to the lack of criteria (Wunder 2011).

Wunder (2007) has analyzed initiatives of PESs in Bolivia and Vietnam and found that none of these initiatives were able to satisfy his five criteria. The 1st criterion,

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<sup>3</sup> In The Problem of the Social Cost (Coase 1960), Coase postulates that an efficient allocation of resources can be achieved in the presence of externalities if transaction costs are low, regardless of the initial allocation of property rights.

that the payments must be voluntary, is not verified because the transactions are part of 'command and control' practices. In the 2nd criterion, it is difficult to define exactly what is being bought, and, worst of all, it is hard to give an exact value for this service.

However, many of the initiatives studied satisfied more than one criterion, showing that it is very hard to put in practice the theoretical model of PESs but there are thousands of 'false-PESs' all over the world. Shigeo Shiki, ex-coordinator of ProAmbiente, a precursor PES initiative in Brazil in the 2000s, in a presentation on "Markets for PES and challenges of rural development policy in Brazil" (Shiki 2011) stressed that in Brazil the recent creation of markets for PES is showing that practice is much more advanced than theory. Thus, it is important to understand how relevant the criteria established by Wunder (2007) are to a PES initiative.

### **Efficiency of PES**

Even if there is not yet a consensus on the definition of PES, all these systems have a common objective which is to provide environmental services, which are getting scarce, as well as to develop mechanisms to offer these services in the long run (Mayrand & Paquin 2004). Overall, PES should be efficient, and in order to make them efficient, implementers need to carefully consider important aspects such as additionality, leakage, baseline, permanence and transaction costs (Wunder 2007), discussed below.

Additionality is the difference that a project makes to a baseline situation. Although it is difficult to analyze because we do not always have the tools to compare how the situation would be without the project, this analysis is crucial for PES efficiency. The reason is it is worth paying for a situation that would not change without the project, but not paying for scenarios that are already changing (e.g., reducing deforestation)

without PES. The latter scenario is represented in the case of Costa Rica. The historical rates of deforestation started to turn around in the 1990s, before the implementation of the first PES system (Wunder 2007). In a simulation study about predicted deforestation in the Amazon region, Soares-Filho et al. (2006) showed that the forested area of northwest Mato Grosso, including Cotriguaçu municipality, is potentially threatened, which means that policies such as PES, command-and-control, or creation of conservation units should be additional to the conservation of these forests.

For that and other reasons, there was much debate about including avoided deforestation in the Clean Development Mechanism (CDM) of the Kyoto Protocol, which aimed to provide more flexibility in the reduction of emissions stated by the Kyoto Protocol by including benefits from not only industrial projects but also from other projects such as renewable energy, sanitary landfill, energy efficiency and reforestation (CGEE 2009). Up to now, improving forest management and protecting natural forests that could disappear without a PES program were not considered, whereas both reforestation and afforestation are eligible for carbon credits (Wunder 2007).

We have room here for a classical debate on PES, which is the confrontation between efficiency and fairness, with the following main question: should we pay people who are a real threat to the forest, instead of rewarding poor rural people who already protect the forest, produce environmental services and are not a real menace to the service provision? Wunder (2007) says that, from an efficiency point of view, only those who represent a threat should be paid for the provision of ES, but he also points out the importance for ethical reasons, of compensating those who are contributing to the preservation of the services. Wunder's position is closely related to the additionality



criteria, in which the allocation of the resources should be greater where the project has more chance to change the context.

Another key concept for efficiency in PES is leakage, which is the impact of the project on neighboring (or even farther) areas. In this case, the project may work really well in a specific area but the damage or threats 'migrate' to other places. A local example of leakage is the implementation of a PES scheme in an area where people were starting to raise cattle. After the prohibition of this activity, people will find ways to keep raising cattle on the borders of the area outside of the project. A global leakage example could be an area where timber was illegally collected and exported. After the implementation of the PES scheme, the timber buyers will find another supply area to buy illegal timber.

Permanence, which is the capacity of a project to extend and maintain the results even after the closure of the project, is also crucial to PES projects. If a reforested area is totally harvested after the end of the PES scheme, the project has not had any permanence.

Finally, but no less important, it is fundamental to look at the transaction costs if we want to obtain efficiency in PES projects. Those are costs to start up (negotiation, design, contract) and run the project (administration, monitoring, enforcement). Both buyers/intermediaries and sellers have transaction costs, which are the costs to implement (search for a site, negotiate and contract with beneficiaries) and run (administration and management of resources, monitoring and enforcement of the site) a PES initiative, and the smaller the costs, the more competitive those projects, especially for the implementers (Wunder 2007). In developed countries these costs tend

to be higher than in developing countries, but this can be different when there are many stakeholders involved (Wunder 2007). This is the case of Canada's land-diversion, where transaction costs are up to 25% of total costs, compared to Costa Rica, a developing country that shelters one of the first PES initiatives in world, where buyers pay 7% of total costs, and participating farmers pay 12-18% to intermediaries to get the certification (Wunder 2007).

According to Wunder (2007), start-up costs are normally higher than running costs, especially in pilot experiences. In addition, high transaction costs are related to the number of smallholders and actors involved in the PES, the weakness of institutions and property rights, and the information related to baseline and to monitoring of land.

### **PES – Linking Conservation and Poverty Alleviation**

In many areas where conservation needs to be enhanced, poverty also must be addressed. Allying conservation and poverty alleviation is not a simple task. Although they overlap a great deal, often there is little integration between the two. Many programs aim to achieve goals of both biodiversity conservation and poverty alleviation; however they are more likely to be successful if they consider both problems together (Agrawal and Redford 2006).

These arguments emphasize the importance of accomplishing objectives of both conservation and development, and trying to minimize potential tradeoffs. Furthermore, these goals should be treated as complementary rather than mutually exclusive. We have to take into account that human beings also have important rights and needs and can be harmful towards natural resources. If both issues are not simultaneously addressed, there will always be contradictions, and potential divergence due to conflicting strategies.

As pointed out by Romero and Andrade (2004), it is time “to find a balance between preservationists ... and ... devolutionists;” the first are strictly concerned with biodiversity preservation, and the latter emphasize empowerment of local communities to have the control of their forests. Considering the need to achieve this balance, PES schemes may represent a middle way. PES have been scrutinized as a potential contributor to poverty reduction (Landell-Mill & Porras, 2002; Pagiola et al. 2002). In some cases PES are seen as a tool to increase the incomes of beneficiaries, and some donors are indeed interested more in its social effects than in the environmental impacts of PES (Wunder et al. 2008).

In this debate about PES and poverty alleviation, Wunder advocates that PES should primarily focus on the environment and never on poverty, even if it is an important side objective (2008). On the same wavelength, Pagiola et al. (2005) express that PES should be “a mechanism to improve the efficiency of natural resource management but not a mechanism to reduce poverty.” However, all of them recognize that PES schemes have the potential to reduce poverty. On the other side, some defend that PES could be a poverty trap since it will restrict beneficiaries’ development because they will have more limitations imposed (Landell-Mills & Porras 2002).

Two different aspects have to be considered in order to make this link between PES and poverty alleviation. The first is related to the reduction of poverty among program participants, and the second is related to the impacts of PES in reducing overall poverty in a country (Pagiola et al. 2005). For logical reasons, researchers are more focused on analyzing the first aspect. Their conclusions are that the impact of a PES program derives from the increase of income it generates; however non-income

impacts are also common such as social co-benefits (Pagiola et al. 2005) or even social and human capital enhancements.

A study conducted by Börner et al. (2010), pointed out that, among all the most common tenure categories in the Amazon region (largeholders, smallholders, official colonization settlements, indigenous lands and extractive reserves), large landholders are the ones that were more prone to reap PES benefits at this time, simply because, from 2002-2006, they were responsible for more than 80% of deforestation (Börner et al. 2010). Nowadays this is slightly different due to increased small owners participation in deforestation. According to IPAM et al. (2013) (Table 1-1), private properties (the majority large landholders) were responsible for 29% of total deforested area in the Brazilian Amazon in 2013. In Cotriguaçu, large landholders were responsible for 10% of deforestation, and small rural settlers settled by the INCRA agrarian reform institute were the ones that contributed more to forest losses, 52% (SEMA MT et al. 2009) (Table 1-2).

Concerning income impacts, at first sight there is an assumption that participants will be better off because participation is voluntary. However the amount of payments is an important issue. It should take into account the minimum willingness to accept (WTA) of providers of ES and the maximum willingness to pay (WTP) of the buyers of the services. The opportunity costs should also be considered in the decision of the amount to be paid, in order not to compromise the success of the program (Pagiola et al. 2005). In top-down projects those questions are not always addressed.

Non-income impacts are normally reflected in the strengthening of institutions, promotion of participation, building of social capital, and land-tenure consolidation,

among others (Pagiola et al. 2005; Wunder et al. 2008). Finally, PES can also bring non-expected negative social-effects such as paternalistic expectations, inequality, jealousies or even tensions between participants and non-participants.

In Cotriguaçu, as in other places in the Amazon, small producers are the less favored portion of the population. The municipal government has difficulty providing them with minimal living conditions. These people have little access to schools and health services, and during large parts of the year many communities are inaccessible due to the bad roads; public transportation does not exist. In sum, they are geographically and socially isolated. Thus, even if a direct PES does not become a reality there, indirect PES and social and economic co-benefits would likely be welcome. On the other hand, large landholders have enough resources to provide a good educational level to their kids in other municipalities or even in Cuiabá, Mato Grosso's capital, and can go to a hospital in Juína if something happens. These people normally have a house in the city and pay someone to manage their farms.

### **Social and Economic Features of PES**

The solution of threats to environmental services therefore goes further than understanding and mapping those services. It is fundamental to understand social and economic aspects such as the logic of culture and society with respect to the preservation of resources as well as the inclusion of those services in the market.

Costanza et al. (1997) estimated that the value of ecosystem services for the entire biosphere was "US\$16-54 trillion/year, with an average of US\$33 trillion/year." Using data from the literature, in this study they valued 17 ecosystem services in all the biomes. They concluded that the real value could certainly be much larger, and suggested the need to give the correct weight to world natural capital in political

decisions. The main point here is not related to the accuracy of the financial value but, rather, it is much more important to know how this information will be considered and used by decision makers in the design of conservation policies. Otherwise, it would be senseless to continue providing numbers and values, while allowing the maintenance of the business-as-usual scenario of degradation, instead of reformulation of policies to lead to proper consideration of the values of ES for human well-being.

Some environmental goods or services are intentionally produced to be sold, such as food, timber, and minerals. However, other ES are also provided unintentionally as externalities<sup>4</sup>. ES also have the characteristic of being public goods, to which everybody has access, and which all people can use. Besides that, since ES are public goods, everybody can degrade the ecosystem that supplies these services (Kumar Duraipah 2006). Hence, from an economic perspective, being an externality and being public goods contributes to the degradation of the ecosystems and does not provide the incentive to preserve them, simply because people see these services as free. Thus, it is crucial to recognize the values of ES as well as their scarce character. Payments for environmental services (PES) are a way to address these issues (Wertz-Kanounnikoff 2006).

According to Bryant et al. (1997), about 50% of primary forests on earth are already gone, especially due to anthropogenic activities over the last 30 years. This reduction represents consequent degradation and losses of environmental goods and services, and reduced potential for economic development. One alternative to tackle

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<sup>4</sup> Externalities are the costs and benefits that result from economic activities for other parties not directly involved in the transaction.

these problems is to consider these services in market transactions. PES is thus a mechanism for transferring resources to providers of environmental services as a strategy to reduce forest degradation. However, many challenges are still to be faced: how to assess the monetary values of ES; how to promote the monetary transactions of them; to whom the ES belong; and who are the buyers.

A recent proposal is the Clean Development Mechanism (CDM), where developed nations are allowed to offset their emissions by investing in projects to reduce greenhouse gas emissions. This proposal is highly criticized by Martínez-Alier & Jusmet (2001), who say that it is much cheaper to promote carbon storage in developing countries than to reduce carbon emissions in rich countries. The reason is that the price of sequestered carbon is heavily weighted by who has the power to pay for it (the rich); thus the price of carbon sequestration is far less than it would be with less income disparity. Rich countries act, therefore, as if they owned a disproportionate share of the absorptive capacity of the carbon dioxide in earth.

### **Pros and Cons of PES**

In short, PES has the principal goal of transferring resources to those who contribute to conserving or producing services through environmentally sound practices and techniques in the agricultural, industrial and urban environments, as well as secondary goals that seek to reduce poverty, make livelihoods more sustainable, and improve broader social well-being. These initiatives are still in a pilot phase, with only a few examples established and many criticisms. PES programs, as they are being implemented nowadays, present many problems, described by Wertz-Kanounnikoff (2006):

- High transactions costs, especially in cases where PES is implemented for conservation purposes in weaker institutional and governance structures, such as in Africa, making it difficult to find a buyer
- They are “time-intensive and less adequate for quick-fix approach”
- Scientific uncertainties of PES, especially related to the difficulties to make a link or trade-offs between ES and how they could be provided
- Difficulties related to the identification of potential buyers of ES

Some authors accuse PES of impeding development because it promotes “rents against development” (Karsenty 2004), forcing people to stay at the same socioeconomic level, especially in cases where the land users are not provided resources to engage in other activities to complement their incomes.

The main goal in the creation of the PESs is to transform the services offered by the ecosystems into “products” that will be traded, such as carbon or clean water. This means that we need to have defined data about the nature of the market, the structure of the demand for these services and a value for this ‘commodity.’ Fortunately the environmental economics field has mechanisms to measure these services. However, Connif (2012) mentions that, in general, “critics are uncomfortable with viewing nature as a service provider fit to be incorporated into the global capital markets.”

Other criticisms of PES are related to the risks inherent in the idea that economic concerns and issues are more important than non-economic arguments (Redford & Adams 2009). The utilitarian views of ES can ignore other fundamental services that are not necessarily directly related to sustaining and fulfilling human life, such as fire, droughts and diseases. Redford & Adams (2009) consider that if this overvaluation of environmental services continues, the tendency will be the maximization of some few services only, which will lead the ecosystems to “increased ecological brittleness.”



There is also a concern that communities will be worse off and marginalized and not equitably compensated in PES schemes, especially if they are implemented inside REDD initiatives. Those critiques are related to the fact that “commodification of life forms and ‘biodiversity’ (biodiversity credits, for example) undermines local (non-monetary) cultural, conservation and sustainable use values” (Griffiths 2007, p. 8). He also argues that recent indigenous case studies of carbon offset plantation schemes confirm that indigenous peoples are often marginalized and fail to receive equitable benefits from such projects.

According to Wunder (2006), some critics fear that PES will be responsible for separating conservation from development, and for communities losing their aspirations of land-development activities (Karsenty 2004, Romero & Andrade 2004). Finally, Kosoy and Corbera (2010) see PES as “commodity fetishism” with three main weaknesses: simplification of complex systems, non-observance of multiple values, and disregard to asymmetries in people’s access to environmental services. Farley et al. (2010), on the other hand, remember that PES, compared to the other four mechanisms to provide ecosystem services (prescription, penalties, persuasion, property rights), is the only one that can be effective at the global level because it does not represents high threats to local sovereignty of poor countries (Farley et al. 2010). However, the impact of PES over ES provision is still negligible due to the large amount of obstacles to implement it, such as the need to develop efficient mechanisms for disbursing payments and the necessity to have defined property rights, a pre-condition for PES.

In sum, there are a lot of concerns with PESs. It is necessary to ask under which circumstances we should apply this tool; who exactly should receive these benefits; how

to evaluate PES efficiency; how to make the payments; how to raise funds; and many other questions. It is important to point out that the practice of PES seems even more complicated than the theory. However, if carefully used – if implemented and designed with an appropriate level of caution and critical enthusiasm - this tool potentially can bring real advantages at local and broader scales.

## **PES and REDD**

In the context of global climate change, REDD (Reduced Emissions from Deforestation and Forest Degradation) emerged after the Kyoto Protocol as a new strategy whereby forest owners could be compensated for preserving their forests (WRI 2008), with the goal of diminishing greenhouse gases emitted from deforestation. The REDD concept came from Brazilian and American researchers who wanted to include in the accountability of greenhouse gases the ones avoided by the reduction of deforestation and of forest degradation (Moutinho et al. 2011). They presented the “Compensated Reduction of Emissions” proposal in the COP-9 in Italy, in 2003, proposing an international financial compensation to developing countries that were avoiding deforestation and, consequently, carbon emission. REDD projects ideally follow the “three Es” criteria, which are: effectiveness, efficiency and equity. The goal is to know if the project is meeting the climate targets (effectiveness), at the minimum cost (efficiency), with a fair allocation of the resources (equity) (WRI 2008).

Originally, the idea of REDD was limited to PES financial transfers. However, due to the fact that REDD is still a concept that is being created, it has the flexibility to have different formats. An example of this is the REDD+, in which REDD is extended by sustainable forest management, conservation of forests and enhancement of carbon sinks. This variation is being discussed as an alternative for developing countries to

reduce gas emissions and invest in sustainable development and low carbon-emission land use practices. The idea is to include REDD+ in the next climate treaty, after 2020; it will be the first mitigation mechanism to reduce emissions proposed by developing countries.

REDD can be seen as an umbrella term, having different meanings according to “different countries, organizations and even individuals” (Angelsen et al. 2009), as well as different arrangements. The main idea behind REDD is related to financial trades in a multilevel system (global-national-local) of financial compensations, i.e., PES, to promote the increase of forest carbon stocks and reduction of carbon emissions (Angelsen et al. 2009). According to Angelsen et al. (2009), REDD has the main characteristic to provide financial mechanisms to developing countries interested in reducing deforestation.

A core issue in REDD is, therefore, to create a multi-level (international and national) “payments for environmental services (PES) scheme,” as shown in Figure 1-4. This figure shows that REDD has at least three different levels – sub-national, national and international – with communications between them. In the top level (developed countries), buyers will buy environmental services from national or sub-national entities in developing countries. Other conditions that enhance the provision of these services, such as tenure reforms or law enforcement, can also be considered and traded. In the middle level, countries will pay local landowners or state governments to reduce emissions. Also in this case, measures to reduce emissions such as forest management or better local management can be considered in the transaction.

At a local level, the more advanced initiatives are in the states of Acre, Pará and Amazonas. In only two one of these initiatives, Acre and Amazonas, is there effectively a PES project. The REDD initiative of Mato Grosso state, focus of this study, along with Pará, are initiatives that will not be using cash incentives, which is a variant mechanism to implement PES (Jindal & Kerr 2007).

## **Research Plan**

Using a case study focus on the REDD initiative being carried out in Mato Grosso state, I use various conceptual and methodological tools to analyze the feasibility and applicability of PES as a complementary tool to promote conservation. The main research question that I propose to answer with this study is: to what extent does *ex ante* assessment indicate that PES is feasible and useful to be applied in the context of the Amazon Forest?

My analysis focused on the two main stakeholders: the rural smallholder settlers, due to their importance for deforestation, and medium and large landholders due to their importance for providing ES as well as their potential to deforest. Findings from this study are expected to be crucial for researchers seeking ways to understand the complex dynamics of PES programs, as well as decision makers interested in implementing a PES project. The study will develop a framework to analyze key pre-conditions that must be considered for a PES to be successful, as well as conceptual tools for analyzing the economic and socio-demographic conditions of the region, and of local household-level production systems, before the implementation of the project. The use of such tools and analyses can help to reduce the risk of failure of a PES project in other sites is minimized.

The study is organized in six chapters. This chapter introduces the study and reviews the literature on core topics related to this study: payments for environmental services (PES), strategies to reduce emissions of greenhouse gases from deforestation and forest degradation (REDD), resilience and the adaptive cycle; and household livelihood diversity and dynamics, and the example of CSV REED initiative in Mato Grosso. Chapter 2 describes the field site for the study, including the rationale to choose this site (municipality of Cotriguaçu), socio-economic features of Cotriguaçu, and the case study example of CSV, a REDD-like initiative.

Chapter 3 assesses the implementation of a hypothetical Payments for Environmental Services (PES) program in Cotriguaçu municipality, from the perspective of a Social Ecological System (SES) and discusses to what extent this tool might contribute to the resilience of this system. This analysis highlights the usefulness of concepts like SES, resilience, panarchy and the adaptive cycle in analyzing the potential for PES implementation in complex SES settings like Cotriguaçu. It also provides decision makers with information about the resilience of the rural settlers system in this municipality, which will support the development and implementation of public policies in areas where resilience should be strengthened. Chapter 4 provides an *ex ante* investigation of PES pre-conditions in Cotriguaçu. It analyzes to what extent the Cotriguaçu Sempre Verde project (CSV) meets the preconditions established by Wunder (2009) for implementing a PES project and which preconditions appear to be most relevant in this analysis, especially among rural settlers and medium and large private landholders, target stakeholders of this study. Chapter 5 uses a modeling tool, the Ethnographic Linear Program, to carry out an analysis of household livelihood

systems with or without different levels of PES. This analysis also verified to what extent families would agree to receive PES smaller than the opportunity costs to preserve the forests. Finally, Chapter 6 summarizes the results obtained and their contributions to the literature, and proposes to decision makers some suggestions for carrying out *ex ante* assessments to implement a viable PES.

Table 1-1. Deforestation in 2013 by land title category in the Brazilian Amazon

| Land tenure category                  | Deforested area in 2013<br>(km <sup>2</sup> ) | % of total<br>deforested |
|---------------------------------------|-----------------------------------------------|--------------------------|
| Indigenous land                       | 148.0                                         | 3                        |
| Conservation unit                     | 312.2                                         | 6                        |
| Environmental protected area          | 234.0                                         | 5                        |
| Rural settler                         | 1,399.9                                       | 29                       |
| Private property                      | 994.0                                         | 20                       |
| Non designated public land            | 665.2                                         | 14                       |
| Land lacking land titling information | 1,121.5                                       | 23                       |
| Total                                 | 4,874.8                                       | 100                      |

Source: Extracted from IPAM (Instituto de Pesquisa Ambiental da Amazônia), ISA (Instituto Socioambiental), and IMAZON (Instituto do Homem e Meio Ambiente da Amazônia). 2013. O aumento no desmatamento na Amazônia em 2013: um ponto fora da curva ou fora de controle? IPAM ISA IMAZON, Brasília D.F., Brasil. Available from <http://www.imazon.org.br/publicacoes/ou-tros/o-aumento-no-desmatamento-na-amazonia-em-2013-um-ponto-fora-da-curva-ou-fora-de-controle> (accessed March 2014).

Table 1-2: Deforestation in Cotriguaçu by land tenure

| Land tenure category         | % of deforested area |
|------------------------------|----------------------|
| Rural settlements            | 10                   |
| Private areas                | 52                   |
| Escondido indigenous land*   | 0.2                  |
| Other land tenure categories | 37.8                 |

Note: \*Rikbaktsa indigenous group



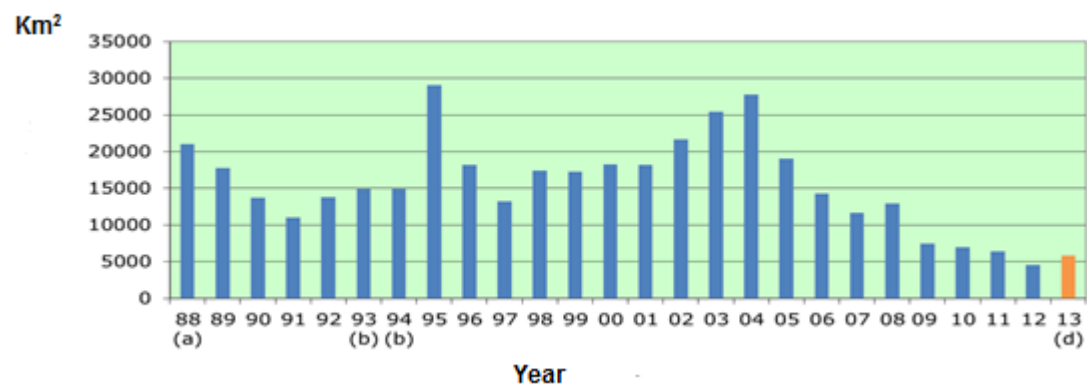


Figure 1-1. Deforestation in the Brazilian Amazon 1988-2014

(Source: INPE (2014)).

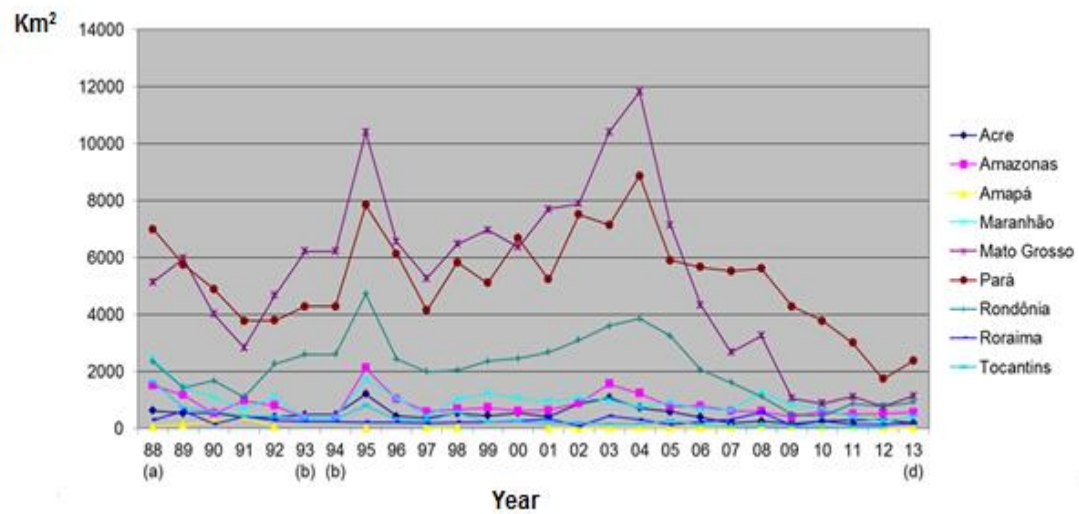


Figure 1-2. Deforestation in the states of the Brazilian Amazon 1988-2014  
(Source: INPE (2008)).

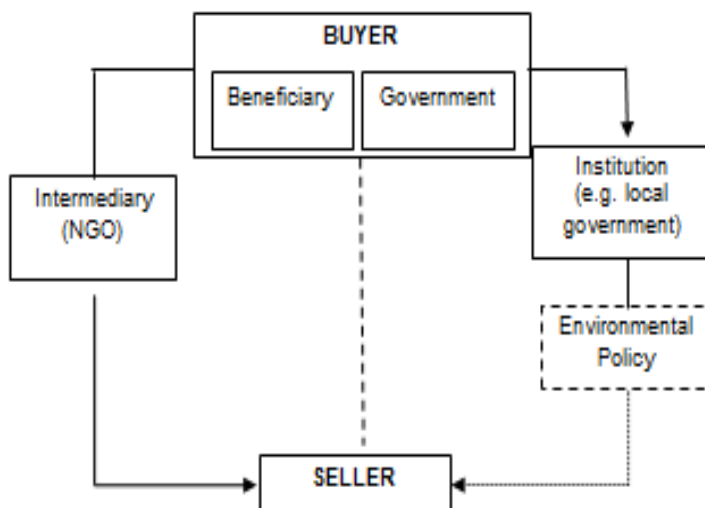


Figure 1-3. Modalities of PESs

Note: Adapted from Wunder et al. (2008).

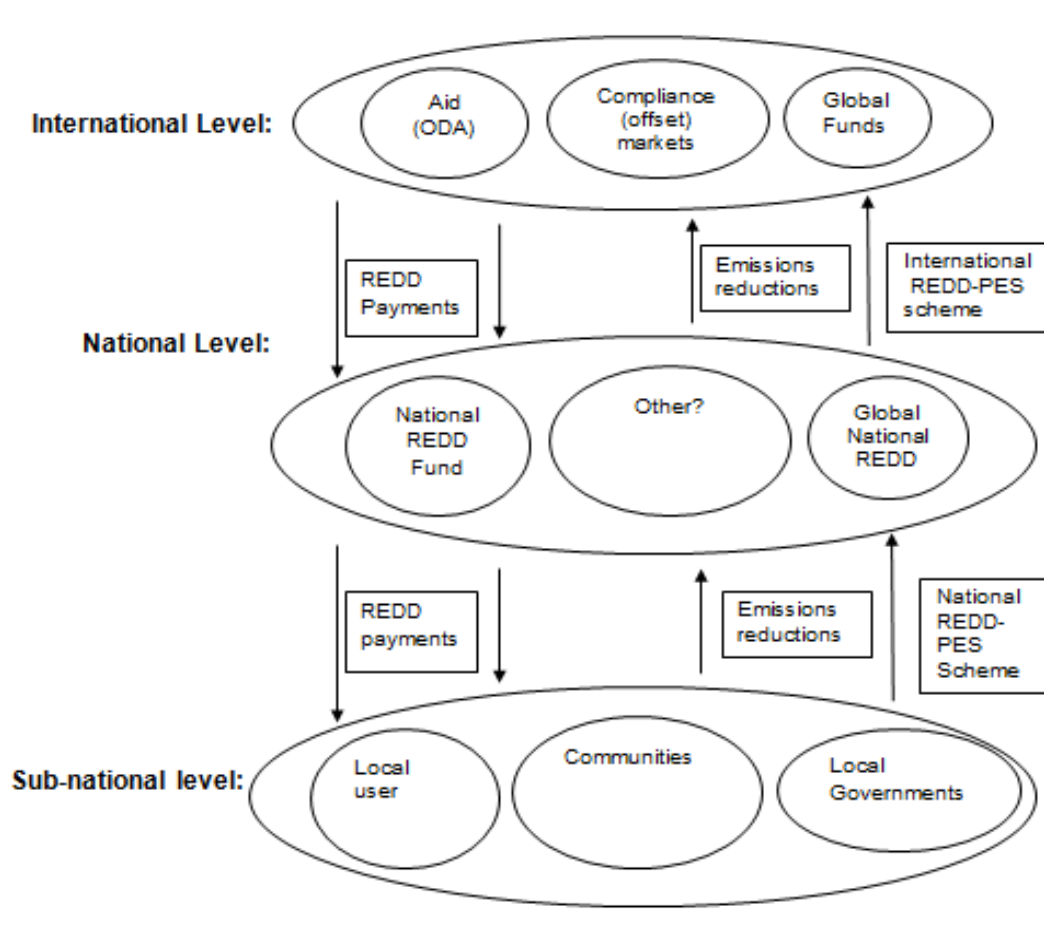


Figure 1-4. Conceptual model of a multi-level REDD "PES" scheme

Source: Angelsen, A., M. Brockhaus, M. Kanninen, E. Sills, W.D. Sunderlin, and S. Wertz-Kanounnikoff. 2009. Realising REDD+: national strategy and policy options. Center for International Forestry, Bogor, Indonesia.

## CHAPTER 2 THE CONTEXT OF COTRIGUAÇU

### **Introduction**

Cotriguaçu makes an appropriate case study of the key conditions and impacts in an Amazonian PES program because it epitomizes export-oriented development on the Amazon frontier and its deforestation consequences. The study region also combines protected areas and diverse smallholder communities in a complex and challenging SES context for PES implementation. Moreover, the existence of a REDD-like program and other concrete efforts provided an opportunity to assess concrete conditions for implementation of an on-going PES program.

An important initiative to reduce deforestation in the state of Mato Grosso was the development, in 2009, of the PPCDQ MT (Action Plan to Control Deforestation and Fire), a state policy whose goal was to reduce up to 89% of deforestation in Mato Grosso (SEMA MT 2009). Also in 2009, the Secretaria do Meio Ambiente (SEMA-MT), Instituto Centro da Vida (ICV), and the Nature Conservancy (TNC) started a discussion about REDD proposals for Mato Grosso; thus, REDD could capture funding from the Amazon Fund for the implementation of the Action Plan.

In 2011, the working group of REDD+ in Mato Grosso designed a state law that has been submitted to the governor in order to be sent to the state legislature for approval. The working group carried out a broad consultation with civil society in order to guarantee the participatory characteristics of the process. The draft of the law that creates the MT REDD system was submitted to the state house of representatives in December of 2012, but still had not been considered two years later. Following these discussions, the state government, in alliance with the NGO ICV, the Instituto

Socioambiental (ISA), and TNC drew up the Cotriguaçu Sempre Verde program (CSV)<sup>1</sup>, a REDD-like initiative in the municipality of Cotriguaçu, Mato Grosso (ICV 2011). The main goal of this project is to promote social and economic development of the municipality through conservation and sustainable management of natural resources among small and large landholders and indigenous people of Cotriguaçu (Micol et al. 2008) and can be considered a REDD-like initiative because, even if this is not explicitly a REDD project, it shares the same final goals to reduce emissions from decreasing deforestation and degradation of the forests.

At the beginning, CSV was conceived to be a REDD initiative with individual cash payment mechanisms, but it has gone through several changes to adapt to the local reality. As is not unusual in Amazonia, the municipality has a multiplicity of stakeholders with particular needs and conditions. To be effective, proponents of the CSV would need to engage all of these stakeholders in the project and think about all those specific issues. Besides, the costs of payments (implementation, transaction and, especially opportunity costs) seemed elevated, thus proponents started to think about other alternative strategies for the project. Presently, CSV does not implement cash payments for program beneficiaries. Instead, it offers technical assistance as well as machines

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<sup>1</sup> The majority of funds for CSV come from Fundo Vale, a Brazilian national entity whose goal is to support sustainable development initiatives. There are also contributions from Climate and Land Use Alliance (CLUA) and Office National des Forêts (ONF), two international institutes concerned about climate change issues. These alternatives of funding sources make the project more independent while still protecting national sovereignty, since the largest part of the funds is national. ICV (Instituto Centro da Vida) is the proponent NGO responsible for the management of funds as well as design and implementation of the project, in partnership with TNC and SEMA.

and tools such as materials for fencing or a freezer for milk storage. In sum, it promotes investments to improve the productive matrix for rural settlers.

In fact, the additionality of the project stands exactly among the stakeholders that have a really high opportunity cost to preserve – the medium and large landholders. These people occupy 54% of the municipality and they are responsible for only 10% of deforestation and still have substantial areas of standing forests. Small rural settlers, on the other hand, occupy only 14% of the municipality but are responsible for 52% of deforestation, and many have already reached their legal maximum of forest clearance. Paying small rural settlers would be certainly less expensive than paying medium and large producers, but might not be worthwhile because additionality of forest-scarce small settlement areas is too low. Thus, proponents of the CSV decided to invest in other initiatives such as the promotion of sustainable livelihood activities, and the strengthening of local governance capacity among more disadvantaged people, among other activities.

ICV has worked in Mato Grosso since 1991 and in Cotriguaçu since 2001, and has produced studies about regional deforestation dynamics. The problem is figuring out exactly how to deal with these different drivers and how to make an attractive REDD proposal. Proponents have listed the main challenges associated with implementing a REDD project in Cotriguaçu, as follows<sup>2</sup>: i) strengthening environmental management at an institutional level; ii) improvement of sustainable forest management practices and spread of the application of these practices; iii) promotion of sustainable cattle ranching in order to avoid degradation of natural resources, namely soil and water due to the low

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<sup>2</sup> Adapted from ICV (2011).

technological level of implementation; iv) improvement of forest governance in rural settlements, which includes support for local political organizations, support for production and trade, and land tenure regularization; v) land tenure regularization within the Escondido indigenous area.

To face these challenges, proponents are designing a participatory plan that will include all the stakeholders in the municipality. The lines of action of the CSV project are distributed in the following components: I) Restructuring of Environmental Management; II) Support for Sustainable Forest Management; III) Support for Best Agropastoral Practices; IV) Strengthening of Governance over Natural Resources in the Rural Settlements; and V) Integration of the Rikbaktsa Indigenous Group into the project.

Component IV, the one directly related to rural settlers, has the goal to provide ways to improve management of natural resources and consequently improve life quality in the rural settlements. The main activities being implemented are related to the promotion of community organization, and environmental regularization, with support for the implementation of CAR (Cadastro Ambiental Rural), a necessary tool to control monitoring and deforestation and also necessary for producers that want to access credit for production.

Especialty after the 1990s, most Brazilian economic development policies were driven by external interests and the potential to export natural resources (Azevedo 2009). Thus, the government implemented “Brasil em Ação” (Brazil in Action, in 1996), “Avança Brasil” (Forward Brazil, in 1999) and the “Programa de Aceleração do Crescimento” (Accelerated Growth Program, or PAC, in 2007), developmental programs



which aimed to promote market integration through the improvement of local infrastructure for export (improvement of ports, roads, among others). In addition, starting in 1998 the federal government allowed small producers to access the PROCERA (Programa de Créditos Especiais para a Reforma Agrária), a special credit line that could be used to buy cattle. In this context, Mato Grosso forests became even more threatened. According to Becker (2007), Mato Grosso followed the export-oriented development model of agricultural frontier expansion with a focus on the external market.

Mato Grosso is one of the most deforested areas in the Amazon region. By 2008, 38% of the previously-forested area of the state had been deforested (Micol et al. 2008). Additionally, 20 out of 43 Amazonian municipalities with highest rates of deforestation are located in Mato Grosso. The state is one of the biggest producers of soybeans in Brazil, and cattle ranching, logging and agriculture are the most important drivers of deforestation.

With 80% of its forests intact, the northwest of Mato Grosso is the last frontier of the state, located in the Amazon deforestation arc. Cotriguaçu municipality, located in northwestern of Mato Grosso (Figure 2-1), had an average rate of deforestation of 1.5% per year between 2000 and 2008, higher than the average of other municipalities in northwestern Mato Grosso (1% per year; ICV 2011). For these reasons, the protection of this forested area is critical.

### **Demographics of Cotriguaçu**

Cotriguaçu municipality has an area of 9,123km<sup>2</sup>, which represents only 1.01% of Mato Grosso's area, and 0.11% of the Brazilian territory (IBGE 2014). Cotriguaçu borders Amazonas state to the north, Juruena municipality to the south, Nova Bandeirantes

municipality to the east, and Aripuanã municipality to the west. The Cotriguaçu area is divided into one conservation unit (The Parque Nacional do Juruena with 14% of the total area and 0.2% of the deforestation of the municipality); one indigenous land (The Terra Indígena Escondido, with 18% of the total area and 0.2% of the deforestation of the municipality); three smallholder rural settlement projects (the Projetos de Assentamento or PAs Juruena, Nova Cotriguaçu and CEDERES II, with 14% of the total area and 52% of the deforestation of Cotriguaçu); and private lands (54% of the total area and 9.9% of the deforestation of the municipality) (SEMA MT et al. 2009).

The Human Development Index of Cotriguaçu is 0.721, which is slightly lower than the Brazilian index at 0.76 (UNDP 2005). In the last census, the population of Cotriguaçu was about 14,965 habitants (IBGE 2009). The economic basis of the municipality, especially in the 1990s, was logging as the main source of employment, followed by cattle ranching.

Cotriguaçu was traditionally occupied by indigenous peoples, especially those who speak the Tupi language, particularly the Rikbaktsa group. In the mid-1980s, the Cooperativa Central Regional Iguaçu Ltda. (Cotriguaçu), a company from Cascavel, Paraná, in the south of Brazil, bought one million hectares of land in northwest Mato Grosso. The idea was to provide land to the small producers of the south, because there was no more available land for agriculture in this part of the country. Some of the original colonists had lost their lands due to the creation of the Itaipu Dam and bought lands from the Cooperativa Central Regional Iguaçu Ltda. with the money from their indemnification.

In 1984, the first immigrants arrived in Cotriguaçu, and in 1986 the city began to be built. In 1988, on the same day, the Juruena municipality and the district of Cotriguaçu within it were created and were legally connected. Finally, in 1991, Cotriguaçu became independent from Juruena, and the municipality of Cotriguaçu was created with only one district. After the mid-1990s, INCRA started the second phase of colonization of the municipality with rural settlers from nearby states such as Mato Grosso do Sul, Rondônia, and other parts of Mato Grosso. This intentional settlement contributed to the rapid population growth in the region.

### **Economic Development of Cotriguaçu**

In Cotriguaçu, cattle production started in 1995, with a boom by 2003. According to the Secretary of Agriculture and Environment, Arnaldo de Campos, this boom was due to the entrance of big cattle ranchers in the region around 1995. Among small producers, especially rural settlers, cattle ranching started in 1998 when this public had access to the PROCERA credit program and used part of this credit to buy cattle.

The current literature places the responsibility for the spread of cattle ranching in the Amazon on governmental policies such as investments in infrastructure, especially the construction of the Belém-Brasília and Transamazônica Highways, the construction of rural settlements and colonization projects, and the provision of credit and federal incentives for cattle ranching (Toni 1999). Additionally, cattle ranching is an advantageous activity due to its low labor input requirements, ease of transport, quick cash flow, and the low marginal cost of forming pasture. Nowadays, almost all loggers in Cotriguaçu are also cattle ranchers to guarantee economic benefits, since logging is considered an unstable activity.

As of 2010, there were approximately 240,600 head of cattle in Cotriguaçu (INDEA 2010), which occupy 185,300 ha of pasture (Ramos Filho 2010). Between 2003 and 2010, the activity grew by 191% and was still expanding, with this expansion dependent on the carrying capacity of pastures to support this growth (Ramos Filho 2010). The current challenge for proponents of the CSV program is to improve cattle management techniques through providing technical assistance and capacity-building to producers, especially small landholders.

In 2005, the Brazilian Ministry of Environment, through the Institute of Environment and Renewable Natural Resources (IBAMA), implemented the Curupira Operation, in which the government dismantled a gang involved in emitting false documentation that authorized the illegal transport of forest products (Moura 2006). In this operation, about 130 people were sent to jail, and 30 IBAMA workers were expelled from the institution. That was the beginning of the decrease of illegal logging in the region, reinforcing the legitimacy of cattle ranching and giving birth to a new trend in Cotriguaçu: Sustainable Forest Management (SFM). SFM promises to provide profits for producers, but it is a long term activity. Despite these advances, there is still a lack of governmental policies (such as credit) to promote and incentivize SFM. Among large private landholders, there has been an increase of Sustainable Forest Management Plans. Cotriguaçu currently has 12 legalized SFM plans which, however, are not 100% transparent, have questionable logging permits, and need improvements in some steps of the exploitation, such as planning and harvesting techniques (IFT 2010).

### **Drivers of Deforestation of Cotriguaçu**

The main drivers of deforestation in Cotriguaçu are, *grosso modo*, the poor management of land use activities (logging and cattle ranching) both by large

landowners and smallholders in the rural settlements. Besides, there is also a political issue that encouraged an illegal occupation of sites in the rural settlements.

During the last 10 years, due to expectations of high income in agriculture and, especially, cattle ranching activities, the municipal forests have suffered great pressures. Currently, the most deforested areas in Cotriguaçu are located in the rural settlements (52% deforested), followed by private areas; (22% deforested) (SEMA et al. 2009). Because of their importance for deforestation and potential REDD programs, this study focused on the analysis of the social-ecological system of rural smallholder settlers as well as medium and large landholders.

Although Cotriguaçu already has a well-defined land regularization situation among private landholders, this is not the case for rural settlements. Not one of the three rural settlements in the municipality (PAs Juruena, Nova Cotriguaçu and CEDERES II) had been fully titled, and some places are in a completely irregular situation. The classic case is in PA Juruena where people invaded the whole common Legal Reserve (in the community of Santa Luzia). According to technicians from the Municipal Environmental Secretary, no legal action was taken against the invaders.

This situation is in fact a political issue because the invasion was stimulated by local politicians who promised to give land to these people, and it is now legally impossible to remove them. Having a piece of land in settlements became a business. Land prices increased a lot and people are illegally selling their sites and buying larger lands in cheaper places. The land tenure insecurity in the settlements has promoted deforestation and depletion of natural resources, because since people do not know for sure if they will have tenure rights, the tendency is to exploit the area as much as

possible through timber extraction and transforming forests into pasture, and then move to a new place.

### **Stakeholders Analyzed in this Study**

The following chapters of my dissertation will focus on rural settlers and on medium and large producers of Cotriguaçu. Table 2-1 presents some demographic information about these stakeholders. All rural settlements were founded in the 1990s, but they vary widely in size and population. By far, the largest rural settlement is the PA Nova Cotriguaçu, with almost 100,000ha and around 1,500 families. This was the last settlement to be founded. On the other hand, the smaller settlement is the first to be founded, the PA CEDERES with 11,359ha and 226 families. In total, these settlements occupy an area of 141,419 ha and the total population of rural settlers in more than 2,000 families (around 8,000 people).

Rural settlements are the most recent land tenure occupation of Cotriguaçu. In the 1980s, the strategy of the military government was to bring landless people to occupy lands in northern Brazil in order to relieve land pressures elsewhere and to protect it from foreign invaders, among other reasons (Hall 2008). Only in the 1990s were these settlements formally founded and, even today, none is titled.

Each rural settlement is divided into a few communities. Four communities in these three rural settlement projects were sampled for this study: Vale Verde, Santa Luzia, Entre Rios, and Nova Esperança (Table 2-2 and Figure 2-2). The initial idea of the CSV project was to involve all 2,100 families of settlers of the municipality. Due to financial restrictions, however, among the four communities visited, Nova Esperança, in rural settlement Nova Cotriguaçu, was the only community attended by the project; the specific actions of CSV for this rural settlement were related to promoting improved

agropastorial practices and governance strengthening. Even if the other communities are not still included in the project, the plan is to include those in the future, when project activities will be internalized as a governmental policy.

I also sampled medium and large private producers, who are distributed throughout Cotriguaçu municipality (Figure 2-1). Below, I will describe the main characteristics and sustainable activities carried out by smallholders and by medium and large producers. An important characteristic of the private producers of Cotriguaçu is the fact that the majority of these properties is titled and has no tenure problems (SEMA et al. 2009). This context is very important in PES initiatives in order to exactly know who detains the ecosystem services.

In each community the sample was of 30 families, randomly chosen in order to avoid bias, making a total of 120 surveys with rural families. In all communities, I managed to obtain a complete list of all households through support of the local health agent and/or other local leaders. After having the complete list, we put the names on small papers, put them in a box, and extracted 30. After selecting the initial 30, we would select 10 extra names to place on a waiting list, in case we could not find respondents or they did not agree to complete the survey. In one community (Linha Entre Rios), we hardly found 30 people living in the site. In this case, we would interview people who had properties both in the settlement project and in the city, but obtained the majority of their income from the rural property..

### **Vale Verde Community**

Vale Verde community is located at the rural settlement Juruena, 50 km from the city of Cotriguaçu, and has an area of approximately 9,324ha, with 21% of it still covered by forests. PA Juruena was owned by the Cotriguaçu Company, (Cooperativa Central

Regional Iguaçu), a company from the south of the Brazil that bought lands in the municipality to be divided and sold to smallholder migrants from southern Brazil. When landless organizations realized that this area would be a rural settlement, people started to invade the area (with the support of local rural workers syndicates). Subsequently, the municipality registered and legalized these occupations.

According to the Brazilian Forest Code (Law 12.727/2012), in the Amazon Region the area of forest to be preserved must be 80% of the whole property. In rural settlements, Legal Reserve can be individual or collective. In the Vale Verde community, lots are 50 ha, and each household has its own Legal Reserve on the landholding. The main drivers of deforestation in the community are agriculture and cattle ranching. Due to the lack of markets for crop production, however, agriculture has decreased over the last few years, giving way to a growth in cattle ranching. Besides, many families got access to credit for cattle raising, which incentivized production.

Two initiatives to promote sustainable activities are being initiated in the community. The first is the development of agroforestry systems, with a mix of peach palm or *pupunha* (*Bactris gasipaes*), and other potential economic fruit species. There are three families (all relatives) engaged in this pilot activity. These families transformed a large part of their pasture areas into a joint plantation of pupunha mixed with other fruit species. They all share the costs of the plantation together and, in the future, will share the profits. This initiative is being subsidized by Gilberto Siebert, an ex-mayor of the municipality, and technically supported by UNDP/GEF.

The second sustainable production activity is being implemented by the international department of Office National des Forêts (ONF-I), a French governmental



institution that has a few initiatives in Mato Grosso such as reforestation, carbon monitoring, and collection of forest products, of which the latter involves settlers from Vale Verde. These settlers collect and sell forest products, mainly Brazil nuts (*Bertholletia excelsa*), as well as seeds of species with commercial timber value, such as mahogany (*Swietenia macrophylla*), *cerejeira* (*Amburana* spp.), *copaiba* (*Copaifera* spp.) and *jatobá* (*Hymenaea courbaril*), from the São Nicolau Farm (managed by ONF-I) and earn an alternative income from this activity.

That said, the above-mentioned initiatives are isolated and only included a minority of the community population. In addition, some of these producers were even criticized by their neighbors for going against the community trends and investing in something so risky and tough.

Compared to the four communities of the study, this one is the most politically organized, which could be a good indication for implementing PES. Talking to some leaders at this site, they showed an interest in trading ES. About six families of this community are working on the seed collection mentioned above, also part of REDD initiatives.

### **Santa Luzia Community**

This community has an area of approximately 7,574 ha and is also located in the rural settlement Juruena, about 40 km away from the city of Cotriguaçu. This community has a different land distribution than the Vale Verde community. Instead of having 50 ha, each family has 25 ha and a symbolic piece of land in the common area, which they are allowed to use minimally as it contributes to their collective Legal Reserve requirement. The problem is that this common Legal Reserve was totally invaded by squatters a few years ago, which has provoked the depletion of natural resources.

The main economic activities in the community are agriculture and cattle raising for milk production. A dairy factory from Nova Bandeirantes, a neighboring municipality of Cotriguaçu, is the main buyer of milk in the community. This factory strategically lent a few milk freezers to settlers, especially in the communities of Vale Verde and Santa Luzia, and twice a week they buy all the milk produced by the settlers. The price paid for each liter is R\$ 0.52. This initiative benefited both the company and the families engaged in the project. To maximize gains, however, households are opening more areas of pasture and raising more cattle than the area can support with current practices, which has provoked deforestation and soil degradation.

The community of Santa Luzia is included in the federal PAA program (Programa de Aquisição de Alimentos) of CONAB (Companhia Nacional de Abastecimento). In this program, producers are asked to sell agricultural products weekly. In return, they are guaranteed a maximum of R\$ 4,500.00 per year on the condition that they are constantly providing agricultural products. This is not a huge amount paid, but helps people to escape from extreme poverty.

This community, compared to Vale Verde, is less prepared to have a PES initiative. According to conversations with some leaders, they have an interest in this type of initiative, but the land tenure situation in this community is still irregular since many landless families invaded the legal reserve, and resources are being quickly depleted.

### **Entre Rios Community**

Entre Rios community is located in the rural settlement CEDERES II and has approximately 6,400ha of area, with 30% still covered by forests. Lot sizes are, on average, 50ha. This area also belonged to the Cotriguaçu Company. Afterwards, the

company hired the Colonizadora do Aripuanã, which sold the plots and took people to the region.

The agreements between the government and the company were that the largest part of the area would be sold to private owners, and some should be given to INCRA to distribute free to smallholders to create rural settlements. For this reason, this settlement is confused in terms of land tenure situation since it is a mix between private owners and official settlers. Due to this confused land tenure situation, it is difficult for outside organizations to engage local people in sustainable development activities.

The community of Entre Rios has three lines of settlement and a population of 35 families (~ 140 people). This settlement does not yet have electricity, and the school is 25 km away from the community center. The community seemed to be socially disorganized. According to local leaders, people never get together and discuss common problems. They have a representative in the municipality (“Vereador”) who served as our guide, but this person never promotes meetings or group discussions due to the fact that people never attend. Our presence in the settlement was considered, by key informants, beneficial to the local people, because through our questions about their livelihood systems, problems, and quality of life, they were able to self-reflect. We were also surprised to realize that both meetings that we promoted (village and women’s) were well-attended and, at the end, they were thankful to us, because they were able to have contact with their neighbors again. Concerning PES, local leaders have almost no knowledge about this tool, but leaders are interested on the implementation of such a project in the municipality.

## **Nova Esperança Community**

Nova Esperança community is located in the rural settlement Nova Cotriguaçu, about 65 km from the city. The community area is 3,500 ha, with 30% still covered by forest. In the 1990s, INCRA distributed the lands of this settlement to smallholders registered in the database. These families were mostly from the south of Mato Grosso and northeast of Brazil. Among the three rural settlements in Cotriguaçu, Nova Cotriguaçu has suffered the least amount of invasions, because INCRA correctly distributed the lands.

In this community, the individual landholdings are larger than in the other communities at an average of 100 ha. The community does not have electricity yet, and they do not know when it will arrive. The main drivers of deforestation in Nova Esperança are agriculture and cattle ranching, which is similar to the other communities sampled.

Even though Nova Esperança was a relatively calm community, it still had some conflicts over land a few years ago. Histories of purposeless violence were common in the past, especially in the extreme north of rural settlement Nova Cotriguaçu, and made this place a lawless land. But the municipal government made efforts to be more present and to calm down conflicts.

There are some interesting isolated experiences in the community to promote more sustainable land uses. As an example, a famous agroforestry system of 15 ha of coffee, pupunha (*Bactris gasipaes*), cacao (*Theobroma cacao*) and a mix of another 50 species, is located in this community. This initiative has been technically supported by UNDP/GEF. Besides being involved in CSV Program, Nova Esperança is one of the communities engaged in the PAA program from CONAB. This is a stimulus for agriculture and could possibly help control the expansion of cattle ranching, because

families in this program have the guarantee of some fixed financial resources from cropping.

Concerning PES, people there are not very aware of this tool. The community has little potential for getting into this sort of initiative due to the large deforestation in this rural settlement and, consequently, low additionality.

### **Medium and Large Landholders**

Medium and large landholders are producers who hold more than 4 fiscal modules of land (~ 400ha). According to INDEA (Instituto de Defesa Agropecuária do Mato Grosso) (2010), there are approximately 80 medium and large landholders in Cotriguaçu. During the field research I randomly chose 40 of these producers to interview.

The majority of these producers are mainly cattle ranchers, but there are also loggers and cattle ranchers at the same time. Some of them have more than one property, where they engage in different activities. These producers arrived in Cotriguaçu at the beginning of the municipality's creation in the 1980s. Their motivation was to purchase cheap lands. With few lands available in the south of the country, and with the creation of Itaipu dam, they had few lands there and the Cooperativa de Triticultores, that brought these people to Cotriguaçu, offered lands for a very competitive price. Due to the difficulties there, however, a lot of families ended up returning to their states of origin. Others were able to overcome the obstacles and stayed in the municipality.

While all producers started with a few hectares of land, some of them had a little more money and bought larger areas. To survive, everybody had to log the area and work with agriculture. Some people specialized in logging. There are even people who originally were loggers in the south of the country and had enough know-how to deal

with this activity. The majority of these former loggers are prosperous now, and have joined the group of medium/large producers. With the crisis in logging due to the overexploitation of the resources, loggers further diversified their activities and started investing money in cattle, especially for meat. Currently, there are producers with more than 10 thousand head of cattle in Cotriguaçu.

Along with the Rikbaktsa indigenous people, large producers are the ones that actually hold the majority of forests in Cotriguaçu. Medium and large producers control 54% of the area and are responsible for 22% of the deforestation in the municipality. By contrast, the Rikbaktsa indigenous people control 18% of the total area and are responsible for only 0.2% of the deforestation in the municipality. Understandably, large producers follow the logic of capitalism, and it is difficult to mention PES to them without bringing to mind the large profits they are able to earn by engaging in other activities. This means that, applying PES to medium and large producers would exclude indirect social co-benefits, such as poverty reduction. Among small properties, by contrast, PES has the capacity to reconcile forest conservation and poverty alleviation.

A summary of demographics, land tenure and causes of deforestation for each community and medium and large producers is presented in Table 2-3. We can observe that cattle ranching is a common activity for both stakeholders. Besides, rancher's' areas are much larger than those of small producers and deforestation is much lower, only 22%, compared to 52% in small producers areas. Finally, ranchers are more likely to have land title than are small producers.

### **Main Sources of Income**

The main sources of income in the four smallholder communities analyzed, and among medium and large producers, are described below (Figures 2-3 and 2-4). In all

communities, cattle ranching was, by far, the land use activity that provided the highest income. The second land use activity with higher income was agriculture.

Vale Verde is the community where people have the highest total income derived mainly from cattle ranching activities. Milk is also important in Santa Luzia, but less important for the other locations. This is due to the fact that a milk company from Juina (*Indústria e Comércio de Laticínios Vale do Juruena*) is regularly buying milk from Cotriguaçu small producers, especially those from the Vale Verde community. The company left some milk freezers (machines that conserve large quantities of milk until the truck collects it) and once a week they buy all the milk gathered during the week.

Figure 2-4 shows the difference in income between cattle ranching for meat and cattle ranching for milk. Santa Luzia is the largest milk producer, and Entre Rios is the smallest one. The majority of milk producers are obtaining little income, and few producers are really making profits from milk. Although the revenue is currently low, given the strategic nature of this product as well as the guarantee of constant source of income, the Mayor of Cotriguaçu is investing in this activity, and the CSV project is promoting good practices in milk production with support from Embrapa. At the time of data collection, the average price of milk was R\$ 0.52/liter.

According to the IBGE 2006 Census (2009), Cotriguaçu produced 2 million liters of milk, on 544 rural properties in 2006. This corresponded to 3,707 liters per year/property, which means 10 liters/day/property. Each cow in Cotriguaçu produced around 1,087 liters/year, higher than the average production in Mato Grosso of 918 liters/year.

Milk production is widespread throughout the Amazon, and may benefit many families due to the production of milk and meat, and manure for gardening, and the low investment in machinery, but small scale producers tend to get lost in their accountability and may not know for sure whether or not the economic returns are worthwhile. Besides, many producers in the field complained that this system was like slavery to them and their families because they were not able to stay away from their properties even one single day due to the need for daily milking.

In sum, Cotriguaçu is a challenging municipality to implement preservation-oriented policies. If on one hand the region shelters the largest forest remaining in Mato Grosso, on the other hand many areas are occupied by rural settlements as well as by medium and large producers. Both stakeholders are interested in cattle ranching and other non-sustainable livelihood activities that are not compatible with the preservation of the forest.

This is the context where CSV, a REDD-like program, will be implemented. CSV appears as an opportunity to decrease deforestation and to include the municipality in a promising market for environmental services. Thus, this study has a special interest in the potentialities for a PES project to reduce deforestation, to improve the rural settlers system resilience, and to contribute to maintaining the livelihood conditions of small producers that still detain forests in their properties. However, many aspects must be assessed in order to understand these potentialities for PES and if this is the best tool to be applied. A better knowledge of the rural settlers as a social ecological system is the first step to cognize the potentialities to insert a PES, and will be assessed with details in the following chapter.



Table 2-1. Demographics of rural settlements in Cotriguaçu

| Rural Settlement   | Area (ha) | Number of families | Year of foundation |
|--------------------|-----------|--------------------|--------------------|
| PA Nova Cotriguaçu | 99,988    | 1,479              | 1994               |
| PA Juruena         | 30,072    | 400                | 1996               |
| PA CEDERES         | 11,359    | 226                | 1991               |
| Total              | 141,419   | 2,105              |                    |

Table 2-2. The four communities studied among the three rural settlements of Cotriguaçu

| Community      | Rural settlement   | Year of establishment | Community area (ha) | % deforested area | Current population | Total population 2 years ago |
|----------------|--------------------|-----------------------|---------------------|-------------------|--------------------|------------------------------|
| Vale Verde     | PA Juruena         | 2004                  | 5,000               | 50                | 400                | 480                          |
| Santa Luzia    | PA Juruena         | 2006                  | 7,296               | 50                | 140                | 190                          |
| Entre Rios     | PA CEDERE S        | 1992                  | 4,500               | 50                | 260                | 300                          |
| Nova Esperança | PA Nova Cotriguaçu | 1993                  | 6,000               | 60                | 450                | 520                          |

Source: Field work (2011).

Table 2-3. Information on four communities of small producers, and on medium and large producers of Cotriguaçu

| Community                               | Vale Verde                           | Santa Luzia                              | Entre Rios              | Nova Esperança        | Medium and large landholders |
|-----------------------------------------|--------------------------------------|------------------------------------------|-------------------------|-----------------------|------------------------------|
| Year settled                            | 2004                                 | 2006                                     | 1992                    | 1993                  | Since 1984                   |
| Community total population              | 400                                  | 140                                      | 260                     | 450                   | 250 (land owners)            |
| Number of families                      | 150                                  | 47                                       | 86                      | 150                   | 80                           |
| Total area (ha)                         | 5,000                                | 7,296                                    | 4,500                   | 6,000                 | 112,669                      |
| % Deforestation                         | 63                                   | 69                                       | 80                      | 60                    | 10                           |
| Land tenure situation                   | Land re-concentration                | No common legal reserve                  | No common legal reserve | Land re-concentration | Areas regularized            |
| Causes of deforestation and degradation | Cattle ranching, land concentration* | Disorganized occupation, cattle ranching | Cattle ranching**       | Cattle ranching       | Cattle ranching, logging     |
| Number of people interviewed            | 30                                   | 30                                       | 30                      | 30                    | 40                           |

Note: \* New settlers are buying lands from people who are abandoning the settlement and combining lots to become larger producers. \*\*The settlement does not yet have electricity, and some people are abandoning the land to move to the city and other places where land is cheaper.

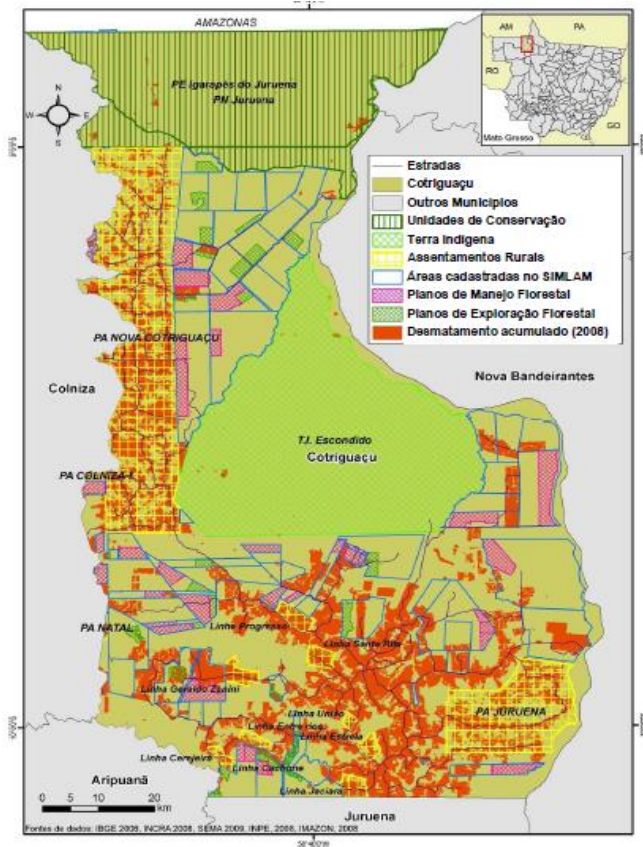


Figure 2-1. Map of Cotriguaçu

(Source: Extracted from SEMA MT et al. 2009)

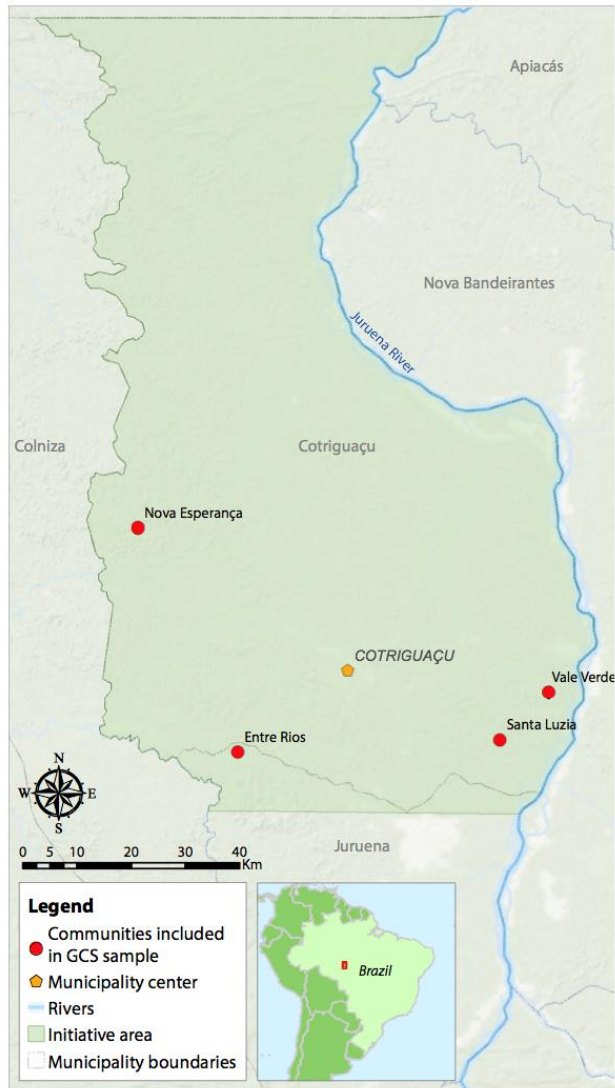


Figure 2-2. Map of location of communities studied

Source: Field work (2011).

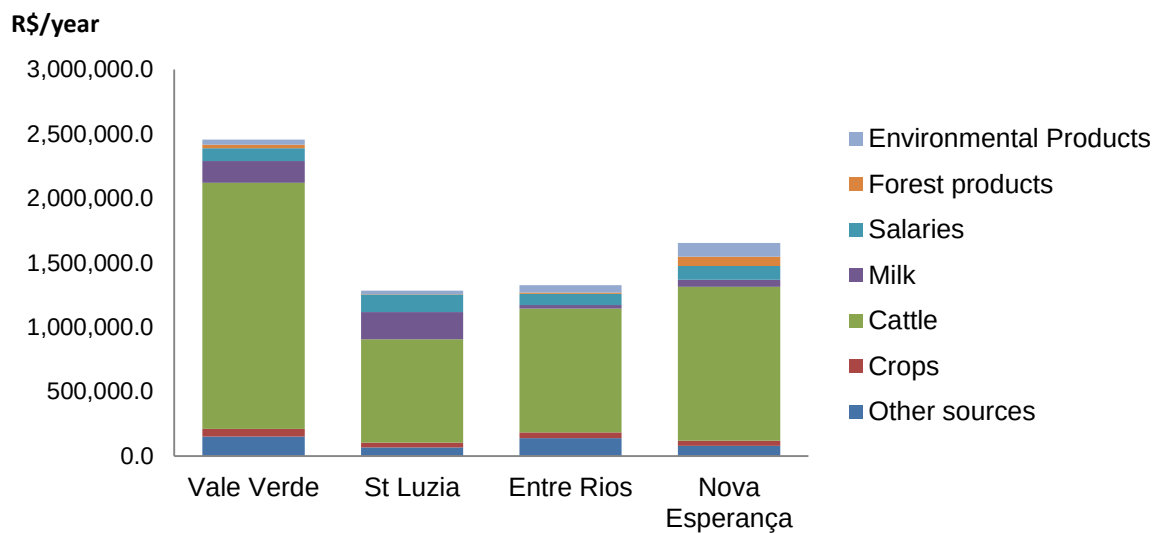


Figure 2-3. Main income source (R\$/year) in the four communities in Cotriguaçu

Note: Sample size n=120.

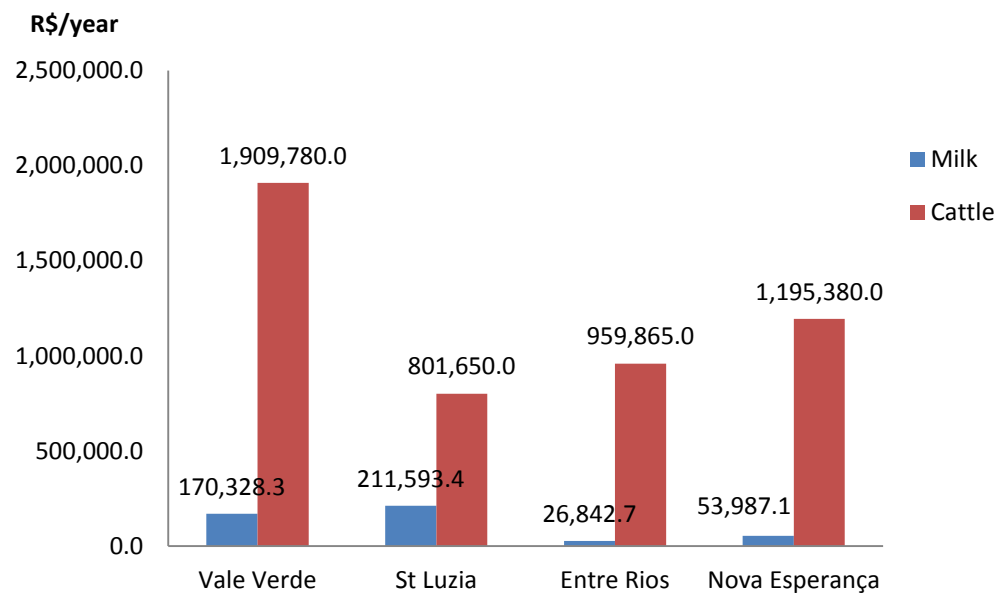


Figure 2-4. Income from cattle and milk in the four communities visited

Note: Sample size n=120. Source: Field work (2011).

CHAPTER 3  
PAYMENTS FOR ENVIRONMENTAL SERVICES AND SOCIAL, ECONOMIC AND  
ENVIRONMENTAL RESILIENCE AMONG RURAL SETTLERS IN COTRIGUAÇU

**Resilience of Social Ecological Systems**

According to the Resilience Alliance (2010), resilience is a property of social and ecological systems (SES) that encompasses both ecosystem resilience, which is the capacity of an ecosystem to tolerate disturbances, and social resilience, with the addition of humans, and their impacts, to these systems. The SES have three main characteristics that determine their potential resilience: “i) the amount of change the system can undergo and still retain the same controls on function and structure; ii) the degree to which the system is capable of self-organizing; and iii) the ability to build and increase the capacity for learning and adaptation.”

Resilience may have many perspectives. One of them is related to the capacity the system has to tolerate disturbances but there is also another one, the speed the system takes to recover after a disturbance (Adger 2000) (Figure 3-1). The fact is that resilience is a term present in many disciplines, from psychology to ecology, and there is not a common concept across all of them (CARRI 2013) and in social sciences and engineering it is widely used to refer to recovery after disasters (CARRI 2013). In 2013, the Community & Regional Resilience Institute (CARRI) identified around 46 definitions of resilience in physical, ecological, community and individual domains (CARRI 2013).

Figure 3-1, extracted from Adger (2000), shows two ways to represent resilience in an ecosystem scale: resistance and recovery. The distance between  $r'$  and  $t$  (threshold) in Figure 3-1a is the system capacity to absorb or resist a perturbation and means that highly resistant systems have a larger distance than lesser resistant ones. Figure 3-1b shows the time the system takes to recover. Resilient systems return



quickly. On the other hand, less resilient systems will take longer to recover (Adger 2000). Some authors or institutes such as CARRI (2013) also consider “creativity” to be an important aspect to promote system resilience (special in social systems) to reach a better situation than before the occurrence of disturbance.

Resilience is a concept commonly used in ecology (Adger 2000), and its application in social systems is contested. In fact, social scientists argue against simply taking the concept of resilience from ecological sciences and applying it into social systems without considering that there are many differences between them (Adger 2000), which may generate distorted analysis. On the other hand, Adger (2000) points out that social resilience might be explored and has many lessons to provide on how society works.

According to WRI (2008), resilience includes three dimensions: social, ecological and economic. Social resilience can be defined as the capacity to face and resolve external and internal crises. This resilience can be supported with the improvement of social capital and social identity. Ecological resilience is related to the ability ecological systems have to overcome natural or human-caused disturbances such as climate change, pollution or over-exploitation of natural resources. Finally, economic resilience is the ability to recover from economic shocks (WRI 2008). In this study, resilience will be considered as the maintenance of systems, in a context of constant changes, due to the fact that complex systems are always reorganizing themselves and adapting to an unstable reality (Buschbacher 2011).

There are many studies linking the relevance of PES to poverty reduction or the provision of environmental services but there are not many studies linking PES to

communities' adaptation to disturbances. Van de Sand (2012) published an interesting study in which she linked PES with adaptation to climate change, using concepts such as vulnerability or socio-ecological systems. Results suggested that PES could enhance the capacity to adapt to climate change and, as a consequence, adaptations supported the reduction of vulnerability of the systems (Van de Sand 2012).

Considering the rural settlements of Cotriguaçu as a SES, this chapter considers the most important disturbances of this SES, and analyzes the capacity of a PES tool (in the context of CSV project) to improve, or build, system resilience. Identifying the weak and strong points of resilience among Cotriguaçu settlers can suggest in which areas public policies have to be implemented both to support the capacity to respond to disturbances, such as natural, social or economic shocks, as well as to make the recovery speed faster. Specifically, there are three research questions:

- Question 1: How have drivers interacted at different historical moments to frame the adaptive cycle of settlers in Cotriguaçu?
- Question 2: Would the resilience of the rural settlers of Cotriguaçu SES be stronger with the implementation of a PES project with individual cash payments in the context of the CSV project?
- Question 3: To what extent could a hypothetical PES project with individual cash payments increase the CSV project's (a REDD+ scheme) impacts?

To answer these questions, the chapter is organized in five parts:

- First, I outline the attributes of SESs and the indicators that point to changes in the attributes.
- Second, I describe specific approaches such as adaptive cycles and panarchy to analyze social-ecological complex systems
- Next, I describe the methodological approach to analyze different aspects of resilience of rural settlements in Cotriguaçu
- After that, I present the results and discussion for each research question

- Finally, I present the conclusions and recommendations

### **Attributes of SESs**

In order to understand SES dynamics, it is necessary to identify the most relevant attributes of it, which are the key characteristics without which the system would be different (Gunderson & Holling 2002). In addition, we also need to investigate what drivers are affecting the system the most, in order to focus on policies and initiatives that may control these effects. Finally, we want to determine if the current situation is the most desirable or if the system needs changes. If changes are needed, it is necessary to focus on what should be done in order to reach the most desirable situation.

Considering that we are working on a system of high complexity, it seems prudent to complement my comprehension of the system's most important attributes with the vision of other researchers on the same SES, such as Olival et al. (2011), who carried out a study about SES applied to the Amazon. These authors proposed a list of four key attributes (relationship with land; labor; social organization; and productive matrix), distributed across three dimensions (social, economic and ecological). This study adapted these attributes and included one more dimension (cultural). The final result was a list of seven main attributes distributed along four dimensions (Figure 3-2), as well as a list of indicators to operationalize these attributes. Thus, in the case of the first attribute, "relationship with land," for example, the correspondent indicator will be a safe and sustainable relationship with land. For the attribute "production," the indicator will be a diversified production, and so on.

System resilience might be understood as the capacity this system has to adapt to internal and external drivers. Depending on the system state, resilience can be

desirable or not (Brand & Jax 2007). Thus, system attributes will constantly be re-organizing themselves in order to maintain (or reach) the desirable characteristics of the SES.

A desirable situation of the SES should be produced by positive values of these indicators. As will be described in detail in the methodological section, **deforestation** and **out-migration** are the current undesirable situations in the rural settlements SES that affects most of the seven attributes. For better system dynamics, this Business as Usual (BAU) scenario should be shifted to a scenario of decreased turnover of people in the settlement, and more sustainable use of forests as indicator of the attribute environmental services.

This may be obtained by changing the conditions of SES attributes, which means that large changes in the SES and the inclusion of new attributes would not be needed. Therefore, in this study we focus on the improvement of the resilience of what is “working well,” such as the diversification of production and enough income to the families, and to promote adjustments in attributes that are not good enough, such as relationship with land and natural resources, which are leaving the system more vulnerable to disturbances.

By doing that, it will be possible to analyze how the SES will behave when a PES is implemented. Knowing to what extent the system supports disturbances, or how quickly the system recovers from disturbances, is fundamental for analyzing whether or not the implementation of a PES can work as a factor to strengthen SES resilience.

For a PES to be implemented, we need to preferably have available original forest cover and enough ES to supply. In addition, providers will have to have other

alternative economic activities; otherwise they will be dependent on the income from PES. Other system attributes, such as income, are totally related to the type of land use in which they will be engaged. Our rural settlers' SES has a diversified production, which helps to guarantee sufficient income. However, ES and forest cover are at serious risk, and the relationship with the land is not sustainable and safe, since it is too oriented to non-sustainable cattle ranching and agricultural activities. Thus, if we intend to analyze feasibility of PES in this context, the analysis should be focused on the attributes directly related to PES, which are: relationship with land; land cover; environmental services; and income.

### **The Logic of the Adaptive Cycle and Panarchy**

In a large system to be analyzed such as the rural settlers of Cotriguaçu, a variety of attributes make up part of the social structure, such as institutional arrangements, migratory flows, and economic decisions, as well as of the ecological structure, related to the use and exhaustion of natural resources. This system also has a particular regime, i.e., “a set of system states within a stability landscape” (Folke et al. 2010).

The regime in this SES is partially stable and is characterized by high deforestation rates, related to non-sustainable livelihood activities, and high out-migration levels, due to lack of infrastructure in rural areas, non-access to market, among other reasons. Some authors would consider as a more desirable regime an opposite scenario where: deforestation rates decrease, with the incorporation of sustainable livelihood activities, and the people are maintained in the rural settlements (Resilience Alliance 2010). Due to this complexity, we will use specific tools designed to analyze complex systems, such as adaptive cycles and panarchy, described below.

The adaptive cycle idea comes from the study of ecosystems and is a tool used to analyze processes of destruction and reorganization (Resilience Alliance 2010). We can represent SES using the adaptive cycle, derived from the dynamics of ecosystems (Gunderson & Holling 2002). The adaptive cycle has four phases: 1.growth (r); 2. conservation (K); 3. collapse (omega) and 4.reorganization; as shown in Figure 3-3.

The growth phase, from r to K, and the reorganization phase, from Omega to Alpha are the most important phases of the cycle. The growth phase is usually slow. Ecological and social systems usually take a long time to increase the amount of nutrients, biomass (in the case of ecosystems), and skills, social capital, money, trust (in the case of social systems) (Gunderson & Holling 2002). The conservation phase tends to be stable and durable but system slowly accumulates vulnerability. Finally, a disturbance occurs – phase 3 – the whole system collapses; resources which were previously locked up in the system's structure get released, and then these resources become available for restructuring and reorganizing – phase 4. The capacity to support or deal with these disturbances, as well as the speed to recover from it are both determinants of the system resilience.

In some cases, a shift in the current undesirable regime (or stability landscape) would be desirable but, because the system is resilient, not possible. The solution would then be to promote changes in variables and attributes to move the system to a new regime. This system capacity is called transformability (Folke et al. 2010).

When we have a nested hierarchy of adaptive systems, we are representing panarchy, which is a group of adaptive cycles representing the interaction of ecosystems, knowledge, power and social organization across diverse scales, from

local to global. This is a way to show how reality works instead of analyzing parts separately. In order to represent the panarchy of a municipality, we need to look at the history of the system, the drivers of the disturbances of the system, and the scales and origins of the many adaptive cycles (Bernasconi 2012). All of these variables will have important implications for the potential implementation of a PES program.

### **Methodological Approach**

The study was developed through the collection and analysis of primary data with rural settlers and key informants of Cotriguaçu as well as secondary data about the region. It was funded by the Center for International Forest Research (CIFOR) and by the University of Florida (UF). Both were implementing activities in the municipality. CIFOR was conducting a study on REDD and I was the field research supervisor of this study. The UF was carrying out a course called “*Gestão Colaborativa de Sistemas Sócio-Ecológicos-Econômicos na Amazônia Brasileira*” (Collaborative Management of Socio-Economic-Ecological Systems in the Brazilian Amazon) in which I participated as a student. This course was implemented from 2010 to 2012, and I was able to carry out open interviews with local stakeholders and to learn about the SES tool.

The field research in Mato Grosso was conducted between March 8 and May 11, 2010, when my team of 6 enumerators and I interviewed medium and large landholders, and between July – August 2011, when we interviewed rural settlers. I conducted 120 interviews in four communities in the three rural settlements (30 interviews per community) of the municipality: Settlement Projects (PAs) Nova Cotriguaçu, Juruena and Cotriguaçu-CEDERES.

Following CIFOR’s method for choosing the communities to be included in the survey, I completed a form called Village Appraisal with a variety of information (such as

deforestation rates, area, number of rural settlers, among others), from 15 communities to select sample villages through an iterative process in collaboration with Dr.

Subhrendu Pattanayak, Professor at Duke University and part of CIFOR's team. In his analysis, I encouraged Dr. Pattanayak to favor the communities of Vale Verde and Nova Esperança in the selection of communities for the study due to their strong likelihood of being included in the CSV project. On the other hand, communities that had a smaller chance to be included in the REDD initiatives were removed from the selection process.

In order to answer the three research questions of this chapter, regarding how the drivers of deforestation and degradation in Cotriguaçu have interacted at different time periods (Q1), the analysis distinguished between the different phases of the adaptive cycle of smallholder settlers and panarchy in the region since 1995. Building on this analysis, to address Q2 we first explored indications of the degree of current resilience of the system, and then analyzed whether or not the insertion of a PES might increase it. In addition, the study sought to investigate whether or not the current situation is ideal, suggesting whether or not resilience should be encouraged in order to maintain the same regime. For Q3, we projected the potential impacts on the system of the insertion of a hypothetical PES in the context of CSV program.

## **Results and Discussion**

### **Framing Cotriguaçu's History in the Adaptive Cycle**

In order to understand the evolution of the social-ecological system of Cotriguaçu's rural settlers, this analysis focused on the principal drivers of deforestation, and the factors at different levels that affected them over the past three decades of the municipality's history. This contextualization is a fundamental tool to frame the SES within the adaptive cycle.



Two moments are key for Cotriguaçu and will be determinants to the building of adaptive cycle: the first is the initial euphoria of occupying this new land, which was quickly overcome by the second key moment when obstacles and disappointments to living in the region emerged, and frequently led to the final abandonment by settlers of their sites. In informal conversations, people stated how difficult it is to live in Cotriguaçu, especially due to the isolation, the mosquitoes (“piuns”) and the lack of social assistance, thanks to an absent state.

Cotriguaçu municipality was established in 1984. Since 1990, three rural settlements were created by the Federal Government: the PA Cotriguaçu – CEDERES (11.359ha, capacity of 226 families, with 207 families settled, created in 1990), the PA Juruena (30.072 ha, 524 families of capacity, 466 settled, created in 1997), and the PA Nova Cotriguaçu (99.988ha, capacity for 1.479 families, 1.282 settled, created in 1995).

These rural settlements of Cotriguaçu were originally occupied by people from Mato Grosso and Rondônia. In the study sample, 24% were from Rondônia, 22% from Paraná, 21% from Mato Grosso and 33% from other states from Northeast Brazil (field data collected in 2011). At the beginning of the rural settlements, the most important livelihood activities were logging and agriculture. Timber was the first resource to be exploited in Cotriguaçu rural settlements and is now almost a scarce resource, especially if we consider commercial timber. In fact, producers deforested these areas to start agricultural activities and to show INCRA that they were making the lands productive, because this was a condition to have a lot. Very few producers were able to make money with timber, and even large loggers had to diversify their production to be in a less risky situation.

Once installed, producers started their first productive activities, such as agriculture, especially coffee, maize, coffee and rice for consumption and trade. In the first years, producers conventionally cut down the native forest and also made some profits selling the logs. As a way to have an immediate income, a lot of families worked in the logging industry, in the urban area.

In 1990, as part of a neoliberal macroeconomic adjustment, the Federal Government cut subsidies to the cooperative that was buying local agricultural production. The producers lost the main buyer of their production and the cooperative failed. The result of that was the abandonment of the area by many families and the land re-concentration by the ones who stayed there and bought these lands. Due to the lack of markets for agricultural products and low infrastructure in the rural settlements, producers started to diversify their production, buying cattle at a very small scale, and looked for new possibilities of markets. In a few years, the system became specialized in cattle ranching. Thus, those producers that survived the removal of the subsidy could be regarded as resilient. Even if there is a turnover of people, there are also some who had been there since the beginning of the settlement. The ones who stayed were able to maintain themselves as rural settlers in spite of a series of stresses. Probably the most important factor that contributed to this resilience was the capacity of these stakeholders to adapt their livelihood activities after disturbances to a previous activity in which they were engaged. This capacity could be associated with their skills, social network, personal interests, among others.

At the end of the 1990s, with the agricultural market failure, and the possibility of accessing credit for cattle, they started to practice dairy farming. Currently, the

government is promoting the strengthening of cattle ranching (especially dairy) among small producers because this is an activity with high economic potential that also guarantees some stability to producers. However, due to the large demand for capital and labor, a lot of producers are dedicating themselves only to cattle, which again, is risky if there is a crisis in the milk market (Blau 2012). Thus, the end of the 1990s was a period of large investment in infrastructure in the municipality and in the rural settlements, as well as investments in social services in the rural settlements (schools in the whole municipality; health center in the PA Nova Cotriguaçu; and building of a hospital in the urban center). Until then, people had to go to Juína, a municipality six hours away from Cotriguaçu, if they needed basic health assistance.

This new level of organization happened from 2000 to approximately 2005 and due to this range of investments, rural settlers had a more viable social and economic situation. However, high rates of deforestation accompanied this investment. Also, due to the increase of land values, greater out-migration rates still occurred because many people sold their lands for a high price and went to other rural settlements in other municipalities.

As a consequence of continuing high rates of deforestation, in 2008 the Federal Government included Cotriguaçu in a list of priority municipalities in the fight against deforestation. For that reason, credit for production was cancelled, environmental monitoring was strengthened, and 17 other policies were implemented, such as bringing electricity to the region, and enforcing the obligation to register all rural properties in the “Cadastro Ambiental Rural” (CAR). This last tool is fundamental to monitoring rural activities and deforestation in the properties. As a complementary policy, and as part of

the international discussions on the need to do something to combat climate change, the design of the REDD initiative started in Mato Grosso. These REDD initiatives were related to, among others, the implementation of the CSV project, as well as command and control tools such as a strong presence of environmental enforcement institutions in the region.

Finally, in 2010 the REDD initiative was redesigned. This included a decision not to implement a PES program with cash payments in the area. Instead, multilevel non-monetary initiatives were implemented among the variety of stakeholders: rural settlers, medium and large private landholders and indigenous people. In addition, the governance capacity of local institutions was also enhanced.

The second most-important economic activity still is cattle ranching due to the climate, soil and land tenure conditions (IBGE 2009). Currently, the base of Cotriguaçu economic activities is still the logging industry, which is one of the biggest sectors responsible for employment and income (IBGE 2009).

Due to the increase in deforestation rates in Cotriguaçu, the state government, represented by SEMA MT, along with ICV and TNC, designed a proposal to implement a REDD initiative in the region as a way to conserve the forested areas and to promote rehabilitation of degraded lands. Table 3.1 summarizes this whole history and points out external and internal events that provoked changes and disturbances in the system studied. These disturbances, classified as ecological, social, economic and political, help to understand the adaptive cycle of Cotriguaçu.

### **Disturbances that Affected the SES**

The majority of the events that provoked disturbances in the system rural settlers (Table 3-1) came from the higher-scale political dimension, and provoked important

positive or negative consequences on the local level, which will be explained below.

There are important linkages among both external and internal levels as well as among social, economic, political and ecological dimensions. For example, the international pressures against deforestation in the Amazon, in the 1990s, were critical to the implementation of national policies that were reflected in local livelihood systems.

Furthermore, international market preferences were also crucial for livelihood choices, which, in turn, provoked an increase in deforestation in some areas.

Table 3-1 shows the most important external and internal events that impacted the rural settlers system. Instead of disasters, the nature of these events is especially related to political and economic interventions originating from outside of the system. Notwithstanding, these events were shocking, and caused determinant disturbances, which, allied to a persistent lack of infrastructure, provoked people's out-migration, a permanent undesirable constant in the system. Below there is a list of the strongest disturbance events in the system.

### **The creation of Itaipu dam**

The first key event was the creation of Itaipu Dam, at the beginning of the 1970s. This dam occupied a large land area in the south of Brazil and promoted out migration to the north. Many families in the south stayed without any productive lands and the possibility to migrate to the north of the country and have areas larger than in the south, became very attractive. Cotriguaçu was, thus, one of these areas that received large waves of migration from people from the south in the 1980s.

### **Cut of subsidies to rural cooperative in 1990**

As mentioned above, in the 1990s, as part of the neoliberal macroeconomic adjustments, and oriented by the World Bank's structural adjustments policies, the

federal government was forced to decrease its participation in the states productive sectors. Without this support, the cooperative that used to buy local products was unable to survive. This cut of subsidies undermined the permanence of rural settlers by the 1990s. Without markets to absorb their production, these people had no way to sell their produce and had two options: to abandon their lands or to change their livelihood activities. Many families engaged in mining activities in neighboring municipalities. Only wealthier families could choose the second option. They were able to start buying cattle and did not need to out-migrate.

### **Subsidies for cattle: the PROCERA program**

In 1998, the government provided small producers the possibility to access credit to buy cattle through the PROCERA program. This was determinant to increase deforestation, since people opened native forest areas to raise cattle.

### **Governance shocks**

At the end of the 2000s, the Federal Government created a blacklist of municipalities with the highest deforestation rates. Cotriguaçu was number 5 on this list. Through this tool, municipalities suffered various sanctions such as credit embargoes, subsidy cuts, and fines. This was, probably, the most important motivation to change environmental BAU context of high deforestation rates in the municipality.

### **REDD discussions**

As a consequence of the widespread, large-scale promotion of cattle ranching in Brazilian Amazonia, the international community started to pressure the government to decrease deforestation in the Amazon and to protect local communities. Discussion of REDD appeared by 2010 as an additional turning point on the scenario of preservation-oriented policies. The implementation of projects like CSV is largely due to the

international community applying pressure to preserve the Amazon, , not because local proponents decided to become a green community and to implement a different kind of development project.

Looking at the important events above, we can conclude that Cotriguaçu's history has been very susceptible to external events, from the creation of the municipality, due to the implementation of the Iguaçu Dam, until the implementation of local preservation policies, as a reflex of global REDD policies. However, there are two events that were the most determinant: the cancellation of subsidies and governance shocks. The system suffered a strong collapse when people were not able to sell their production and replaced these activities with more market-oriented ones such as raising cattle.

Another important aspect is that both events are contradictory. One regards development, the other regards conservation. The equilibrium between both could have been a good start to promote the conditions for retaining people and preventing their migration to other places.

### **The Adaptive Cycle of the Rural Settlers SES in Cotriguaçu**

Based on these historical facts, on stakeholder's perceptions and on my own perceptions, I designed the Adaptive Cycle of the SES Rural settlers Cotriguaçu (Figure 3-4) and illustrated its history using the cross-scale concept of panarchy in Figure 3-5.

In the Adaptive Cycle, the four phases were distinguished in an attempt to understand the system's process of growth, stabilization, collapse, and reorganization. Certainly, this exercise was challenging due to the fact that the system had a level of complexity that was not totally absorbed by this tool. However, the exercise was rewarding in terms of reflecting about all the phases.

Panarchy is a group of adaptive cycles in the SES that interact among different scales (Holling 2001). The scheme shows interactions among public policies at international, federal, state and local levels and illustrates the “bouncing” effects of these interactions between scales. The effects are stronger from higher (larger and slow), to local (smaller and faster) scales but also happen in the opposite sense, according to Holling’s theory. The scheme also shows that in the past decade the Cotriguaçu SES has been vulnerable to environmental policies and external market demands and it is now at a crossroads with a possible transition to a low-carbon economy as the alternative to the BAU.

### **Phase I (1984-2000)**

In the adaptive cycle, this is the “r” phase (growth) of the system (Figure 3-4). This phase happens from the creation of the municipality (1984) until the beginning of cattle ranching livelihood activity among rural settlers in the beginning of the 1990s. Itaipu Dam construction, migration of displaced farmers to Cotriguaçu and creation of the rural settlements are important facts of this phase.

### **Phase II (2000- April 2005)**

“K” is the phase of system stabilization: the “golden years” of Cotriguaçu. It is debatable whether or not there was a true stabilization in this period because of the lack of real balance between development and preservation of natural resources; life was still hard in the settlements, which made people continue to abandon the areas. In fact, there was a high turnover of people in the rural settlements since people would come and go, and few of them were able to stay for a long time. In other words, this was also the beginning of the collapse phase, leading us to believe that the system never really



got stable. It is sort of a “jump” or fast transition from  $r$  to “ $\Omega$ ,” system collapse (Figure 3-4) especially in terms of social and economic development.

### **Phase III (May 2005- April 2010)**

“ $\Omega$ ” is the phase of decadence of the system due to continued high pressure on forest resources, triggering a reaction from the government. The situation in this phase was of high rates of deforestation, allied to the restricted access to exploitable resources, and low economic circulation. What happened was the continued abandonment of the rural settlements and land concentration because some people bought lands from people who were abandoning their plots, a continuity of what was happening in Phase II. Another important factor in this period was the pressure by the environmental institutes (IBAMA and SEMA) to protect natural resources, including the Curupira operation against illegal logging (see Chapter 2, Economic Development of Cotriguaçu), and the prohibition of the use of fire.

### **Phase IV (2010 to present)**

The “ $\alpha$ ” phase of system reorganization, control and environmental regulation is currently beginning in Cotriguaçu and the adaptive cycle has not been completed yet – we are still preparing for the start of the new phase of growth ( $r$ ). In 2010 the municipality was identified as one of the priority conservation areas in Mato Grosso.

Putting these phases together gives us (see Figure 3-4) the shape of the adaptive cycle of the rural settlements of Cotriguaçu. This adaptive cycle is distorted and does not fit Holling’s theory very well. Even if I tried to define the phases and to insert them in the cycle, it was hard to frame reality in the methodology. The main difficulties here were related to finding the most relevant turning points of the system, since the historical facts and long discussions with local stakeholders showed a large

variety of troubles and disturbances. Thus, selecting and prioritizing these main difficulties was an issue. The solution was to first understand the context as a whole, and, at the moment of prioritization of the main difficulties, choose the facts that were pointed by local people as the most significant.

Along with opinion from local people, the prioritization was complemented by literature consulted and was systematized in Table 3-1. Otherwise, there was a risk to include an exhaustive list of facts that would not necessarily be relevant to shape the adaptive cycle and analyze system resilience (Figure 3-4). Following, there are key observations that emerged from this analysis:

### **The Most Affected Attributes of the System**

All of the pre-defined attributes of the system (land use, labor force, land cover, environmental services, income and production) are negatively affected by disturbances over time, resulting, sometimes, in a complete degradation of these attributes. This degradation happened especially with labor force (when people out-migrated), environmental services (destruction of it in many parts of the rural settlements), and land cover (in places where deforestation or degradation are high). Income and production were affected by the lack of possibility to reach the market and to sell their production.

### **The System Never Reached Stabilization**

As mentioned in the description of phase 2, the system of rural settlers never reached a real stabilization phase due to the fact that people never had the ideal conditions for living and working. So, the K phase is dynamic and fast, impeding the system to stabilize or to resist disturbances. This shape of the adaptive cycle could have been different if the new activity in which people got engaged after the failure of

the cooperative – cattle ranching – was not so environmentally harmful and, moreover, out-migration was controlled, especially in the sense of maintaining people on their rural plots, instead of abandoning their areas and contributing to land re-concentration.

### **Resilience is Slow or Non-Existent**

The system is constantly trying to recover from disturbances, in many cases with overlapping impacts in more than one attribute, such as the subsidies to cattle that negatively affected both land cover and the provision of environmental services. The system is unable to quickly compensate for repeated external shocks suffered.

In short, it is possible to say that, looking at the adaptive cycle and at the historical facts, the SES regime, or stability domain, as explained by Folke (2010), is not ideal and, therefore, resilience becomes irrelevant when we do not intend to recover to a state before the disturbance. Besides, this was confirmed in many conversations with local stakeholders both those from the rural settlements as well as decision makers. According to local people, the municipality was not a good place to live because of the lack of opportunities and conditions. Thus, people were obliged to abandon their areas. Informants stated that this could be different if government invested more in the well-being of people in rural areas.

The scenario is not completely catastrophic since there are people, subsisting from their production, recovering from disturbances, but, as a whole, the lack of infrastructure and continued out-migration are a constant in all the phases of the cycle, and this brings land re-concentration and more deforestation because people do not intend to stay there for long. Younger producers, especially, want to deforest and use the land to raise cattle, and then look for better living conditions in other places.

It is unfortunate that SES stakeholders do not have the necessary conditions to change the system regime, and it does not evolve in the direction of a better situation. Resilience for this system would mean the persistence of the current negative situation of out-migration and deforestation. Some local stakeholders forecast that in a few years all of the remaining rural settlements will be transformed into large farms if the government does not make efforts to provide suitable conditions for them to stay on their lands.

### **The Potential Role of the CSV Project in the Resilience of the System**

Considering that the current regime of the system is not the ideal, as discussed above, and resilience can be analyzed in specific situations, it is important to see the system through two different perspectives: i) from the perspective of changing the regime; ii) from the perspective of improving general system resilience. The first perspective requires large efforts from the government and significant investments in infrastructure and improvement of living conditions. The second perspective is more practical and involves the strengthening of weaknesses in the adaptive cycle and will be discussed in this section.

According to CSV proponents (field data information 2011), the CSV project had the potential to enhance the system's capacity to support disturbances and to become more sustainable, reducing deforestation and stabilizing out-migration. This could be reached with the main goals of the project and could provide a different future for the rural settlers.

The project had little acceptance at the beginning, because producers were afraid of promoting changes in their productive systems. For example, in the rural settlement of Nova Cotriguaçu fewer than 5% of families got involved in these activities.

But their participation, nonetheless, made a large difference, especially when these families were able to show results of these activities to other families in interchange events.

CSV could potentially contribute to the strengthening of the resilience of the system, particularly because one of its goals is to keep producers efficiently in their lands, and it is concerned about land tenure regulation, reduction of poverty, and income generation, among other important constraints to resilience. Besides, this project contemplates the four resilience dimensions: social, economic, environmental and cultural. As we saw above, economic and social dimensions are key issues in this region, which have to be considered by this kind of project.

We will finally discuss now the inclusion, or not, of a PES in this whole context of SES, resilience, and CSV. As stated above, one of the concerns of the chapter was to investigate if the SES resilience would be stronger with the implementation of a PES initiative (Research Question #2). Thinking about the CSV project as a REDD policy with or without a direct cash PES, raises the question of what would be the benefits of this tool, basically an economic tool with the main benefit to protect natural resources and with co-benefits such as poverty alleviation, increase of social capital or even increase of self-esteem, that could largely promote positive impacts in the region.

As mentioned, the main issues of the SES are out-migration and deforestation, which are provoked basically by lack of infrastructure, non-access to markets, agricultural pests, no government support, and inefficient monitoring. The conceptual base of PES states that this is a tool that enhances the provision of environmental

services, and consequently reduces deforestation, as well as many co-benefits such as poverty reduction.

Therefore, in the Cotriguaçu context, PES is able to address only part of the problems. There is no way for PES to improve local infrastructure of market access, for example, showing that much still needs to be done by local institutions before considering its implementation.

Preliminary results of the second part of the CIFOR Global Comparative Study on REDD in May 2014 showed that more recently, out-migration has decreased in the rural settlements of Cotriguaçu. One of the reasons for this decrease could be the improvement in local social conditions, such as the arrival of electricity with the “Luz para Todos” (Light for All) federal program.

### **Hypothetical PES Project with Individual Cash Payments and Potential Increase in the CSV Projects’s (a REDD + scheme) Impacts**

To what extent could a hypothetical PES project with individual cash payments increase the CSV project (a REDD scheme) impacts? It is difficult to say whether implementation of a hypothetical PES would generate positive impacts, especially if decision makers follow the traditional definition of PES that does not consider a social interest in the management of natural resources Wunder (2005). As mentioned in the analysis, the context is against PES due to the high level of obstacles such as: high opportunity costs to preserve the forest instead of engaging in cattle ranching activities; local culture not prepared for participating in a PES project, because people think that they will be able to use these payments for investment in their livestock; and the low level of environmental services to provide due to high deforestation in the rural settlements.

The most challenging of these factors is related to the local culture of dealing with payments. If producers who received extra income from PES started investing in inappropriate activities, PES could cause the system to collapse because this would put pressure on forests and provoke disintegration of the system. On the other hand, additional conservation tools used in the CSV project, such as transfer of technology, and governance support, among others, might actually work more effectively than PES in making system resilience more robust. In this context, it is difficult to predict positive impacts of a PES here, unless it comes accompanied by other strong policies to protect the environment and promote development.

### **Improving System Resilience**

We saw above that the system has two main issues: high out-migration and deforestation. In order to solve the first issue, infrastructure should be improved, pests need to be controlled and markets need to be created to buy local production. The second issue is related to the maintenance of biodiversity and environmental services. In this scenario, PES appears a “half-solution” since it is only able to support the maintenance of biodiversity and to maintain environmental services. Thus, PES is not recommendable in this system, at least in this current situation. For rural settlers to stay in their sites, many conditions are needed, and these conditions require far more than the coverage impacts of a PES. As an example, people need a better local infrastructure, with involves better roads, better schools, and improved health conditions. Moreover, people need to have markets to absorb their production, and technologies to fight against pests in their fields. PES programs will not be able to provide these conditions. Instead, PES can guarantee a better provision of biodiversity

and ES. These other conditions must be provided and subsidized by local government, something that has not happened so far.

Therefore, it is difficult to implement a PES with direct cash incentives in the settlements. However, in its strategy the CSV project is using a series of activities that could promote the reduction of deforestation in the rural settlements, as well as strengthening productive activities in this group. The strengthening of sustainable activities in the productive matrix, such as incentives for good agro-pastoral practices (cattle for milk and meat) and the strengthening of governance in the rural settlements, seek to improve the collective and rational management of natural resources to promote a better life quality in the rural settlements (ICV 2012). However, the project still has a low capillarity among the rural settlers: there are almost 2000 families of rural settlers but only 70 (less than 1%) take part in project actions.

### **Conclusions and Recommendations**

Based on the concept of resilience used during the chapter -- the tolerance to disturbances and also the speed of recovery from those disturbances -- and also based on the analysis of drivers at different levels, livelihood activities and disturbances, within a panarchy and a nested approach, results show that the resilience related to tolerance to changes in SES rural settlements in Cotriguaçu is extremely low and fragile. This is due to the fact the system suffered repeated disturbances in a small amount of time, and the capacity to avoid or tolerate these externally-induced changes was insufficient. On the other hand, some people in the system are somewhat resilient to these changes. Moreover, results indicate that the incorporation of a PES tool in the system would not strengthen system's resilience, supporting the argument that this is not the most



appropriate tool to solve issues that are provoking the system's inability to reach resilience.

The first step of the study was to understand the SES, describing its drivers and attributes, in order to understand on what exactly to focus the analysis. The following step was to draw the timeline of Cotriguaçu with its historical facts both for local and external levels. Through this tool, the main events, all of them from outside of the municipality, were highlighted as the most important in terms of provoking impacts at the local level: the creation of Itaipu dam, cut of subsidies to the Cooperative, the PROCERA program for buying cattle, governance shocks and the international discussions on REDD.

The third main step was to trace the adaptive cycle of Cotriguaçu. This step showed that the SES still has not reached a stabilization phase. The main point here is to realize that the system is in a constant collapse and to detect what is needed for it to reach the stabilization phase (K).

Finally, the panarchy scheme (Figure 3-5) showed the interaction among the levels (from international to local). The panarchy scheme showed that International pressure was key to this SES balance in terms of maintaining forest stocks. Without this influence, which converged on a series of federal policies (such as PPCDAm, CAR, REDD, cut of subsidies, among others), as well as state policies to improve control and monitoring such as the Curupira operation. These events definitely contributed to decreasing the rate of deforestation. Discussing resilience is not an easy task. The first trap was related to the adaptive cycle that showed that the system is still in a preliminary stage and has not attained real stabilization. Therefore, even minor

disturbances will abruptly provoke high changes. If the system is not stable yet, is it not possible to think about returning to a stage before the disturbance. On the other hand, it is necessary to reflect about how to reach the stabilization phase and what tools are necessary for that.

In this whole context, CSV appears as a potential driver of more sustainable changes in the sense of increasing the potential resilience of this system. This project, considered to be a REDD initiative, has the potential to promote a series of positive changes in this system such as the strengthening of productive systems and governance in the rural settlements, and promotion of productive sustainable activities, which correspond to the final goals of reduction of deforestation and degradation from REDD. However, this potential is still to be realized due to the small scope of pilot efforts and the continued lack of supporting public policy.

The fact that the proponents of CSV project did not include a PES in these actions is somewhat polemical, particularly because at the beginning proponents wanted to include a PES tool and beneficiaries were keen to this idea. Due to the fact that the rural settlements, in general, present high deforestation rates, the implementation of PES in these areas risks not being efficient because of the low amount of forests available and lack of other preconditions. Therefore, the tools used by CSV (transfer of technology, support to governance, among other), more than a PES, have the potential to make the resilience of the system more robust. On the other hand, it is important to analyze the degree to which each stakeholder contributed to the accumulated deforestation in the municipality. Seeing the historical deforestation rates (flow of carbon) is as important as analyzing the current amount of forests (stock) each

stakeholder has. This stock and flow approach proposed by Moutinho et al. (2011) is used in the phase of benefits sharing because it avoids the perverse effects of compensating stakeholders that, historically, deforested more.

One more point that should be stressed is related to the limitations of the methods used. Application of adaptive cycle and panarchy concepts maybe too reductionist and simplistic. If on one side, we always need to frame and reduce reality in complex systems in order to better understand them, on the other side this simplification risks putting aside other important aspects of the system. Thus, a balance between reducing complexity and considering all important aspects is required to compensate for this modeling oversimplification.

Given that PES is not a fundamental tool for the CSV project to succeed, it is necessary to think about better ways to keep these people in the rural areas of the municipality with good life conditions to survive, develop and protect their natural resources. Below we list some suggestions of actions that could be included in the project in order to improve system resilience. Some of these needs are being focused and implemented by the CSV project, but the scale is still too limited. If this project were broader, it would definitely provide a large benefit to the rural settlements in the municipality, and increase the resilience of the whole system.

### **More Presence of the State to Improve the Living Conditions in the Settlements**

There is a limited presence of the state in the rural settlements. People complain a lot about lack of health assistance, schools, roads, and honest politicians, among others. More recently, access to electricity due to the Light for All program had very positive impacts on living conditions.

## **Provision of Technical Assistance to Producers**

Again, CSV is providing technical assistance, but the scale is too small. In the sampling universe of 2000 families, distributed across three rural settlements, they were active in only one settlement, and in only four communities.

## **Diversity of Products**

Production diversity does not depend on a project or on the government initiatives; instead, it depends on market demands. The government can, however, subsidize some products, or stimulate others. This already happens in Cotriguaçu with the PAA (Food Acquisition Program) initiative from CONAB (Companhia Nacional de Abastecimento, National Supply Company), a federal government institution that buys food products to distribute in schools or hospitals. Producers who are in the program have the security of selling their products every month but they also complain a lot since they need to work with many different products that, according to them, make their lives difficult. But this project is only focused on agriculture. Other products could be also explored such as forest products, or even babaçu (*Orbignya phalerata*), a palm that is locally viewed as a pest because it invades the pasture areas but which could potentially have a large economic potential for small producers since it can be transformed into many different commercial products such as oil (from the fruits), charcoal (from the coconut cover), and handicrafts (from the palm leaf), among others.

One way to provide many benefits to the rural settlements would be local industries to create employment for local people. In Apiacás, a municipality close to Cotriguaçu, it is common to see processed local products such as coffee in the local supermarket. Cotriguaçu also has the potential to integrate people to the local market but it needs some subsidies and technical assistance at the beginning.

## **Improve Governance in the Rural Settlements**

The CSV project already supports capacity building for local governance through political organizations, support for production and trade, and land tenure regularization. The main issue here is that, according to informal conversations with some leaders in the settlements, there is a lack of interest and trust in these political organizations. This is the case of a local organization of a group to collect and sell Brazil nuts that tried to get organized but failed due to bureaucratic obstacles and lack of external support, which demotivated people. The government did not provide conditions to support local organization. According to people interviewed, they need help to be able to carry out basic tasks such as printing documents of proposals. However, the local government institutions are not always willing to provide support. This lack of autonomy can be considered as one of the biggest challenges for the improvement of local governance.

## **Capacity Building for Local Leaders**

Capacity building of the current as well as potential future leaders of the settlements in the municipality is an important task, considering the low level of political organization detected in the field data collection, as well as the necessity to improve governance by this stakeholder group. Courses related to simple aspects of political and social organization are needed, as well as more specific and technical training such as support to write proposals, create email or Facebook accounts.

To conclude, PES might be a tool to solve Cotriguaçu's problems only if applied with other command and control tools. The next chapter (Chapter 4) will provide a more detailed discussion about the conditions for implementing a PES and to what extent Cotriguaçu is able to meet those conditions.

Table 3-1. Timeline of historical facts that promoted changes to the system

| Internal Events                                              | Year | External Events                                                                           |
|--------------------------------------------------------------|------|-------------------------------------------------------------------------------------------|
|                                                              | 1964 | Military Government                                                                       |
|                                                              | 1974 | <i>Creation of Itaipu dam* (colonization of Cotriguaçu)</i>                               |
| Creation of PAs Juruena and CEDERES                          | 1984 | Increase of international market preferences for national commodities (wood, meat, crops) |
| <i>Creation of Cotriguaçu*</i>                               | 1988 | N/A                                                                                       |
| Production is stuck, since there is no market to sell them** | 1990 | <i>Subsidies to cooperative cancelled **</i>                                              |
| High deforestation rates in the Amazon                       | 1992 | ECO-92                                                                                    |
| High dry period in Cotriguaçu**                              | 1997 | Kyoto Protocol, Luz Para Todos                                                            |
| Cattle ranching expands in settlements                       | 1998 | PROCERA                                                                                   |
|                                                              | 2005 | Curupira operation                                                                        |
|                                                              | 2006 | <i>Governance shocks*</i>                                                                 |
|                                                              | 2007 | <i>Discussions on REDD *</i>                                                              |
| Local working group on REDD                                  | 2010 |                                                                                           |
| CSV project                                                  | 2011 |                                                                                           |
| Electricity in the rural settlements                         | 2012 |                                                                                           |

\* *Significant events that caused large impacts at local level.*

\*\* *Significant events that caused impacts related to out-migration.*

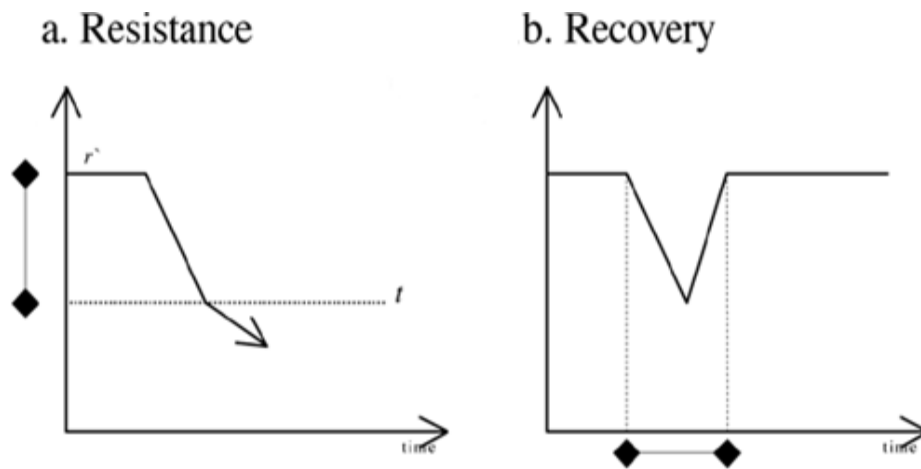


Figure 3-1. Two types of resilience: on the left, disturbance that can be absorbed before state change; on the right, rate of recovery from perturbation

Source: Adger, W.N. 2000. Social and ecological resilience: are they related? *Progress in Human Geography* 24:347–364.

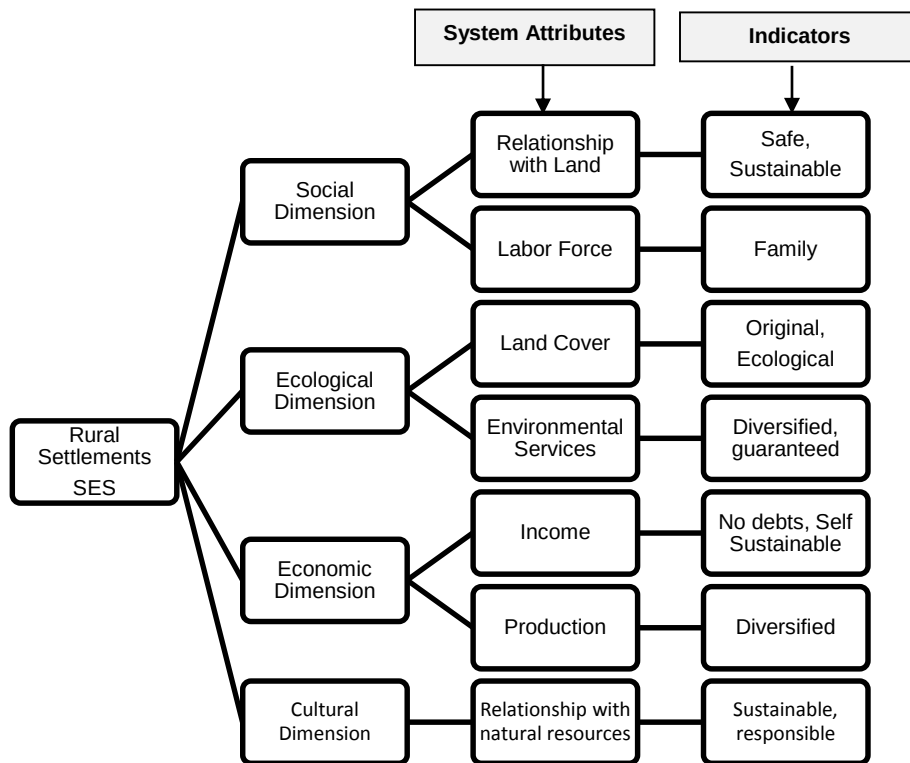


Figure 3-2. Key attributes and indicators of the Cotriguaçu rural settlers SES

Note: adapted from Olival et al. (2011).



1. Growth or exploitation (r)
2. Conservation (K)
3. Collapse or release (omega)
4. Reorganization (alpha)

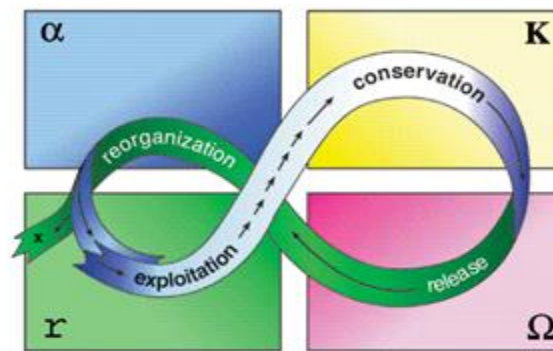


Figure 3-3. Adaptive Cycle

Source: Gunderson & Holling (2002).

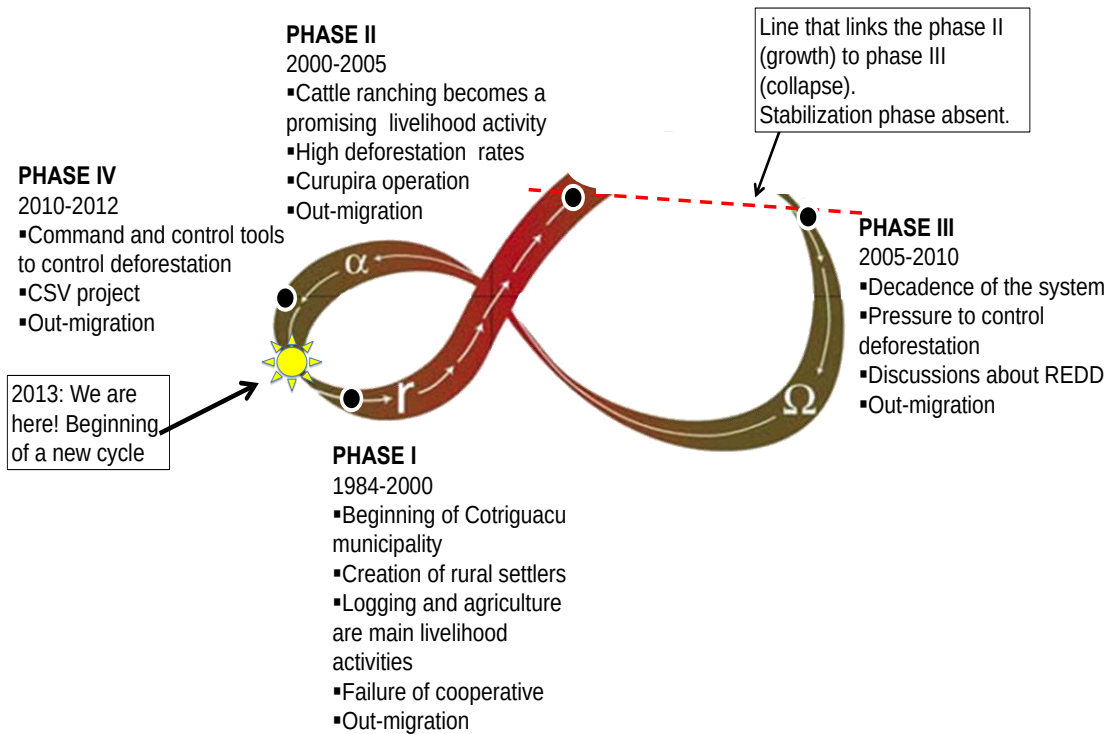


Figure 3-4. Adaptive cycle of Cotriguaçu

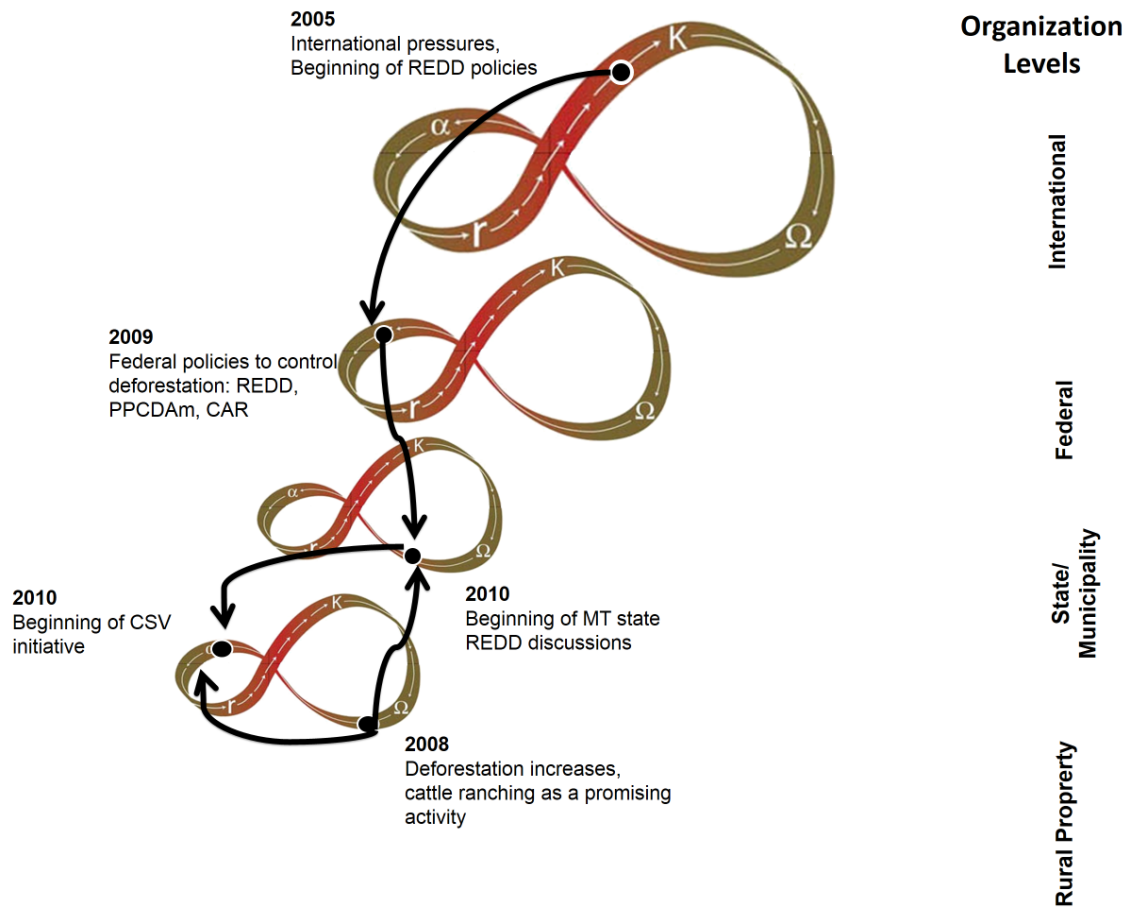


Figure 3-5. A panarchy scheme of Cotriguaçu rural settlements SES

## CHAPTER 4 ASSESSING PRECONDITIONS FOR A PES PROJECT IN COTRIGUAÇU

### **Introduction**

PES success currently is being potentially evaluated from many different perspectives, and clearly success will vary from context to context, and for different stakeholders within each context – at different moments in time. The most common approaches analyzing PES take the perspective of understanding its impacts on poor rural people (Forest Trends et al. 2008), or on deforestation reduction, after the implementation of the project (*ex-post* studies). This chapter is concerned about doing *an ex ante* investigation of PES. It will analyze to what extent the REDD+ project of Cotriguaçu (Cotriguaçu Sempre Verde – CSV) meets the preconditions established by Wunder (2009) for implementing a PES project. Evidence shows that, under certain conditions, well designed PESs are much more successful in the sense of efficiency, equitability and efficacy than badly designed PES (Wunder 2009) and some conditions are necessary for it.

Proponents of the CSV project probably did not think about all the necessary criteria when they considered implementing a PES initiative in Cotriguaçu. The eventual decision to not implement it was based just on economic conditions (high opportunity costs, start-up costs, transaction costs) but there are also other aspects that must be considered in all PES projects. In this study we propose to systematically analyze some of the key pre-conditions that have been identified as relevant to the stakeholders in Cotriguaçu, as a framework to assess to what extent each pre-condition exists for producers of different sizes and origins, as well as which conditions might be improved, and at what cost. This practical application of the framework permits us to assess the

overall environment for a PES program, identify differences among potential PES program beneficiaries, and assess the extent to which pre-conditions might be addressed or changed. The findings show the importance of financial costs for some stakeholders, but show that it also is necessary to look at other aspects before designing PES.

This analysis will contribute knowledge to stakeholders that are designing and implementing PES projects. Even if a PES was not included in REDD project of Mato Grosso, it is important to better understand the local context and to verify whether or not the implementation of a PES in the context of CSV project could be viable. Another option is to ask if PES could be implemented among some stakeholders and not among others, considering the multiplicity of actors in this area. Therefore, research questions in this chapter are:

- Question 1. To what extent does the Cotriguaçu municipality meet the four preconditions established by Wunder (2008, 2009), along with other key preconditions, to implement a PES initiative?
- Question 2. How do the effects of these distinct preconditions vary by stakeholder groups of different sizes and communities of producers, comparing four rural smallholder settlements with large/medium producers?
- Question 3. Which preconditions might be changed in order to support a PES initiative, and at what cost?
- Question 4. What does this practical application of the framework of analysis suggest about the pre-conditions for PES success, and how they vary among sites and social groups?

The chapter is organized in four parts:

First, I will describe the preconditions necessary to implement a PES (Wunder 2008, Wunder 2009), as well as others deemed especially relevant for the Cotriguaçu context and complementary to Wunder's view. Then, I will describe the methods

employed to determine the suitability of PES for the stakeholders in Cotriguaçu. In the third part I will examine to what extent Wunder's and other key pre-conditions for successful PES were found in Cotriguaçu, and how these varied by type of producer, as well as which preconditions might potentially be changed, and at what cost;

Finally, in the conclusions I consider what these findings suggest for those interested in implementing, or in participating in PES programs, as well as to the literature related to the pre-conditions for PES success among diverse stakeholders in a dynamic frontier environment.

### **Preconditions for Implementing a PES project**

A PES project, to be successful, must meet some criteria related to the additionality of the project, the provision of well-defined services and the existence of buyers and sellers (Wunder 2008). Wunder developed not only a definition for PES (page 8) but he was also the first to discuss the criteria for a PES to be successfully implemented. According to him (Wunder 2008, 2009), there are five preconditions that should be achieved and respected to implement a PES inside a REDD initiative, which are:

1. **Informational preconditions:** It must have transparency of what environmental services are being provided;
2. **Economic preconditions:** the economic value of REDD should be higher than its costs (opportunity + transaction);
3. **Cultural preconditions:** a pre-established culture of payments must exist both from the side of providers of environmental services as well as from the users of these services;
4. **Institutional preconditions:** PES requires trust between providers and service users; thus mutual contracts are expected to be complied;
5. **Competitive preconditions:** contrary to universal perception, PES does not require perfect markets to function. This precondition relies more on contracts

between actors with asymmetric information than traditional markets (contract theory, instead of marketing). Most PES initiatives are monopsonies, in which there is only a single buyer, e.g. a hydro-electrical plant, or oligopsonies, when few large buyers dominate the commercial negotiation.

Pirard & Billé (2010) have also discussed the conditions for implementing a PES. According to them, a PES project needs to define the following six elements: “(i) the role of economic assessment for the determination of the relevance and content of contracts; (ii) the nature of service providers, particularly the dichotomy that exists between the local rural populations and industrial companies (ii) the nature of service beneficiaries, who may be localized or widespread; (iv) the nature of PES, which can be viewed as a means either to trigger transition toward activities compatible with the maintenance of services or to assist the resource user indefinitely or in order to compensate for constraints; (v) interactions with public authorities, particularly in relation to the danger of undermining the strength of state authorities in countries where governance remains deficient and is in need of improvement to ensure long term environmental management ; (vi) the relevance of reversing the “polluter pays principle,” however widely accepted, with potentially adverse consequences in terms of strategic behavior.”

Wunder’s preconditions are extremely relevant if we are concerned about a more careful design and implementation of PES. Compared to the others researchers, which are more focused on the criteria to design the PES itself, Wunder is more cautious: he expects that decision makers must analyze the pre-existing conditions of the area where the project will be implemented before thinking about the design itself. Only after verifying these conditions, is it possible to think about the best design for this specific site. To draw my analysis, I will follow Wunder’s perspective of analyzing not the design

of the PES but rather, the conditions of the site before the PES is designed. In order to do that, I will check to what extent Cotriguaçu municipality, with its focus on small, medium and large stakeholders, fits the following criteria, and explain what information is necessary for this analysis.

Considering that we do not have sufficient information about the market or the value of the payments, the competitive precondition will not be analyzed. However, besides these four established requirements, two other relevant features for PES initiatives will be analyzed: additionality and social responsibility.

These two conditions were chosen because they are constantly mentioned in the literature as key conditions for a PES success. Additionality, which can be conceptualized as “the process of determining whether a proposed activity is better than a specified baseline” (Gillenwater 2012), should be considered in carbon offsets programs because it helps identify which policy interventions will produce good or harm to an area (Gillenwater 2012). It consists of the positive changes the project was able to provoke compared to a certain business as usual (BAU) situation. A PES project must be implemented in situations where we can guarantee that the tool will promote positive changes.

Social responsibility, on the other hand, is a concept that comes from business and is related to the fact that an organization must act in benefit of the whole society, especially in terms of reducing social inequality. Although poverty reduction is not a primary goal of PES, it could be one of its co-benefits (Wunder 2006; Pagiola et al. 2005).



The accomplishment of all of those conditions means that the chances for a PES to be successful are high. In a study about the application of PES such as REDD mechanisms, Börner et al. (2010) analyzed some economic and institutional preconditions for PES feasibility. Their results showed that, if carbon prices remained the same as at the time of the study, economic preconditions would be satisfied in more than half of the threatened areas in the Amazon. On the other hand, institutional preconditions did not have the same performance and would be a large bottleneck for PES, mainly due to “land grabbing, insecure tenure, overlapping claims, and lacking information on private tenure” (Börner et al. 2010). Thus, I also analyze whether PES can achieve the preconditions for Cotriguaçu municipality, to see if results reflect Börner et al.’s conclusions.

### **Methodological Approach**

#### **Sample**

Methods used to collect data consisted of interviews with 40 medium/large landholders and 120 rural settlers, respondents from both categories randomly chosen in order to avoid bias. For the medium/large landholders, I had access to a list of the properties above 400 ha (the main criterion for not being a small producer). This list had 80 names of landowners, and 40 names were drawn at random. Some producers were not in town (6) and others were afraid of answering the survey and/or refused to participate (7). Thus, after one more raffle, 13 more names were included in order to complete the sample.

For rural settlers, interviews were conducted with 30 families in four communities. I also had access to a list of all small producers of each community and raffled 40 families, thus we had 10 names on a waiting list in case some producers did not want to

participate in the interview. The four communities chosen were: Vale Verde, Santa Luzia, Entre Rios and Nova Esperança.

It is relevant to better understand the differences between small and medium/large producers in order to draw specific policies for each group to reduce emission. Cotriguaçu is a young municipality that was created in the 1980s. The municipality experienced two different periods of colonization over its short history. First, in the 1980s people with some economic resources left the south of the country and bought areas larger there than they had owned in the south. One of the interviewees said that the slogan at this time was: “sell 10 ha in Paraná and buy 100ha in Cotriguaçu.” Some of these people became successful medium and large-scale producers a few years later. A second colonization involved landless farmers primarily from the north and northeast of Brazil. These people became rural settlers in the three PAs.

All of the 40 large landholders of the sample were originally from the south of the country, mostly from Paraná (40%), Santa Catarina (28%), and Rio Grande do Sul (25%). As already mentioned in Chapter 1, from 120 rural settlers interviewed, 24% were from Rondônia, 22% from Paraná, 21% from Mato Grosso and 33% from other states of Northeast Brazil.

The majority of rural settlers in the sample declared agriculture (35%) as their main activity, followed by cattle ranching (28%). Large producers, on the other hand, declared their main activity as cattle ranching (85%) followed by logging (5%) and store owner (5%). In all small producers' communities, as well as among medium and large producers, cattle ranching is the activity that provides the most income. While 120 small

producers earned about R\$ 4.8 million trading cattle, the 40 large producers interviewed earned around R\$ 34.7 million. Considering that the Gross Domestic Product (GDP) of Cotriguaçu is R\$ 161,684 million (IBGE 2014), the amount produced by my sample (50% of large producers and 1% of rural settlers), is a huge amount. In relation to forest products, medium and large producers of the sample interviewed traded R\$ 4.3 million, an amount much inferior to cattle ranching.

For each precondition analyzed, I used a different tool such as opportunity costs calculations or basic statistics, carefully described in the results part. For the informational precondition, secondary data was the basis for the analysis. For the cultural, institutional, economic, additionality and social responsibility conditions, both primary and secondary data were used. The “economic preconditions” were the ones analyzed with more accuracy, due to the good availability of primary as well as secondary data.

### **Opportunity Cost**

The concept of opportunity cost was first developed by John Stuart Mill, a political economist from the 19th century, to designate the value associated to the opportunities that are forgone as the result of making a decision and not investing the resources on another best alternative (McConnell and Stanley 2005; Pindyck & Rubinfeld 2009). This concept is key in environmental issues as it can inform the choices concerning the use of resources. An example of opportunity cost is the decision of a farmer to use his land to engage in forest management instead of cattle ranching. The benefits of changing land uses to invest in cattle ranching are higher, and he will be losing the opportunities to earn more, so these are the opportunity costs.

The measurement of opportunity costs is a way to compare scenarios and decide which would be the best alternative from the perspective of the landowner. There are different ways to calculate opportunity costs, either by modeling the general balance of a specific area (Cattaneo 2002), or by using broader economic models (Carpentier et al. 2002). Other authors estimate the opportunity costs based on the price of the land (Micol et al. 2008) or by simulations of economic incomes from soybean cultivation, logging and cattle ranching (Nepstad et al. 2007), but all methods should include the lost benefits over time, using economic instruments such as discount rates to calculate net present values of forgone options.

There are not too many studies that have explicitly calculated opportunity costs of areas that are included in PES programs, but the few studies that exist detected that voluntary PES schemes are producing benefits beyond the opportunity costs of the participants (Wunder 2008). Those are the cases of the PES scheme in the Bolivian Los Negros and the Noel Kempff Mercado Climate Action Project (NKMCA), both in Bolivia. In this last one, a logging concession was incorporated into a national park. North-American companies interested in investing in carbon-emission offsets bought the credits from the avoided-logging operation. The local communities that lost their jobs at the timber-companies received, in compensation, economic benefits to manage the park, as well as improving their agriculture and ecotourism activities (Wunder 2007). The gross-benefits (for 3 years) estimated in this area were US\$ 128,580 (Asquith et al. 2002). After the deduction of opportunity costs, 1,050 community members received net per-capita gains of US\$ 100-250.

## Results and Discussion

### Does the CVS Project Meet the Preconditions for a PES?

In 2008, ICV developed a study about the potentialities of PES in MT. The study was based on the following criteria: Potential Value = Potential Additionality – Potential Cost, where Potential Additionality (U\$/ha) = Probability of Deforestation (% of remaining area) \* Carbon Stock (CT/ha) \* Carbon Market Value (iU\$/CT), and Potential Cost = Opportunity Cost of Conservation (U\$/ha).

In this study they pointed out that the largest part of Cotriguaçu was an area of high potential for implementing a PES (Micol et al. 2008) which means that the area has a high potential value of land (U\$/ha) and probability of deforestation from 30-60% in the next 10 years, forest biomass of 80-130 CT/ha and low to moderate opportunity costs (Micol 2008).

In order to make future analyses of the region, and thinking about the CSV project as a REDD+ policy where we might opt to include a PES project, it was necessary to verify the viability to implement this tool among the different stakeholders there. PES is basically an economic tool; however, it can deliver social co-benefits such as “poverty reduction, a transition to more profitable and resilient land-use systems, strengthening of social capacity and supportive institutions” (Greiner & Owey 2013, p. 1-2), self-esteem increase by being proud of the attitude to make part of such relevant programs (Greiner & Owey 2013) as well as governance improvement (UN-REDD 2011) that could promote changes in local relationships between humans and forests.

The analysis of the pre-conditions is not always considered in PES projects, but can raise some important issues about what could be changed to create the opportunity

for a PES to be implemented. All of these conditions will be graded in a scale from 1 to 5, in which:

- **Grade 1:** the municipality/stakeholder does not meet the precondition at all and is not able to change the scenario
- **Grade 2:** the municipality/stakeholder does not meet the precondition but is able to change the scenario in order to accomplish the condition
- **Grade 3:** the municipality/stakeholder partially meets the precondition
- **Grade 4:** the municipality/stakeholder totally meets the precondition

The last step of the analysis will present a graph showing the situation of Cotriguaçu for all the conditions together.

### **Informational Preconditions**

Wunder (2008, 2009) affirms that informational preconditions are closely related to the deep knowledge of the environmental services that will be traded, as well as the transaction costs of all this operation. For him, in order to be successful, it is fundamental for buyers to know exactly what is being traded, and this implies high transaction costs for assessing the environmental service baseline and for designing the program, among others (Wunder 2008). Monitoring, communication and verification are also examples of transaction costs in a PES or REDD system (Azevedo et al. 2013). According to him, transaction costs are higher at the beginning of the project than in the following phases, when the project is already operating. In 2008, Boucher made an estimation of transaction costs for the Amazon region at \$0.38/tCO<sub>2</sub> (Azevedo et al. 2013).

In 2013, The Amazon Institute for Environmental Research (IPAM) developed a study in which costs for implementing a REDD initiative in Mato Grosso were calculated. In this methodology, IPAM calculated the costs related to the startup and transaction

costs (management, registration, monitoring, communication and verification) of a REDD initiative from the perspective of the Federal Government. Concerning the transaction costs, they were calculated for three different scenarios: 1.conservative (current situation, considering actions already included in the government actions), 2.ambitious (the costs were considered in a scenario of better technologies and governance) and 3.hybrid (same topics as both, but prioritization of efficiency).

The result was that, for the implementation of a REDD project for the next 10 years (not considering inflation), the cost was R\$ 22.4 million (scenario 1); R\$ 89.3 million (scenario 2) and R\$ 35.8 million (hybrid scenario). Thus, scenario 2 was the one with higher costs but, overall, for the three scenarios, the implementation of a REDD initiative for the whole state would not be that high (Azevedo et al. 2013) compared to the whole budget of the state. Furthermore, these values are even lower than the ones expressed in the literature (US1,00/ tCO<sub>2</sub>) (Azevedo et al. 2013). Besides, MT State has many opportunities to access funds such as GCF, Carbon “Bundling,” donations, Fundo Amazonia, among others (Azevedo et al. 2013). Thus, the startup and transaction cost for carbon storage could be viable in a REDD context.

Cotriguaçu municipality, being one of the last remaining forests of MT State, also has other ESs that need to be valued such as the ones provided by the Juruena River. In Cotriguaçu, baseline quantification of carbon storage is being raised (L. Micol, personal communication 2012) with support from ONFI. However, other potential ES are not included in this measurement, which means that the study will be incomplete. Thus, from the perspective of information about some environmental services,

especially carbon storage, we consider the CSV successfully meets this first precondition but not for other ES. It would be almost impossible to do that for many ES.

**Grade:** due to the considerations stated above, for medium and large producers, as well as small producers, this precondition will be graded as 3 on a scale of 4 (Figure 4-4), partially meeting this precondition, since carbon stocks can be accounted and probably climate regulation due to the Juruena river or the forest cover could be accounted for, but certainly other ES will not be correctly registered.

### **Economic Preconditions**

According to Wunder (2009), the value received for providing environmental services should exceed the total cost of this provision (i.e., opportunity costs + protection costs + transaction costs of conservation or costs to start up and run a PES project). The opportunity costs are by far the largest (UN-REDD 2011). A PES program has to exceed the sum of these three costs to be successful. This amount is variable and conditioned by the characteristics and needs of each site (Wunder 2007).

The analysis of this precondition suggests that the service buyer's Willingness to Pay (WTP) for PES must be higher than the total cost of providing environmental services. In general, it is well accepted that the WTP for PES must be higher than the opportunity costs since it is the main component of the total costs, and the protection and transactions cost are only relevant just at the beginning of the project (UN-REDD 2011). The opportunity cost approach says that if a producer is compensated for the financial amount that is forgone by not cutting down the forest, he/she will maintain the forest standing. This is currently the approach most adopted among researchers, although there is plenty of debate about the appropriateness of this approach since this proxy could be an oversimplification of the reality. Thus, more rigorous methods ought



to be developed to calculate REDD costs (UN-REDD 2011). However, for Mattos & Hercowitz (2010), it is better to make an indirect valuation of natural resources using opportunity costs as a proxy than to put direct and general prices on natural resources, because we are able to determine the economic value of these particular situations, instead of a general price used for different cases.

The opportunity costs for preserving the forest for producers were determined based on the premise that the benefits from activities such as cattle and agriculture (main income sources for producers) must be foregone to avoid deforestation. This was applied to the four communities visited and also with medium and large landholders.

### **Determining the Opportunity Cost**

The opportunity costs of the main local activities were calculated based on the method suggested by Wunder et al. (2008) in which the assessment is mainly based on 1) information about land distribution and expansion (ha/year); and 2) deforestation rates of the region (ha/year). The estimation will be done for a 10-year period since the beginning of the PES project, using a discount rate of 10%. It is necessary to know these values in order to decide if the PES program will have the expected results in this area. Sites where the opportunity costs are too high or too low should not be considered in PES programs because of additionality issues.

Considering that cattle and agriculture are the activities that provide the most income, the opportunity costs of small and large producers were determined by calculating the annual benefits from cattle and agriculture at the hectare level, since those are the activities that directly change the land cover.

In order to calculate the Present Value (PV) of the OC, we followed Wunder's method and proceeded as follows: first, the total annual net income generated by cattle

ranching, crop and milk production for each community were obtained from the 120 families interviewed from July/August 2010 to July/August 2011. Second, for each community, the total incomes were divided by the area devoted to each activity to obtain the net annual benefits per ha. Third, the present annual values per ha were then calculated over a 10-year period (Equation 4-1). Finally we averaged the sum of the present benefits of each activity over the time period (cash flow) to obtain an annual estimate of benefits per ha. Thus, to be part of a REDD initiative, the payments received by these families should be similar to the income earned due to these activities.

$$\sum_{t=1}^{t=10} I_{ij} e^{-rt} \quad (4-1)$$

Where  $I_{ij}$  is the annual income per ha for community  $i$  and economic activity  $j$  and  $r$  is the real discount rate.

Table 4-1 shows the average opportunity cost of preserving the forests for each community and activity. By far, cattle ranching for meat is the activity that generates the highest income. On average, cattle ranching represents 82% of the total income for all communities. The community with highest average opportunity cost (OC) in terms of foregoing cattle ranching for meat is the Vale Verde (R\$ 1,670/ha/year), followed by medium/large producers (R\$ 1,299/ha/year). From a PES perspective, considering only cattle production, in order to be successful, PES should at least be higher than income derived from this activity in Cotriguaçu, in the range of R\$ 621 – 1,670/ha/year depending on the community. Since families generate their profit from multiple production activities, PES should be higher than the range of R\$ 865-2,026/ha/year to

account for all their foregone income if they engage in these activities instead of preserving the forest.

Although communities have much more gross income from cattle than from agricultural activities (in many cases, agriculture is just an activity for subsistence), the value per month of the former activity does not seem too high. Three of the four communities have average incomes varying from R\$ 52 to 108/month from cattle ranching. The Vale Verde Community is the one with highest income, R\$ 139/month. If we consider the total number of families for each community (Chapter 2, Table 2-3) a REDD project should pay annually around R\$ 250,522, R\$ 57,799, R\$ 59,705, R\$ 93,171, and R\$ 103,908 for communities Vale Verde, Santa Luzia, Entre Rios, Nova Esperança, and Medium and Large Producers, respectively, to avoid deforestation by foregoing cattle ranching.

In sum, opportunity costs for small landholders in this site are high, particularly compared to other places in the Amazon. For example, (Cromberg 2012) calculated the OC for not raising cattle in the Transamazon region and the results were an average of R\$ 351/ha/year, a figure significantly lower than the values obtained for Cotriguaçu.

Furthermore, the Brazilian Forest Code obliges producers in the Amazon to preserve 80% of their forests. Land use in rural settlers' communities in the sample showed only 34% of land is still forest (Figure 4-1). Among large producers this amount is 83% (Figure 4-2). Since only the areas preserved above 80% would have the potential to be included in a PES initiative, most of the rural settlers are out of compliance. Large producers, on the other hand, are inside the law.

As 17% of the land is used by large landowners for activities other than forests (Figure 4-2), and 80% is the threshold to be considered for PES program, the opportunity costs should be calculated only for the 3% of remaining forests. This represents for medium and large producers, an area of 3,380 ha, which they still have the right to deforest. Considering the average OC for medium and large producers preserving 1ha is R\$ 1,299/ha/year, for this whole area this value is R\$ 4,390,620/ year. A high amount, definitely, considering that the initial CSV project budget was of R\$ 500,000.

In the case of small producers, where they are not able to provide ES since deforestation has passed the minimum admitted by law, before implementing a PES, other initiatives such as reforestation and restoration might be considered. In brief, the value of opportunity costs is not the only barrier to implement PES programs among rural settlers. The scarce availability of forests and, consequently, the few environmental services provided, are determinant conditions.

In places where opportunity costs are high, UN-REDD (2011) suggests the design of a PES with social-benefits since this can reduce these opportunity costs (UN-REDD 2011). Thus, it would be a priority to detect the most important social benefits these communities need if we really want to reduce the costs to implement a PES.

On the other hand, in places where ES are not quantitatively or qualitatively enough, the strategy must be focused on improving them. Therefore, CSV proponents are using a series of activities that could improve forests, such as reforestation and recuperation of degraded areas. Besides, they are also working to reduce deforestation in the rural settlements as well as to strengthen productive activities. The strengthening

of sustainable activities in the productive matrix, such as the incentive for good agro-pastoral practices (cattle for milk and meat) and the strengthening of governance in the rural settlements to improve the collective and rational management of natural resources to promote a better life quality in the rural settlements (ICV 2012) are also priorities. The project expects to improve productivity/ha of cattle ranchers, which will contribute to a decrease in deforestation. Unfortunately, the project still has a low adhesion among rural settlers. There are almost 3000 families of small producers and less than 100 are included in project actions as a whole.

So, for the analysis of this precondition, we are considering that PES is not viable among rural settlers. In conversation with ICV staff (R. Farias, personal communication), and according to opportunity costs analyzed, a PES project, at least in Cotriguaçu, a priority area from Mato Grosso, would not be economically viable especially due to the large amount of financial resources needed for implementing it. With the current available resources, proponents are not able to finance adequate payments for medium and large landholders, the main providers of ES. Furthermore, although it is unfair, financing small landholders would not be a wise strategy because additionality in rural settlements is low.

**Grade:** From the perspective of economic conditions, particularly related to the opportunity costs to implement a PES, we consider Cotriguaçu as a difficult place due to the high costs. Therefore, for medium and large producers this precondition will be graded as 1 in a scale of 4 – the stakeholder does not meet the precondition, and rural settlers will be graded as 2 – the stakeholder does not meet the precondition but has the conditions to change this scenario (Figure 4-4). This is because proponents can

raise funds to spread their activities to improve production practices and ES, and can also attract more participants.

### **Cultural Preconditions**

According to Wunder (2008, 2009), motivation of ES providers is fundamental for a successful PES. On the other hand, in situations where non-traditional economic systems function well, which is the case of indigenous communities that revere their natural resources, there will be resistance to PES and the results could even be catastrophic.

In Cotriguaçu, the relationship of small, medium and large producers with natural resources is basically mercantilist. With the exception of the Rikbaktsa (the indigenous group), local producers are not attached to traditions related to natural resources. One of the large landholders declared that for him it was disappointing to have all of these commitments with land preservation, required by the Federal Forest Code, because he came from the south of the country with the goal to financially prosper (E.M. Schuster, personal communication). According to Schuster, this was the motivation for 100% of the people who moved to Cotriguaçu, and engaging them in conversations about forest preservation was time-wasting.

Thus, for key local stakeholders interviewed, PES could be an accepted practice, is applied among small rural settlers and medium and large private landholders. Therefore, they would accept to preserve only if they were somehow financially compensated for it. This inquiry did not include the Rikbaktsa since, in case of implementing a PES, they will probably be outside of this strategy. Because PES could be harmful for their culture, this tool is not recommendable for them (TNC, personal communication).

In conversations with small landholders in 2011, it was clear that, in case of a hypothetical PES, even if financial compensations for preserving would not cover the opportunity costs of engaging in other activities, they would prefer to receive this extra amount and protect their areas, if they can get paid to do it. On the other hand, this logic does not work among medium and large landholders, who would be willing to receive financial compensations for preserving only if these compensations would cover the opportunity costs of more profitable activities.

As a complementary analysis related to cultural precondition, I asked small landholders what they would do if they received an extra monthly amount of money (three categories of payments, R\$ 200, R\$ 500 and R\$ 1,000) every month. I provided four options of answers: 1.investing in their production, 2.leisure (traveling), 3.investing in family well-being (such as putting kids in a better school or buying clothes), and 4.other. The number of answers in which people would apply the money on their productive activities was, by far, the largest, and the category was especially related to buying more cattle and clearing more forest areas to produce pasture (Figure 4-3). A key condition here is the conditionality of the payment, because in order to receive the compensations of PES, producers have the commitment to not deforest. I did not state this conditionality in my inquiry; however, controlling and monitoring producers' activities are still an issue in this region and, in case of implementing a PES, should definitely be strengthened in the whole municipality.

This example shows how risky it could be to implement a PES among these stakeholders, principally because people are used to seeing the forest as an obstacle to their development, not as a good that should be preserved in order to maintain the

ecological balance of the region. Probably, the regional origin of people is a driving factor for this entrepreneurship mentality. All of these producers are from the south or northeast of the country and have little attachment to the region and to the forest. If people living there were originally from this region, this vision could be different. Mattos (2010) says that native people from the Amazon dominate the tradition to use the forest and have pro-forest cultural values in their traditions.

**Grade:** Considering the fact that the large landholders are prepared for trading ES for financial compensations due to their little attachment to land and to their entrepreneurship culture, this precondition will be graded as 4 in a scale of 4 for them. On the other side, small producers will receive a grade of 3 (Figure 4-4) because of their intentions to buy even more cattle once they are compensated. It is important to stress that, once implemented, monitoring should be applied or the project will not subsist.

### **Institutional Preconditions**

In PES projects, trust between ES providers and recipients is fundamental, and this trust does not rise naturally. As a result, an external intermediary is frequently necessary (Börner et al. 2010). In cases where conflicts exist or land tenure is not clear, PES cannot be implemented and institutional limitations occur, especially related to the monitoring of the provision of the services.

According to Forest Trends et al. (2008), each PES project will be designed to better fit its own context, where efforts will be done to satisfy the necessities both from the providers and sellers of ES. As a result, a group of private and public institutions will be necessary. Obviously, a minimum of organization and governance of all these institutions is a prerequisite for the good functioning and interaction of all these



institutions. Besides, an effort in monitoring and control of benefits is also needed in order to avoid misuse of resources.

Wunder (2009) divided institutional preconditions in three factors: markets, trust and land tenure. Concerning the market factor, he states that, for PES to exist, it is necessary to have a market for it, and in general these markets are not competitive since the majority of contracts are bilateral or multilateral. Trust, the second factor, refers to the fact that services buyers have to trust that services providers are taking care of the provision of ES and that they are respecting contracts; negotiation processes and adaptive management are ways that institutions can use to overcome mistrust. Concerning land tenure, it is mandatory that people be the steward of their lands or, at least, have the rights and control over it, which is an obstacle in the Brazilian Amazon (Börner et al. 2010).

## **Market**

There are four different types of markets for PES in Brazil: public, private regulated, private voluntary, and eco-products (Shiki 2011).

- Public: the government uses public funds to pay ES providers, such as the Bolsa Verde and the Proambiente initiatives (Shiki 2011);
- Private regulated: oriented by specific rules, such as some local spring water protection projects (Extrema and Rio Guandu initiatives) (Shiki 2011);
- Private voluntary: oriented by business and ethics which is the case of Surui, Juma and Amazon Funds for REDD Projects (Shiki 2011);
- Eco-products: oriented by eco-friendly products such as organic and agro ecological products (Shiki 2011).

Currently in Brazil the most important financial resources for implementing PES are the funding from the Amazon Fund (created by the Decreto 6.527/2008) and the Bolsa Verde (Law Project 3.398/2012). The first is composed of international donations,

especially from Norway, and the latter is from Brazilian national funds. Aiming to reconcile poverty reduction and forest conservation through the payment of R\$ 300.00 every three months, Bolsa Verde has the goal of including 74,000 families of beneficiaries by 2014. Thus, the REDD project of Cotriguaçu would probably be able to access these two funds. However, there are other available sources that also could be captured by the project such as funding to support REDD Early Movers from the Kreditanstalt für Wiederaufbau (KfW), a German government-owned development bank.

### **Trust**

Trust is a key issue for PES because buyers need to be sure that ES provision is guaranteed, at least during the lifetime of the contract. Consequently, a contract and a posterior monitoring of these lands are necessary.

Mato Grosso is one of the Amazon region states where monitoring is more advanced. They were the first state to implement the SLAPR system (Sistema de Licenciamento Ambiental de Propriedades Rurais do Estado do Mato Grosso) that was later adopted by other states. This system consists of the registration of all the properties of the state followed by the monitoring of all the activities related to forest cover change. The tendency is the improvement of this system, which will make it even more difficult to engage in activities that provoke illegal deforestation.

### **Land Tenure**

In Cotriguaçu, there is the constant concern about *de-facto* rights over ES-producing assets. Despite the efforts of the Federal Government to legalize all lands through the CAR, which is part of the SLAPR system, mentioned above, there are still a few land conflicts inside the rural settlements that must be resolved. However, more

power of command and control tools still is needed in order to solve these issues. Thus, this part of the precondition is difficult to be met.

**Grade:** The institutional precondition will be graded as 2 in a scale of 4 – the municipality does not meet the precondition but has the conditions to change this scenario (Figure 4-4). Large producers will be graded with 2 (the stakeholder group does not meet the precondition but has the conditions to change this scenario) and small producers will be graded with 1 (the stakeholder group does not meet and does not have the condition to change the scenario).

### **Additionality**

As mentioned above, additionality of a project is the difference that a project makes to a baseline situation. In other words, additionality is the benefits that would not have happened if the project had not been implemented. The analysis of the business as usual situation (BAU) is the first thing to be considered. In Cotriguaçu, deforestation is relatively controlled since 2006 (Baby 2012), when the government improved command and control tools. This region follows the same deforestation pattern of the whole Amazon region where deforestation rates had a boom in 2004-2005 followed by a decrease until recently. Thus, for the municipality as a whole, deforestation seems to be controlled, but activities such as cattle ranching, that generate high income, are still a concern and are a large threat that can increase deforestation again, such as happened in 2013.

It is also possible to analyze additionality specifically for the different stakeholders studied, rural settlers and medium/large producers. Rural settlers occupy 14% of the area against medium/large producers that hold 54%. On the other hand,

rural settlers are responsible for 52% of deforestation, and medium/large deforested 9.9% (Figure 4-5).

In other words, medium/large landowners are the ones that hold more pristine areas. Consequently, these areas can be threatened if landholders decide to engage in non-sustainable activities. A PES project, if well applied, would probably be able to control this threat and to solve land conflicts.

**Grade:** This additionality precondition will be graded differently for both stakeholders: 0 for small producers and 4 for medium/large producers (Figure 4-4).

### **Social Responsibility**

In a few words, social responsibility corresponds to an ethical behavior towards social, cultural, economic and environmental issues. Therefore, PES appears as a tool with the goal to reconcile economic development with environmental protection and social justice (Wunder 2006; Pagiola et al. 2005). For some researchers, compensations and rewards for environmental services can stimulate some power imbalances between forest-resource dependent people and potential service's buyers (Swallow et al. 2009).

An example where PES contributes to improved social justice is the Nobility of Nature program in Sichuan Province, China, where Marriott, in partnership with Conservation International (CI) is investing \$500,000 to protect fresh water to be used by more than 2 billion people. Marriott sells honey produced by partnership, income from which is invested in the program that uses this funding to monitor water conditions and to train organic bee farmers.

In a performance evaluation study conducted by Vivan (2012), the researcher found that the majority of PES initiatives in Brazil were implemented in areas larger than

1,200 ha. This was mostly due to the lower transaction costs of implementing PES in large areas with few people involved. This shows that social justice is rarely considered among PES projects in Brazil. In Cotriguaçu, PES should be applied to rural settlers, the less advantaged group. The strategy for paying large producers seems to be very expensive due to the high opportunity costs, as well as not being able to contribute to social justice.

**Grade:** The social responsibility precondition will be graded differently for both stakeholders: 4 for small producers and 0 for medium/large producers (Figure 4-4).

### **All Conditions Together**

In order to have a better visualization of the stakeholders and the preconditions, all the grades were put on a table as well as in a radar graph (Figure 4-4). Putting all the preconditions together we attest that, on the whole, they are poorly met, especially among rural settlers. Medium and large landholders have a relatively good institutional level of organization, but the economic condition is very difficult to meet since high opportunity costs are definitely difficult to be covered.

On the other hand, rural settlers have a low cultural condition for dealing with PES and no additionality because the majority of the forested areas of this group are already deforested. Informational and social responsibility conditions are the strong points for implementing a PES among this group.

The whole analysis leads us to conclude that medium and large landholders are the ones who met the majority of these preconditions, particularly related to cultural, additionality and institutional conditions. However, from the point of view of economic conditions, PES would not be viable due to the high opportunity costs. Besides, it does not seem appropriate to financially compensate stakeholders that traditionally are

financially successful, considering that small producers are the most disadvantaged social group, especially in a region where social inequity is extremely evident.

### **Conclusions**

As discussed during this chapter, small, medium and large landholders of Cotriguaçu municipality are far from satisfying all the preconditions established by Wunder (2008), necessary for implementing a PES initiative. These results support the decision of proponents to not engage in a cash payments compensation mechanism. Their main motivation was related to the lack of financial resources to implement it since implementation, transaction and opportunity costs would be very high and they were not able to afford it with funding they had at the time of the beginning of the project.

While this was a key consideration, other criteria are also important in the decision to implement a PES, i.e. the other necessary preconditions for implementing it and the extent to which the municipality might be able to change their BAU situation and satisfy these conditions.

Thus, some conclusions can be drawn about the PES tool itself, as well as its appropriateness. First of all, PES is not a panacea. If the social, economic, institutional, cultural, and environmental context of the region is problematic, PES will only contribute to worsen it. PES will work really well in places where it is likely not even necessary. However, in places where many actions are necessary, PES will not be the most appropriate tool. Furthermore, PES was shown to work better as a complementary tool. Only with a well-established suite of command and control working well, will a complementary PES function well.

One more inference that can be drawn is that PES works better locally than globally. Because of the large number of criteria and preconditions that need to be

satisfied in order for PES to work, it is not a simple tool to be implemented. PES is complex and expensive and will work only in specific situations. A PES in large territories or with a high number of people involved will be very difficult to manage and the probability to fail is high.

In the specific region studied, PES definitely is not the most appropriate tool to be implemented, at least in the current situation where small producers' areas do not contribute to the additionality of the CSV project nor satisfy enough cultural, economic, and institutional conditions. Moreover, even with medium and large producers, preconditions are only partially met, it being necessary to promote changes on the economic and social preconditions. All these changes would incur in large efforts and would require financial resources.

For that and other reasons, based on this study as well as on previous studies about PES, we conclude that PES is difficult to be applied in practice, and even more difficult if real projects follow the theoretical criteria established for their implementation. There are, however, several PES-like projects all over the world that do not satisfy the criteria but are operating well. The suggestion is to leave the PES conception more open and let reality tells us what is really necessary for a PES. Thus, a re-think of PES concepts and criteria would be a first initiative to permit a dialog between the concept's theory and the reality on the ground. This contradicts Shiki's opinion (Shiki 2011). According to him, PES, in Brazil, is much more advanced in practice than in theory. Whichever opinion is right, both show that it is important to establish a better dialog on both levels.

Considering all the limitations above, the final issue is to consider whether or not it is worthwhile to make the necessary investments and interventions to satisfy these preconditions in Cotriguaçu. The answer is probably not, since it would be necessary to fight against many obstacles such as high opportunity costs, fragile institutions and deficient monitoring. Besides, even if a PES would partially solve economic issues, situations of conflicts for land could become even worse.

In sum, PES theory and practice do not interact too much. In theory, PES has many requirements and conditions to be effective but its potentialities to function on the ground are limited and the ideal scenario of meeting all of the necessary preconditions is too idealized for most developing countries.

For those reasons, it is time to adapt the totally market-oriented PES concept, to a more flexible one where social and even cultural aspects are considered, as suggested by Muradian et al. (2010). Doing that, there will probably be less contradiction between the theory and practice of PES.

The next chapter (Chapter 5) is concerned about the changes a PES could provoke at a household level in terms of an association of reducing deforestation and improving people's income. The main tool used to this analysis is the Ethnographic Linear Programming (ELP).



Table 4-1. Average opportunity costs (OC, R\$/ha/year) for foregoing meat and crops over a 10-year period

| Community                  | OC cattle<br>(meat) | OC crops | Total |
|----------------------------|---------------------|----------|-------|
|                            | (R\$/ha/year)       |          |       |
| Vale Verde                 | 1,670               | 355      | 2,026 |
| Santa Luzia                | 1,229               | 180      | 1,410 |
| Entre Rios                 | 694                 | 345      | 1,040 |
| Nova Esperança             | 621                 | 243      | 866   |
| Medium and large producers | 1,299               | N/A      | N/A   |

Source: data from field work (August 2010 and July 2011)

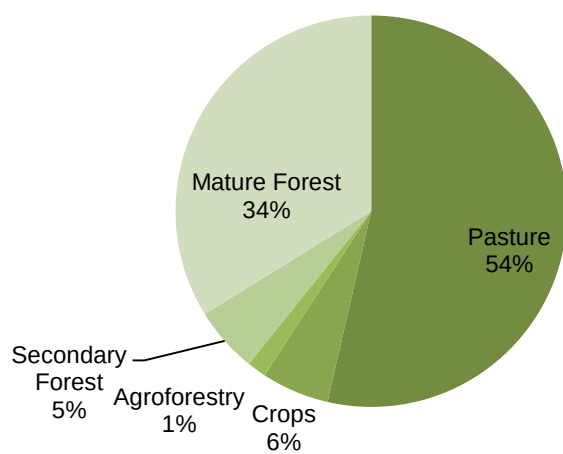


Figure 4-1. Land use by rural settlers in Cotriguaçu

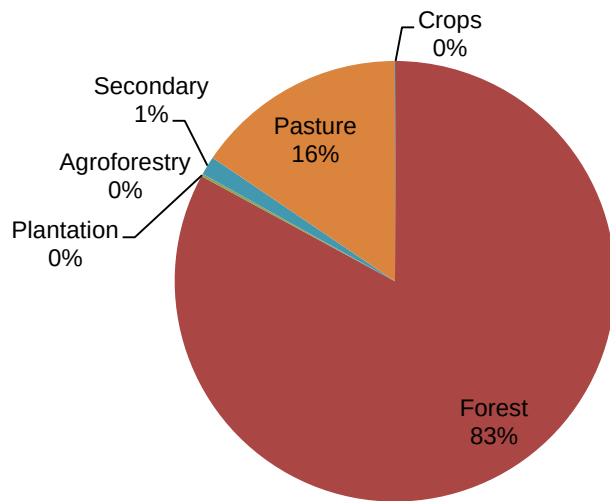


Figure 4-2. Land use by large landholders in Cotriguaçu

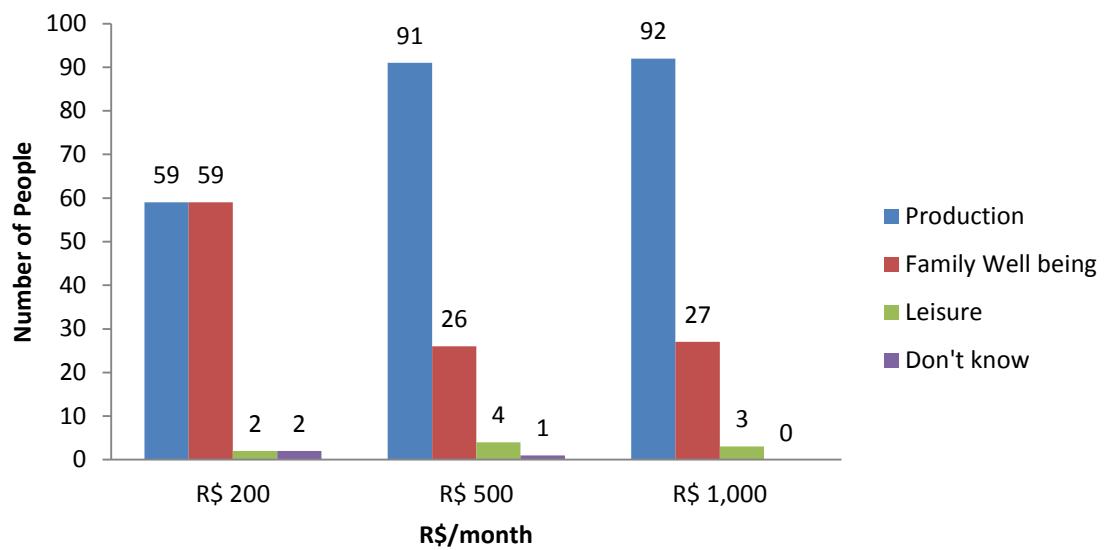


Figure 4-3. Resource allocation of hypothetical income (R\$/month) to rural settlers

Source: field work data (2011).

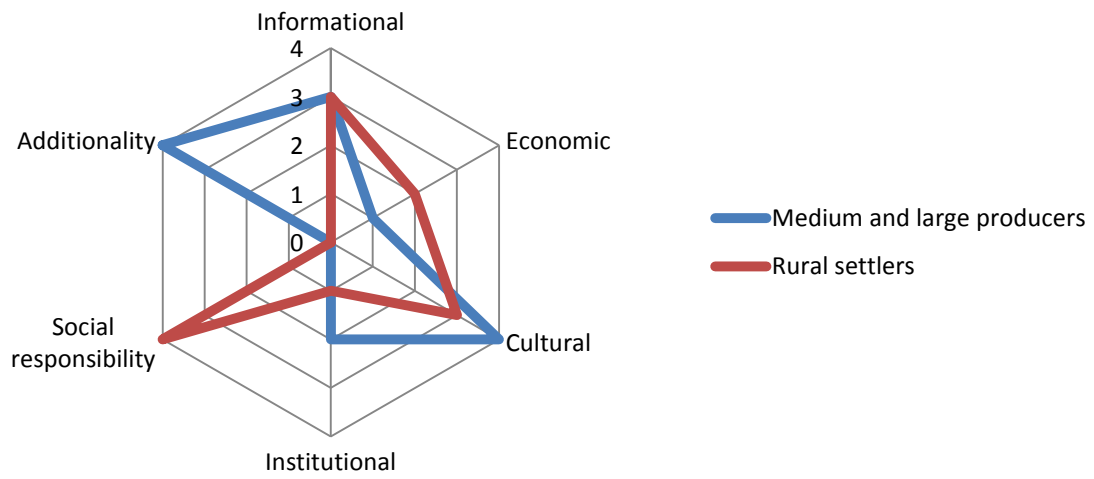


Figure 4-4. Radar graph showing to what extent the pre-conditions for PES are met in the four communities visited, as well as among medium and large producers

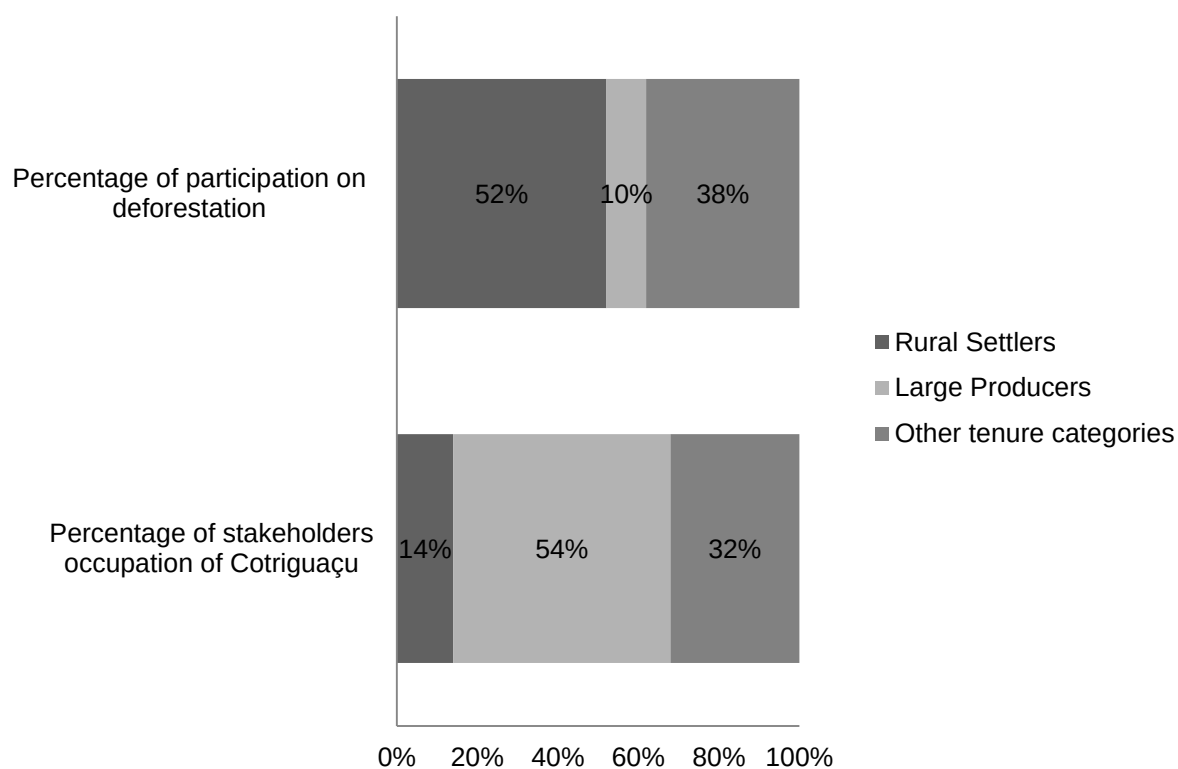


Figure 4-5. Percentage of area occupied by private producers and rural settlers vs percentage of participation in deforestation in Cotriguaçu

## CHAPTER 5

### THE POTENTIAL IMPACTS OF PES ON HOUSEHOLD PRODUCTION SYSTEMS, USING ETHNOGRAPHIC SYSTEM MODELLING

#### **Introduction**

There are several studies about the impacts of PES in the ecosystem as well as in the livelihood system (Viana et al. 2012; Yanez-Pagans 2013). These studies are normally conducted after the implementation of the project and help decision makers evaluate the success of the project in terms of achieving goals such as decrease of deforestation, and determine to what extent this goal is due to the implementation of the project, among others (Viana et al. 2012). However, there are also other ways to predict impacts even before the program implementation through the use of modeling tools. Although these models are not able to incorporate all the variables and visualize drastic changes such as technological improvements, their uses are widely accepted in the case of a system with complex interactions (Evans et al. 2001).

Many researchers consider PES as a suitable tool that combines forest conservation with economic gains and with a high potential to reduce poverty (Wunder 2006). However, it should not be a mechanism to reduce poverty since its main goal is to improve the efficiency of natural resource management (Pagiola et al. 2005; Wunder 2006). On the other hand, there are limited studies about poverty reduction after the implementation of PES programs (Pagiola et al. 2005).

In order for PES to work, there is a general consensus in the literature that compensations must cover all the costs, especially the opportunity costs, of other land uses. However, this opinion is refuted by Muradian et al. (2010) who state that landholders could be convinced to accept an amount smaller than the opportunity costs, because they could receive other compensations such as land property rights.

The objective of this study is to understand the potential impacts of PES, from a livelihood perspective, on forest conservation as well as on income increase among rural settlers in the northwest of Mato Grosso state, in the Brazilian Amazon, and to project the effects on the households of a smaller than opportunity cost payment. In addition, the study analyzes to what extent the PES amount should cover the entire project costs, particularly the opportunity costs for beneficiaries to preserve the forest.

The hypothesis of this analysis is: The potential impacts of a PES in the REDD-like initiative in Cotriguaçu would be able to increase the annual income of small landholders and to make them switch their livelihood activities into sustainable activities.

The research questions of this chapter are as follows:

- Question 1. What are the impacts on rural lands deforestation when producers are paid to preserve their forests?
- Question 2. How could PES be viable in order to improve social and economic impacts of households?
- Question 3. Would landowners accept payments to not deforest lower than opportunity costs to engage in other activities?

In order to answer to these three questions, I will employ a systems modeling tool from a household perspective (developing a livelihood system model projection, or Ethnographic Linear Programming - ELP) to understand the impacts of PES on the household dynamics.

## **Methods**

### **Ethnographic Linear Programming**

Linear Programming models started to be used by agricultural economists in the 1950s. In the 1980s, ELP, with the addition of the ethnographic aspect, was proposed by Professor Hildebrand and associates. That included in a perspective of a household



system, the economic, socio-cultural and biological variables to explain resource management decisions (Hildebrand et al. 2003). Since then, it has become widely applied by University of Florida students among others (P. Hildebrand, personal communication 2014).

The idea behind ELP was to inform farmers about the best way to allocate their resources in order to maximize their profits (Hildebrand et al. 2003), by testing different scenarios, or livelihood strategies, over the years. It also informs others about potential impacts of proposed policies, infrastructure and technology. In fact, ELP is a way to include the real conditions of actual farmers into the model and to provide realistic answers to producers because we can better understand the interactions among resources and decision making over these resources (Hildebrand et al. 2003).

Several researchers used ELP in an attempt to understand variations in household cash income and in land use patterns after different interventions such as changes in livelihood practices or livelihood patterns, due to diseases or adhesion to PES programs. This is the case of Mello (2012), who analyzed the potential effects of carbon trading policies in the households; of DiGiano (2006), who studied the potential impacts of Proambiente, a PES program in the Brazilian Amazon; and of Thangata et al. (2007), who assessed the impacts of HIV/AIDS on household food security in Malawi.

There are, however, some criticism related to participatory methods and linear programming. The most common are related to the fact that these methods require community participation and therefore require a long time to collect data (Wilsey et al. 2012). Another criticism is related to the fact that considering that the model reflects a system, instead of the whole community, the efforts of increasing peoples' participation

must be arranged with the efforts to better understand the livelihood system. However, benefits of using this tool seemed evident to most of the researchers who used it (Wilsey 2012).

### **The Survey**

One hundred twenty people were interviewed in communities Vale Verde, Santa Luzia, Nova Esperança, and Entre Rios as part of a larger study conducted by CIFOR, which served as the main source for this research (see survey in Appendix A). A few more questions were included in the survey in order to complement data collection to design the ELP models based on actual family production systems. These questions supported the design of the calendar of activities as well as the labor needs and gender labor divisions, requirements to design the model.

Besides, two other complementary surveys were implemented: one for the whole community, where the idea was to understand the use of natural resources from the community in a general perspective; and the other just for women, in order to capture information that they probably would not feel comfortable to share if men were present. Finally, some families were more closely observed considering that, in several cases, we stayed a few days in local people's houses.

### **Model Specification**

#### **Livelihood System Models**

This model focuses on learning the effects of Payments for Environmental Services on rural small producers in Cotriguaçu rural settlements. This section will, then, describe the system of production of the households studied as well as the calendar of activities (Appendix B), based on a gender division of labor activities.

A PES program can bring positive effects for a community in terms of cash income and for the whole problem of depletion of natural resources. However, it can also bring negative effects especially if there is no correct monitoring and control and people do not respect the rules of the program. Thus, this analysis is necessary to predict the effects that these payments could bring at household and local levels. Families were chosen for the model based on the premise that PES beneficiaries must have more than 80% of their areas forested, in respect to the percentage the producers ought to preserve by law (Brazilian Forest Code). People would only receive payments if they were able to preserve above 80% of their forests.

From the 120 families interviewed for the Global Comparative Study from CIFOR, only four were eligible to participate in a hypothetical PES program because only these four families had less than 20% of their areas deforested<sup>1</sup>. Another four families had between 70-80% forested lands. Although they are out of the range to receive payments, they might receive some incentives such as seeds or seedlings, or tools, to reforest their lands and contribute to the provision of environmental services, and become eligible for a hypothetical PES. Thus the model will include them, totaling eight families for the analysis. Figure 5-1 shows the percentage of forests for each of the households selected. If there were more households respecting the premise to have more than 70% of forests, these families would definitely be included in the model and this would increase the accuracy of the analysis. On the other hand, it would not be

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<sup>1</sup> According to the Brazilian Forest Code (Law 12.727/2012), in the Amazon Region the area of forest to be preserved must be 80% of the whole property. In rural settlements, Legal Reserve can be individual or collective. For the purpose of this study, Legal Reserve will be individual.

appropriate because these families are not likely to be potential beneficiaries of a PES initiative due to the low amount of forests they have. Still, other strategies seem important to be designed in order to provide these people with the basic conditions for the recuperation of their areas.

### **Description of the Household Systems of Interviewed Families**

Although the economy of the families studied is based on the sale of agrarian products, an important portion of their income comes from salaries and social programs. These families work with annual crops (rice, maize, beans and cassava) and also with cattle ranching (see livelihood systems in Appendix C-1), especially to sell milk. In general, families consider cattle as a fast way to have cash money in case they need it for some emergency. Therefore, they compare it to a “savings account.” In this context, two variables are added to the system: command and control tools, and PES, both having as a goal to reduce the rising deforestation in the rural settlements.

Families in the settlement have a well-defined gender-based division of labor, but in homes in which the man is absent, it is common to see women engaged in heavy work. Both men and women are able to receive salaries. Women normally work at schools and men are more engaged in activities on farms close to the rural settlement. Concerning social programs, only women are eligible to receive this benefit because the Brazilian Government believes that, if men receive this amount, they will not share it with their wives and kids and will spend on alcohol consumption (Jones & Molyneux 2014). If PES is implemented, the logic will be the same and this benefit will be incorporated to the female income. Table 5-1 shows the activities engaged in by men and women.

## **Household Information**

Table 5-2 shows information concerning size of the family, time living in the settlement, main activities, mean income (2010-2011) and percentage of forest in the area controlled by the households. The main criterion for choosing these households is the fact that they have the highest percentage of forest among the 120 households interviewed.

## **Basic Assumptions and Scenarios Projected for ELP Analysis**

The Amazon region has two seasons: a dry season, from November to April, and a rainy season, from May to October. Labor activities are different in both seasons and require a specific labor distribution. For example, concerning agricultural activities, forests must be clear-cut at the end of the rainy season, planted at the beginning of the dry season, and harvested by March-April. Each of these periods will require a specific labor effort, according to the activity implemented, e.g. during the dry season people will be more engaged in clear-cutting the areas and preparing the land for planting crops or pasture. On the other hand, in the rainy season they will be more involved in harvesting. Cattle ranching is an activity carried out during the whole year. However, it is necessary to clear the areas of forest and other vegetation during the first year of the activity. Thus, the analysis considered this seasonal division for agriculture and cattle ranching activities in order to capture labor requirements.

I conducted the analysis of the ELP models with a focus on the maximization of end cash (male cash + female end cash), comparing the amount of labor that males and females dedicate to different tasks, and the land use with or without the payments.

Four distinct scenarios for rural settlers in Cotriguaçu were generated and projected (Table 5-3). The livelihood systems were projected only for one year, because

the intention of the model was not to see system behavior over time. Instead, there was a concern to include crucial variables into the analysis of livelihood dynamics that are considered by decision makers and researchers as keys to decrease deforestation.

Therefore, the first scenario is the current situation of the households, and included the main components of the actual livelihood activities related to land use and income generation. This scenario reflects the Business as Usual (BAU) situation. Producers currently do not experience strong federal enforcement of laws regarding use of forest resources; they are not effectively obliged to respect the Forest Code because monitoring is weak. That is the reason why the majority of the producers interviewed do not have 80% preserved.

In the second scenario, called “CC tools (20% land use)”, the variable command and control was added to the models, requiring producers to respect the Forest Code and use only 20% of their lands for agropastoral practices. However, they were able to use products from the forest insofar as as they did not clear cut trees.

In the third scenario, “CC tolls + PES (10% land use),” a PES variable was added to the second scenario. If people preserve more than 80%, they will receive financial compensation for this preserved area. Therefore, in this scenario people are working only on 10% of their lands.

Finally, the fourth scenario is called “CC+PES smaller than OC.” This scenario is exactly the same as the previous one, except that the amount paid is 10% smaller than the opportunity costs of PES paid in scenario three. The idea is that in this scenario PES will not cover all the opportunity costs. Table 5-3 shows a summary of the four scenarios.

The opportunity costs of scenarios three and four are based on the costs of protecting the forest instead of raising cattle. As shown in Chapter 4, due to the different benefits per ha, each community had a specific opportunity cost that was calculated per ha/year (Table 5-4). In the PES scenarios, I multiplied the opportunity cost for the number of ha each family was allowed by law to deforest but left aside in order to receive PES.

Models were built with inputs and constraints. The inputs included were: biophysical characteristics of the land property (i.e. size of the land, type of land cover), amount of labor (i.e. household composition and labor days available), and other sources of income (i.e. salaries, retirements, social programs) per each family. The constraints included the maximum availability of these characteristics.

Three variables were changed in order to model the changes that would result under each of the scenarios in Table 5-3. These variables are: percent change in forest cover, percent change in pasture, and the minimum household income. Each household was modeled separately based on its actual characteristics, and then subjected to modifications specified for each scenario.

The following variables were used to compare the outcomes after running the model in the four scenarios: forest use above legal limits (ha); area of crops (ha); area of pasture (ha); and household income in Brazilian Reais (R\$). In each scenario the families were provided a defined land amount they could use for their activities.

Therefore, they could not use more forests than the amount permitted by law. When in the outcomes there were lands being used from forest above those permitted, that

meant the scenario was not viable (example of outcomes in Appendix D). Results for each household are explained below.

## **Results**

Results are shown individually for each household analyzed. The analysis especially considered the degree to which each household refrained from opening more forest areas than permitted by law, in each of the four scenarios.

### **Household 1**

Results showed that in this household context it was not viable to have “CC tools+PES” and “Smaller than OC” scenarios because, even with PES compensation, it would be necessary to deforest more than permitted in order to satisfy family needs (Appendices E and F – Household 1).

This household is the one with the largest amount of forest in the whole sample of 120 households; about 97.8% of the site is still forested. The family is large, composed of a couple and four children, which means that their consumption needs are large and will be even higher as children grow up.

Household income does not depend entirely from land use since the husband has a job on a farm and the wife receives monthly income from the Bolsa Familia. Even with these cash sources, the model results showed that in “BAU” scenario they would have to clearcut some areas in order to satisfy consumption needs, especially to raise cattle (Appendix G-1).

In the “CC tools” scenario, they would have the right to use 20% of their forest, and monitoring would be applied. Even then, results show that they would have to use slightly more (0.22) than the 20% allowed in order to satisfy their consumption needs.



In the scenario “CC tools + PES,” this family was not able to succeed by working in only 10% of their property, even receiving PES. They would have to deforest 5.76 ha more than allowed.

Finally, results of the scenario “PES Smaller than OC” are similar to (“CC tools+PES). They receive less money but this does not change land use, which means that it does not matter too much if PES is similar or smaller than the opportunity costs of raising cattle.

## **Household 2**

Results show that in this household context all the four scenarios are viable. The family will not need to deforest more than permitted in order to satisfy family needs. However, with both PES scenarios other sources of income would support the family to satisfy their needs. Even in the fourth scenario, where PES does not cover the opportunity costs, results of the model show that this would satisfy all the family consumption requirements (Appendices E and F – Household 2).

This household is the second-most forested in the sample, with a small family (a single man), which means that family needs are low and land use is not pressuring the forest. The household head has a fixed salary, which also means that he does not need to cut trees to survive. Still, he raises some cattle to save more money.

In the four scenarios he will not need to clear more forest areas than he is already permitted, and crops and pasture areas remain the same in the four scenarios. When PES are included, this household has an income increase, which could guarantee the preservation of this site (Appendix G-1). When PES are lower than opportunity costs, the results are still satisfactory, which indicates that PES does not need to cover opportunity costs to be accepted.

### **Household 3**

Results show that in this household context it is not viable to have “CC tools+PES” and “Smaller than OC” scenarios due to the fact that even with PES compensation, it would be necessary to deforest more than permitted in order to satisfy family needs (Appendices E and F – Household 4).

Household 3 is a large one, formed by a young couple, three children and an old man. Currently, this household keeps 85.6% of its land in forest, and the family has a small income (salary and social program), which means that they need to use the land in order to satisfy their basic consumption needs. PES benefits provided a small increase in family income.(Appendix G-1).

### **Household 4**

Results show that in this household context it is not viable to have “CC tools+PES” and “Smaller than OC” scenarios because, even with PES compensation, it would be necessary to deforest more than permitted in order to satisfy family needs (Appendices E and F – Household 4). Household 4 has a single man. Especially when PES is included, the household member is not able to survive without deforesting more than permitted. In terms of income, PES provided a small increase in their income (Appendix-G-1).

### **Household 5**

Results show that in this household context it is not viable to have “CC tools+PES” and “Smaller than OC” scenarios. Even with PES compensation, it would be necessary to deforest more than permitted in order to satisfy family needs (Appendices E and F – Household 5).

This household has a large family (a medium age couple, two teenagers and a child) but, by far, they have the highest income due to the fact they have a timber saw company. Their annual income is about R\$ 200,000. This household is already in an illegal context because it has more than 20% deforested. Compensation from PES would have the conditionality to reforest the areas in order to reach the small amount permitted. Scenarios with PES still would not be viable because the family would need to open more forest areas than permitted to meet their needs. PES benefits increase even more the income of the family (Figure G-2).

### **Household 6**

Results show that this household context (like Household 2) fits the ideal because it is viable on the four scenarios because it does not need to clear forests above than permitted (Appendices E and F – Household 6). This household is composed by only one person, who also works outside of his property and receives a small salary. However, he depends on the farm to complement his income. When PES is included, like in other households, it provides an increase in the income (Figure G-1). Hence, he raises cattle and plants some crops. Scenarios without or with PES look similar in terms of land use.

### **Household 7**

Results show that this household context also seems to be ideal, even more than household 6, because the family is larger and they do not need to open more forest areas than permitted in the four scenarios (Appendices E and F – Household 7).

Six people make up this relatively large household. They have income from activities outside of the site (R\$ 22,890), which means that they do not totally depend on the site to satisfy their consumption needs. Scenarios without or with PES are similar in

terms of land use. They will not be pressured when command and controls tools are implemented, nor when they receive PES (and are allowed to work on even less land). When PES is implemented, the only variation is in the income, which had a significant increase (Figure G-1). This compensation will control the opening of more areas.

### **Household 8**

Results show that in this household context it is slightly non-viable for the four scenarios, due to the fact that the family deforests above the permitted, even if only a small area (Appendices E and F – Household 8).

A single man forms this small household. He receives extra income due to some occasional labor, which means that he needs to work in his property in order to satisfy his consumption needs. Land use in the four scenarios is similar. He still needs to open more areas in the four scenarios. Concerning PES, it helps to increase income (Figure G-1) but it is not sufficient to control the opening of more forested areas.

### **Discussion**

The main goal of this chapter is to understand the impacts of a (hypothetical) PES in the use of forestlands and in the families' income, through the ethnographic linear program modeling tool. Depending on the impacts of PES on forests use and income, it is possible to assess the viability of it among the properties chosen.

The basic criterion to include a household in the model was to have at least 80% of area covered by forest (in order to respect Brazilian Forest Code Law). From the eight households detected, only four perfectly fit this condition. Thus, to increase the sample, four more households with almost of 80% of forests were included with the condition as potential beneficiaries for PES (for the purpose of this modeling exercise) to reforest their areas. Four scenarios were considered:

1. BAU Scenario: the current condition of the households;
2. Command and control tools (CC tools) Scenario: in this scenario people were allowed to use only 20% of their areas and monitoring and enforcement tools were able to punish the ones that did not respect the law;
3. Command and control tools combined with PES Scenario: in this scenario, people were allowed to deforest only 10% of their areas and could trade the ES of the other 10% they were allowed to deforest but preferred to be compensated;
4. Command and control tools combined with a lower than OC PES Scenario: this scenario is similar to the last one but payments are smaller than the OCs of preserving the forest instead of engaging on other activities.

The analysis of the four scenarios clearly shows that PES is, in most of the cases, non-viable among the 8 properties of the model (Table 5-5). What was considered to make a scenario viable or non-viable was the capacity of the family to satisfy their consumption necessities to survive, without opening forested areas more than permitted. The model considered only basic consumption necessities.

Three aspects are relevant for these outcomes: the first is related to the efficiency of the command and control tools. In 6 out of the 8 households, this tool worked as a good solution to control deforestation and also allowed people to produce. It is, however, relevant to point out that in all these households people had an alternative income and therefore they were not dependent or at least, totally dependent from the forest to fulfill their consumption needs.

The second important aspect is related to the extra income as a driver to support deforestation control. In most of the cases, HHs that had enough areas to trade environmental services had also large extra income from activities outside of their areas. Probably for this reason, they did not engage in clearcutting the forest.

Finally, a third aspect is related to the fact that it seems irrelevant if PES received is equal or smaller than the opportunity costs to preserve. In all the cases PES that was

viable, it was also viable if a smaller payment was offered.

These findings pointed out that PES is not a good tool to promote forest preservation in the rural settlements of Cotriguaçu. First, because there are not many households that fit the criterion for being beneficiaries. Second, many families that have this potential are not forest dependent and PES would be just an additional income. These people normally have salaries or receive benefits from social programs, which decrease the pressure on the forests. However, PES can be a good tool to promote forest preservation if used as a complement to other tools, such as command and control instruments; and PES can also work in situations where the families had already deforested larger areas and needed this financial support to recuperate these areas.

In households with slightly less than 80% of preserved forests, the models showed that people would necessarily be forced to reforest their areas, which means that they would have to employ more labor efforts for this. Still, the program accepted these conditions well and showed that it was possible to satisfy families' consumption needs with the support from PES.

In all the cases, command and control tools (e.g. monitoring and fines) were shown to be effective and sufficient in the models, although in practice this monitoring does not occur in perfection. According to the last report about deforestation and degradation in Cotriguaçu developed by ICV using data from IMAZON (ICV et al. 2012), illegal deforestation is still a reality in the municipality, principally in the rural settlements. Thus, the implementation of a more efficient monitoring system is urgently needed. Only after having total control over lands and deforestation, could a PES tool be implemented.

Other reflections related to the value of the payment can be made. In exactly all the eight cases modeled, the results of the model that ran with a full PES (equal to the opportunity costs) and a smaller PES (below the opportunity costs) were similar. This could be telling us that cash from PES is not a golden bullet to reduce deforestation. Instead, command and control tools seem much more effective for that. PES can more appropriately be seen as a prize for the good behavior of providing environmental services. Besides, PES is fragile in the sense that it does not guarantee a land use scenario for a long term. PES is not eternal and, as soon as payments stop, chances for engaging in deforestation are increased in the absence of other measures.

To establish effective PES policies for the Amazon Forest it is recommendable that the government pays a reasonable amount, at least close to the opportunity costs of engaging in non-sustainable activities. Besides, it is important to give ways for the communities to engage in sustainable alternative activities, such as offering courses that would improve their capacity building and other even more important conditions such as: feeder roads, land regularization, technical assistance and credit.

Finally, a smaller than the opportunity costs payments might be acceptable. However, there should also be incentives to engage in sustainable extractive activities that could guarantee greater monetary income for the family even after the end of the program of PES, as well as such inputs as assets and tools. The model confirms the idea that PES does not need to cover the whole cost, especially the opportunity cost, to be successful.

Another important consideration is the fact that all the households studied had alternative activities and extra income, which means that these families were not

dependent uniquely from forest resources. If the situation were different, in a context in which people depended exclusively from the forest to survive and to satisfy their consumption needs, very probably they would not be able to leave areas aside from production. This is the case of many households in all the rural settlements in the whole Amazon forest.

Failure to consider these findings in the design and implementation of REDD programs may result in serious damage to the concept of the project and among potential beneficiaries. Moreover, it can provoke perverse side effects, such as increasing the risk of more deforestation by those trying to receive cash benefits because if monitoring fails, people will want to invest resources from PES in their production systems to have more profits.

### **Conclusion**

PES is seen by many researchers as a suitable tool to conserve forest while reducing poverty (Wunder 2006). However, there are not sufficient studies focused on PES impacts to confirm the efficiency of this tool in terms of reconciling preservation and poverty alleviation.

In the context of Cotriguaçu, PES is partially a viable solution for small producers in the rural settlements. The first obstacle for a PES among this public is related to the low additionality of the project because most forests in rural settlements are already deforested or degraded. However, there are still some producers with highly forested areas and these were the people included in this ELP modeling.

Results showed that when PES is applied, people will have their capacity of land use reduced and some will still need to open forested areas to satisfy their consumption needs, pointing out that PES is not always viable, even if PES was covering opportunity



costs. An additional analysis was related to the amount paid. Comparing PES in a normal situation with PES smaller than the opportunity costs, this will not change too much in terms of the need to open more areas. This indicates that PES can work in situations where the financial compensation does not cover the opportunity costs. However, negotiations should be needed at some point in order to convince people to receive less than the costs of raising cattle.

In short, PES is viable where people have the conditions to live without opening more forested areas. In all the households studied, families were involved in other activities and had alternative income. For that reason, they were not totally dependent on the forest and could leave areas aside to preserve and still survive. Unfortunately, this is not the case of many other families in the rural settlements in Cotriguaçu and in all other places in the Amazon.

Table 5-1: Main activities engaged by male and female in the households interviewed

| Activities      | Responsible         |
|-----------------|---------------------|
| Crops           | Male and Female     |
| Cassava flour   | Male and Female     |
| Cattle          | Male                |
| Small Animals   | Female and children |
| Salaries        | Male and Female     |
| Social Programs | Female              |

Table 5-2: Socio-economic information of the eight households (HH)

| HH  | Family members<br>(gender and age)                                                | Time living<br>in the settlement<br>(years) | Main activities                    | Income from salaries/<br>social programs? | Annual amount of extra<br>income (R\$/year, 2010-2011) | % forest |
|-----|-----------------------------------------------------------------------------------|---------------------------------------------|------------------------------------|-------------------------------------------|--------------------------------------------------------|----------|
| HH1 | Man (30)<br>Woman (27)<br>Child (11)<br>Child (9)<br>Child (4)<br>Child (1)       | 10                                          | Agriculture, cattle ranching       | Yes                                       | 1,608                                                  | 97.7     |
| HH2 | Man (34)                                                                          | 17                                          | Cattle ranching                    | No                                        |                                                        | 86.7     |
| HH3 | Man (65)<br>Woman (52),<br>Man (31),<br>Women (17)<br>Child (2), and<br>Child (1) | 16                                          | Agriculture, cattle ranching       | Yes                                       | 8,352                                                  | 85.6     |
| HH4 | Mand (30)                                                                         | 6                                           | Cattle ranching                    | Yes*                                      | 8,240                                                  | 80.0     |
| HH5 | Man (40)<br>Woman (38)<br>Woman (19)<br>Child (15)<br>Child (12)                  | 17                                          | Timber company;<br>Cattle ranching | Yes**                                     | 200,000                                                | 79.2     |
| HH6 | Man (54)                                                                          | 16                                          | Cattle ranching                    | Yes***                                    | 3,300                                                  | 78.8     |
| HH7 | Man (54)<br>Woman (54)<br>Woman (20)<br>Man (24)<br>Woman (22)<br>Child (2)       | 20                                          | Cattle ranching (milk)             | Yes****                                   | 22,890                                                 | 75.5     |
| HH8 | Man (63)                                                                          | 16                                          | Cattle ranching                    | Yes*****                                  | 2,500                                                  | 73.8     |

Note: \*He works in a farm with cattle ranching and sells timber; \*\*They own a saw company; \*\*\*He is truck driver and transports log; \*\*\*\*They own a sawmill; \*\*\*\*\*He occasionally works in construction.

Table 5-3: Four scenarios considered for running the ELP model

| Scenarios       |                |             |                                       |                                       |
|-----------------|----------------|-------------|---------------------------------------|---------------------------------------|
|                 | BAU            | CC tools    | CC tools + PES                        | CC tools + PES smaller than OC        |
| Forest Area     | Between 70-90% | 80% or more | Above 80%                             | Above 80%                             |
| Deforested area | Between 10-30% | 20% or less | Below 20%                             | Below 20%                             |
| PES             | No             | No          | Yes, if forest is preserved above 80% | Yes, if forest is preserved above 80% |

Table 5-4: Opportunity costs (OC) to protect the forest instead of raising cattle in the four communities studied in Cotriguaçu

| Communities    | OC protect forests<br>(R\$/ha/year) |
|----------------|-------------------------------------|
| Vale Verde     | 1,670                               |
| Santa Luzia    | 1,229                               |
| Entre Rios     | 694                                 |
| Nova Esperança | 621                                 |

Table 5-5. Viability of PES in the four scenarios

| Households | Scenario<br>BAU | Scenario<br>CC tools<br>(def. 20%) | Scenario<br>CC tools<br>(def.10%)<br>+ PES | Scenario<br>CC tools<br>(def 10%)+<br>PES<br>smaller than<br>OCs | Observations<br>about salary and<br>main activity |
|------------|-----------------|------------------------------------|--------------------------------------------|------------------------------------------------------------------|---------------------------------------------------|
| HH1        | Non-Viable      | Non-Viable                         | Non-viable                                 | Non-viable                                                       | Low income; cattle                                |
| HH2        | Viable          | Viable                             | Viable                                     | Viable                                                           | No income; cattle                                 |
| HH3        | Viable          | Viable                             | Non-viable                                 | Non-viable                                                       | Medium income; cattle                             |
| HH4        | Viable          | Viable                             | Non-viable                                 | Non-viable                                                       | Medium income; cattle                             |
| HH5        | Viable          | Viable                             | Non-viable                                 | Non-viable                                                       | Extremely high income; cattle and timber company  |
| HH6        | Viable          | Viable                             | Viable                                     | Viable                                                           | Low income; cattle                                |
| HH7        | Viable          | Viable                             | Viable                                     | Viable                                                           | High income; cattle (milk)                        |
| HH8        | Non-viable      | Non-viable                         | Non-viable                                 | Non-viable                                                       | Low income; cattle                                |

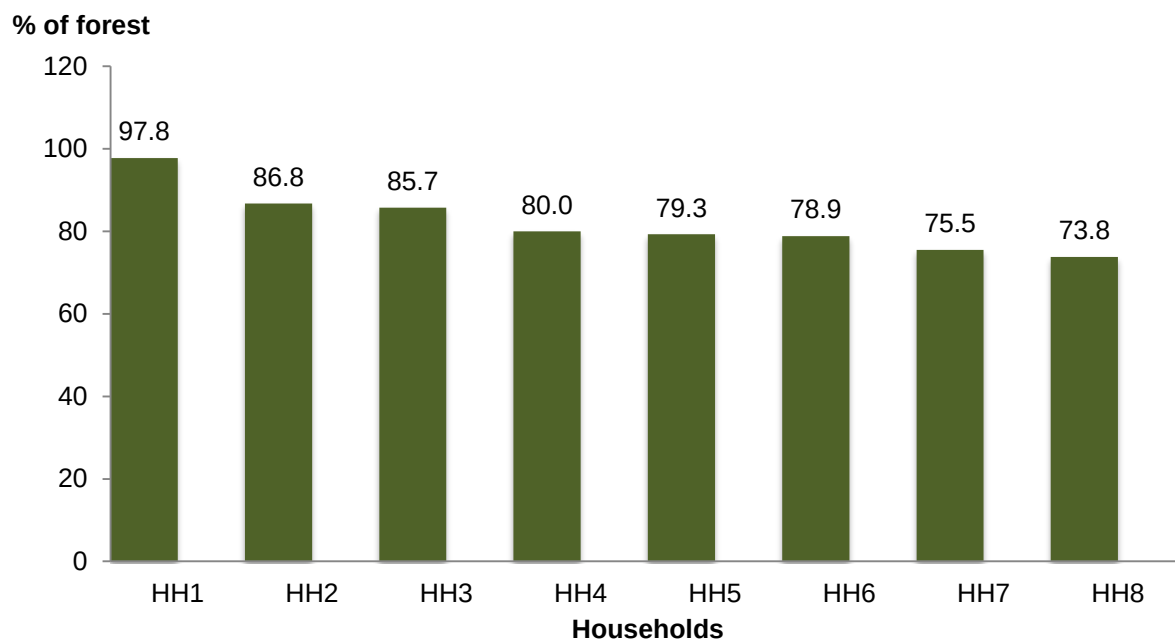


Figure 5-1: Households with the highest percentage of forests among the universe of 120 households interviewed

## CHAPTER 6 CONCLUSIONS

It is a capital mistake to theorize before one has data.  
Insensibly one begins to twist facts to suit theories,  
instead of theories to suit facts.

—Arthur Conan Doyle, *Sherlock Holmes*

Based on the analysis of this whole study, it is possible to draw conclusions about the following aspects of PES:

1. from a perspective of Social Ecological Systems (SES) and systems resilience, to assess a hypothetical Payments for Environmental Services (PES) program in the municipality of Cotriguaçu discussing to what extent PES can contribute to the resilience of rural settlements in Cotriguaçu municipality as a (SES);
2. to develop an *ex ante* investigation of PES analyzing to what extent Cotriguaçu meets the preconditions established by Wunder (2009) for implementing a PES project; and
3. using the Ethnographic Linear Program modeling tools, to analyze household livelihood systems with or without PES, as well as to analyze the ability of families to accept PES even if proposed payments are smaller than the opportunity costs to preserve the forest instead of engaging in cattle ranching activities.

The study showed that the resilience related to tolerance to changes in SES rural settlements in Cotriguaçu is extremely low and fragile. This system suffered a lot of disturbances in a small amount of time, which caused many changes in the system dynamics. Thus, the system has a minimal capacity to avoid or tolerate changes. Even though the system had a reasonable ability to recover from these disturbances in a short amount of time, showing that it is somewhat resilient to the impacts of these disturbances, this is due to the fact that disturbances are related to people's livelihood conditions. Since people need to survive in that area, they have two possibilities: to abandon the area or to recover from those changes. Many people preferred the second



option even if the resilience is not truly fast or the new situation after the changes is not stable or ideal.

The CSV project, a REDD-like initiative, appears as a driver of changes towards a more sustainable socio-ecological system in the sense of increasing the potential resilience of this system. CSV has the potential to promote a series of positive changes in this system such as the strengthening of productive systems and governance in the rural settlements, and promotion of productive sustainable activities, which correspond to the final goals of reduction of deforestation and degradation from REDD. However, this potential is still to be realized.

The issue related to the proponent's decision to not include a PES into CSV lines of actions remains polemical. At the beginning of the project, proponents wanted to implement a PES tool, which pleased many potential beneficiaries. However, rural settlements, in general, present high deforestation rates, and the implementation of PES in these areas risked not being efficient because additionality would be low. Therefore, proponents favored other tools (transfer of technology, support to governance, among other), which, more than PES, have the potential to make the resilience of the system more robust.

The fact is that PES is not a fundamental tool for rural settlers of Cotriguaçu in the context of the CSV project just because it will not contribute to system resilience. Other more emergent actions stand to contribute more to SES resilience, such as infrastructure improvements, which happen through a stronger presence of the government; the provision of technical assistance to rural settlers; the diversification of

their production; the improvement of governance in rural settlements; investment in capacity building of local leadership; and, finally, the creation of local small industries.

Concerning the necessary preconditions established by Wunder (2008) to implement a PES initiative, results of the study pointed out that Cotriguaçu is still far from satisfying those conditions. The criteria evaluated were the informational, economic, cultural, and institutional (market, trust, and land tenure) preconditions as well as additionality, and social responsibility.

In the specific region studied, PES definitely is not the most appropriate tool to be implemented, at least in the current situation where small producers' areas do not contribute to the additionality of the CSV project nor satisfy enough cultural, economic, and institutional conditions. Moreover, even with medium and large producers, preconditions are only partially met, it being necessary to promote changes in the economic and social preconditions. All these changes would demand large efforts and would require financial resources that probably could not be afforded.

Thus, the conclusion for this topic of satisfaction of preconditions is that PES is difficult to be applied in practice, and even more difficult if real projects follow the theoretical criteria established for their implementation. A re-think of PES concepts and their criteria seems necessary in order to permit a dialog between the concept, theory, and the reality on the ground.

Regarding the analysis of PES from a household perspective using the ELP modeling tool, results show that PES is only part of the solution for small producers in the rural settlements in the context of CSV project. The main obstacle is that there are not enough forests in the rural settlements, which would make it difficult to find eligible

families as potential beneficiaries of this initiative. In the cases of families who do get involved in a PES initiative, results of the modeling showed that people will have their capacity of land use reduced and some will still need to open forested areas to satisfy their consumption needs, pointing out that PES is not always viable, even if PES was covering opportunity costs.

In terms of the amount paid in such projects, a comparison between PES in a normal situation with PES smaller than the opportunity costs, results showed that this would not change too much in terms impacts on the households. In other words, PES is viable in situations where payments are lower than the opportunity costs to preserve the forests. Besides, PES is viable in areas where people have other alternative incomes and are not totally dependent from forest resources.

Considering all of the stated above, PES is not suitable to be implemented in Cotriguaçu, for three main reasons. First, PES will not improve the system resilience. Other tools are more imperative and urgent such as more presence of the state, improvement of local infrastructure and strengthening of local governance among others. Second, Cotriguaçu does not fit all the criteria to implement a PES initiative and it seems still more difficult to provide all the conditions to fit these criteria. Third, PES does not improve livelihood systems and does not contribute to deforestation control. With PES, families risk being stuck, and will still need to open forested areas to satisfy their survival needs.

However, these results do not mean that PES is not an appropriate tool for other parts of the Amazonia region. Proponents of PES should assess the potentialities of the region to shelter this initiative. Therefore, this study provides contributions both for the

literature of PES, as well as practical recommendations for the proponents that are implementing PES.

Regarding the contributions to the literature, it is important to realize that there is still a large gap between reality and theory of PES. Even after almost a decade of being discussed and already implemented in many countries, the PES concept is still under discussion. If on one hand it is necessary to reflect about a broader concept that embraces also cultural and social aspects, on the other hand, it is important to have more rigor in PES projects, otherwise this tool will risk becoming “nothing and everything” at the same time.

Moreover, PES should have to be more flexible and respect local aspects. As mentioned in Chapter 2, Muradian et al. (2010) have proposed a concept of PES where PES would be “the transfer of resources between social actors, which aims to create incentives to align individual and/or collective land use decisions with the social interest in the management of natural resources” which Wunder promptly rejected since it was too unspecific and could provoke even more confusion on PES initiatives of fitting in this definition.

Notwithstanding, it is still necessary to think about ways to consider social and cultural aspects in order to minimize the issues of having many initiatives that does not fit the concept. Thus, I propose that the traditional concept of PES established by Wunder (2005) could be complemented by Muradian et al. (2010) vision which will culminate on the following: “Based on the local context and social and political conditions of the social actors, PES is (1) a voluntary market transaction, in which (2) a well-defined environmental service (or a land use likely to secure that service) (3) is

“bought” by a (minimum of one) buyer (4) from a (minimum of one) provider (5) if and only if the provider continuously secures the provision of the service (conditionality) (6) and if these transactions will benefit the management of the natural resources.”

Although it is a large concept, it embraces important perspectives of how environmental services should be traded as well as considering that, most of all, natural resources must be benefited and cultural aspects must be respected.

In terms of practical recommendations, proponents of PES must have in mind some steps before applying it. As said before, normally projects are implemented considering, above all, the availability of financial resources of proponents, but this single criterion is not enough to opt for its implementation.

Due to different approaches of PES and the necessity to better understand the local context before applying them, more *ex ante* studies are needed as well as more systematic follow up of projects already implemented. Such studies should be focused on the Willingness to Accept (WTA) of potential ES sellers and the (WTP) of potential buyers, as well as the use of other valuation methods to find out the correct value of these services. The accurate calculation of the ES provides as well its value, a consensus about sharing benefits distribution, in order avoid conflicts among the stakeholders as a side effect. Riyong Bakkegaard, a researcher from the University of Copenhagen, recently developed a study where she assessed the WTA of the beneficiaries of the Bolsa Floresta program, using two methods: a standard one, where they inquired what people were willing to accept, and an auction, in which they played a game in order to detect the right amount of money people wished to receive from the PES program. Interestingly, results showed that the WTA was lower than the

opportunity cost estimates of preserving the forestland (Riyong 2013). All of these studies can be expensive in terms of time and money, but it seems more reasonable doing this than just investing resources in an area where risks of PES not being successful are high.

Another crucial step is the analysis of the pre-conditions of the region where PES will be implemented. Otherwise, proponents risk “a shot in the dark” because they will be investing in an initiative that might not be suitable to the area. One more practical recommendation is that it is fundamental to understand the dynamics of the system (most important disturbances and weaknesses) where PES will be applied and check if PES can be an appropriate tool to increase the capacity this SES has to tolerate disturbances. Moreover, proponents must be sure that other policies such as command and control tools will be combined with PES. Finally, at a household level, it is necessary to understand to what extent families are dependent on the forest and assess if PES will be viable to halt deforestation and increase families’ income.

PES is an appropriate complementary tool when the necessary base is set up and should be applied with precaution; otherwise its effects risk not being positive. Furthermore, it is especially important to conduct more studies about the appropriateness of PES as well as the impacts of this tool in projects that are already implemented.

## APPENDIX A HOUSEHOLD SURVEY

Questionário de Unidades Familiares  
Estudo Comparativo Global de REDD  
Componente 2: Locais de Projetos REDD  
Centro Internacional de Pesquisa Florestal (CIFOR), Bogor, Indonésia

| <b>Informação básica</b>                                                   |     |                                 |    |
|----------------------------------------------------------------------------|-----|---------------------------------|----|
| Código do país                                                             |     | Código do local do projeto REDD |    |
| Distrito                                                                   |     | Nome da localidade              |    |
| Comunidade de intervenção?                                                 |     |                                 |    |
| Código da unidade familiar                                                 |     |                                 |    |
| Coordenada GPS                                                             | UTM | X:                              | Y: |
| Distância do centro da localidade em minutos de caminhada e em quilômetros |     |                                 |    |
|                                                                            |     |                                 |    |

| Cumprimento das funções de pesquisa                |          |                    |
|----------------------------------------------------|----------|--------------------|
| Função                                             | Por quem | Data (dia/mês/ano) |
| Realização da entrevista                           |          |                    |
| Preenchimento do questionário durante a entrevista |          |                    |
| Codificação do questionário após a entrevista      |          |                    |
| Revisão pelo entrevistador                         |          |                    |
| Revisão pelo FRS                                   |          |                    |
| Digitação no país da pesquisa                      |          |                    |
| Checagem da entrada de dados pelo FRS              |          |                    |
| Checagem de dados na sede do CIFOR                 |          |                    |

**Tabela 1A. INFORMAÇÕES BÁSICAS SOBRE OS MEMBROS DA UNIDADE FAMILIAR**

Nós queremos perguntar informações básicas sobre todos os membros da unidade familiar.

| 1.<br>No. | Nome | 2.<br>Parentesco<br>com o<br>chefe da<br>família<br><i>Códigos<br/>abaixo</i> | 3.<br>Gênero<br><br>0 =<br>masculino<br>1 =<br>feminino | 4.<br>Idade | 5.<br>Anos de<br>educação | 6.<br>Principal<br>ocupação<br>nos<br>últimos<br>12 meses<br>(para<br>aqueles<br>com ≥16<br>anos de<br>idade) | 7.<br>Ocupação<br>secundária<br>nos<br>últimos 12<br>meses<br>(para<br>aqueles<br>com ≥16<br>anos de<br>idade) | 8.<br>Dias<br>doente<br>nos<br>últimos<br>12<br>meses<br>(para<br>aqueles<br>com ≥16<br>anos de<br>idade) |
|-----------|------|-------------------------------------------------------------------------------|---------------------------------------------------------|-------------|---------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1.        |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |
| 2.        |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |
| 3.        |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |
| 4.        |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |
| 5.        |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |
| 6.        |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |
| 7.        |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |
| 8.        |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |
| 9.        |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |
| 10.       |      |                                                                               |                                                         |             |                           |                                                                                                               |                                                                                                                |                                                                                                           |

**Códigos para coluna 2: Parentesco com o chefe da unidade familiar**

|                                 |                     |                                             |
|---------------------------------|---------------------|---------------------------------------------|
| 1 = chefe/a da unidade familiar | 6 = pai/mãe         | 11 = sobrinho/sobrinha                      |
| 2 = esposa/o                    | 7 = sogro/sogra     | 12 = enteado/criança adotada                |
| 3 = filho/filha                 | 8 = irmão/irmã      | 13 = outros familiares                      |
| 4 = genro/nora                  | 9 = cunhado/cunhada | 14 = membros não<br>aparentados com o chefe |
| 5 = neto/neta                   | 10 = tio/tia        |                                             |



**Tabela 1C. INFORMAÇÕES SOBRE O/A CHEFE DA FAMÍLIA E CÔNJUGE**

Nós gostaríamos de fazer algumas perguntas sobre o/a chefe/a dessa família e seu cônjuge (se aplicável).

|                                                                                                                                                                                                                                                      |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1. Qual é o estado civil (legal ou lei comum) do/a chefe/a da família?<br><i>Códigos: 1=casado/a e morando junto; 2=casado/a, mas com o/a esposo/a vivendo/trabalhando fora; 3=viúvo/viúva; 4=divorciado/a; 5=nunca casou; 9=outro (especifique)</i> |   |
| 2. Há quantos anos atrás essa família foi formada pela primeira vez?<br><i>Indicar a resposta em anos</i>                                                                                                                                            |   |
| 3. O esposo/homem nasceu nessa comunidade?<br><i>1 = sim; 0 = não</i><br><i>Se a resposta é 1 (=sim), vá para 5.</i>                                                                                                                                 |   |
| 4. Há quantos anos o esposo/homem da família vive nessa comunidade?<br><i>Indicar resposta em anos</i>                                                                                                                                               |   |
| 5. O esposo/homem da família pertence ao grupo étnico desta comunidade?<br><i>Nota: Se existe apenas um grupo étnico na comunidade, responda 'sim.'</i><br><i>1 = sim; 0 = não</i>                                                                   | 1 |
| 6. A esposa/mulher nasceu nessa comunidade?<br><i>1 = sim; 0 = não; -8 = não se aplica (não há esposa)</i>                                                                                                                                           |   |
| 7. Há quantos anos a esposa/mulher vive nessa comunidade?<br><i>Indicar resposta em anos -8 = não se aplica (não há esposa/o)</i>                                                                                                                    |   |
| 8. A esposa/mulher pertence ao grupo étnico majoritário desta comunidade?<br><i>Nota: Se existe apenas um grupo étnico na comunidade, responda 'sim.'</i><br><i>1 = sim; 0 = não; -8 = não se aplica (não há esposa/o)</i>                           | 1 |

**Raissa:** Trajetória Familiar:

R1. De onde o Sr. Veio?

R2. Com que o Sr. Trabalhava antes?

R3. O que o trouxe para cá?

## 2. BENS DA FAMÍLIA

Preencha a área total, em hectares, para cada uma das quatro categorias de uso da terra na Tabela 2A abaixo.

| Tabela 2A. ÁREAS DE TERRAS DA FAMÍLIA, POR CATEGORIA DE USO DA TERRA |                                                                                                            |                                          |                                                                      |                                                                                 |                                                                              |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Tipo de cobertura da terra                                           | Categoria de uso da terra<br><i>Veja o caderno de códigos: Classificações do uso da terra (definições)</i> | CONTROLADA PELA FAMÍLIA                  |                                                                      | NÃO CONTROLADA PELA FAMÍLIA                                                     |                                                                              |
|                                                                      |                                                                                                            | 1. Área de terra usada pela família (ha) | 2. Área de terra arrendada ou cedida em empréstimo pela família (ha) | 3. Área de terra arrendada ou recebida em empréstimo e em uso pela família (ha) | 4. A família usa terra comum dessa categoria?<br><i>1 = sim;<br/>0 = não</i> |
| 1. Agricultura                                                       | 1. Cultivos agrícolas (roçado, laranja, cacau, banana, pimenta do reino...)                                |                                          |                                                                      |                                                                                 |                                                                              |
|                                                                      | 2. Agrofloresta (árvores frutíferas, madeira, quintal)                                                     |                                          |                                                                      |                                                                                 |                                                                              |
|                                                                      | 3. Pastagem (campo plantado)                                                                               |                                          |                                                                      |                                                                                 |                                                                              |
|                                                                      | 4. Silvo-pastoril (silvicultura + animais domésticos – benefício mútuo)                                    |                                          |                                                                      |                                                                                 |                                                                              |
| 2. Floresta                                                          | 1. Grandes plantações (florestas plantadas com poucas espécies)                                            |                                          |                                                                      |                                                                                 |                                                                              |
|                                                                      | 2. Capoeira nova ( $\leq 4$ anos)                                                                          |                                          |                                                                      |                                                                                 |                                                                              |
|                                                                      | 3. Capoeira mais velha ( $> 4$ anos)                                                                       |                                          |                                                                      |                                                                                 |                                                                              |
|                                                                      | 4. Mata bruta                                                                                              |                                          |                                                                      |                                                                                 |                                                                              |
| 3. Outra                                                             | <i>Veja definição abaixo*</i>                                                                              |                                          |                                                                      |                                                                                 |                                                                              |
| TOTAL                                                                |                                                                                                            |                                          |                                                                      |                                                                                 |                                                                              |

\* A categoria 'outra' de cobertura da terra é composta por áreas de moradia, infra-estrutura, cerrados, campos ou áreas úmidas.

**Tabela 2B. POSSE DA TERRA POR TIPO DE CONTROLE E CATEGORIAS DE USO E TIPOS DE COBERTURA DA TERRA**

Quais são os arranjos de posse das terras que a sua família usa, tanto aquelas terras controladas como as não controladas pela família, e qual o grau de segurança dessa posse? Por 'segurança de posse' nós queremos dizer a confiança que a sua família será capaz de continuar usando as terras que vocês têm atualmente por pelo menos os próximos 25 anos.

*Certifique-se de que as áreas totais nas Tabelas 2A e 2B correspondam! Note que nessa tabela as quatro categorias excluem o 'outro' tipo de uso da terra!*

| 1. Categoria de posse da terra                                                                             | 2. Tipo de cobertura da terra | 3. Área do terreno (com unidade!!!) | 4. Tipo de posse<br>Use o código de 3 dígitos<br>Veja o caderno de códigos:<br>Posse | 5. Nível de segurança de posse<br>Códigos:<br>1 = seguro<br>0 = inseguro | 6. Razão para insegurança de posse, se insegura<br>Liste até 3 razões.<br>Veja o caderno de códigos:<br>Razões para segurança de posse. |
|------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1. Terra controlada e usada pela família<br><br>(coluna 1 na Tabela 2A)                                    | 1. Agricultura                |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            | 2. Floresta                   |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
| 2. Terra controlada e não usada pela família<br><br>(coluna 2 na Tabela 2A)                                | 1. Agricultura                |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            | 2. Floresta                   |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
| 3. Terra não controlada mas recebida em arrendamento ou empréstimo pela família<br>(coluna 3 na Tabela 2A) | 1. Agricultura                |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            | 2. Floresta                   |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
| 4. Terra comunitária compartilhada pela UF com outras UFs<br>(Col. 4 na Tabela 2A)                         | 1. Agricultura                |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            | 2. Floresta                   |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |
|                                                                                                            |                               |                                     |                                                                                      |                                                                          |                                                                                                                                         |

|                                                                                                                              |                                                                                    |    |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|----|
| 3. Quais são as áreas de terra da família, se houver, <b>FORA DA COMUNIDADE?</b><br><i>Se nenhuma use -8 = não se aplica</i> | 1. Terra controlada e usada pela família                                           | Ha |
|                                                                                                                              | 2. Terra controlada pela família e cedida em arrendamento ou em empréstimo         | Ha |
|                                                                                                                              | 3. Terra não controlada pela família mas recebida em arrendamento ou em empréstimo | Ha |
|                                                                                                                              | 4. Uso de terras comunitárias. 1= sim; 0= não                                      |    |

Código Posse: XYZ

1-Estado 2-Comunidade

3-Privado/Indivíduo (X = usuário de direito)

1-Estado 2-Comunidade

3-Privado/Indivíduo 4-Empresas (Y = usuário de fato)

### 3. RENDA DA FAMÍLIA

*Produção e custos devem ser calculados nas tabelas A e B (C e D, se necessário) para um período de 12 meses. Alternativamente os dados podem ser apresentados de acordo com as estações, para o período de 12 meses.*

| <b>Tabela 3A. PRODUÇÃO AGRÍCOLA PARA O PERÍODO DE 12 MESES ANTERIORES À ENTREVISTA</b>                                                                                                                                                                         |                    |                         |                              |                   |                      |                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------------|------------------------------|-------------------|----------------------|------------------------------------------|
| Nós gostaríamos de calcular a sua produção agrícola durante os últimos 12 meses.                                                                                                                                                                               |                    |                         |                              |                   |                      |                                          |
| Essa tabela inclui informação para o período completo dos últimos 12 meses antes da data da entrevista ( <b>marque SIM ou NÃO</b> ). Se NÃO:<br>Essa tabela inclui informação para apenas para uma estação a partir de _____ (mês / ano) até _____ (mês / ano) |                    |                         |                              |                   |                      |                                          |
| No.                                                                                                                                                                                                                                                            | 1. Produto         | 2. Produção total (4+5) | 3. Unidade (para a produção) | 6. Preço unitário | 7. Valor total (2X6) | Raissa Mão de obra. Dias trabalhados/ano |
|                                                                                                                                                                                                                                                                | <u>CEREAIS</u>     |                         |                              |                   |                      |                                          |
| 100.                                                                                                                                                                                                                                                           | Milho              |                         |                              |                   |                      |                                          |
| 101.                                                                                                                                                                                                                                                           | Arroz              |                         |                              |                   |                      |                                          |
| 120.                                                                                                                                                                                                                                                           | Mandioca           |                         |                              |                   |                      |                                          |
| 130.                                                                                                                                                                                                                                                           | Macaxeira          |                         |                              |                   |                      |                                          |
| 128.                                                                                                                                                                                                                                                           | Farinha            |                         |                              |                   |                      |                                          |
| 133.                                                                                                                                                                                                                                                           | Farinha de Tapioca |                         |                              |                   |                      |                                          |
| 129.                                                                                                                                                                                                                                                           | Goma               |                         |                              |                   |                      |                                          |
| 122.                                                                                                                                                                                                                                                           | Batata Inglesa     |                         |                              |                   |                      |                                          |
| 123.                                                                                                                                                                                                                                                           | Batata doce        |                         |                              |                   |                      |                                          |
|                                                                                                                                                                                                                                                                | <u>LEGUMES</u>     |                         |                              |                   |                      |                                          |
| 161.                                                                                                                                                                                                                                                           | Feijão preto       |                         |                              |                   |                      |                                          |
| 162.                                                                                                                                                                                                                                                           | Feijão vermelho    |                         |                              |                   |                      |                                          |
|                                                                                                                                                                                                                                                                | <u>VERDURAS</u>    |                         |                              |                   |                      |                                          |
| 200.                                                                                                                                                                                                                                                           | Tomate/Cubil       |                         |                              |                   |                      |                                          |
| 201.                                                                                                                                                                                                                                                           | Cenoura            |                         |                              |                   |                      |                                          |
| 203.                                                                                                                                                                                                                                                           | Pimentão           |                         |                              |                   |                      |                                          |
| 204.                                                                                                                                                                                                                                                           | Pepino             |                         |                              |                   |                      |                                          |
| 205.                                                                                                                                                                                                                                                           | Alho               |                         |                              |                   |                      |                                          |

|      |                                     |  |  |  |  |  |
|------|-------------------------------------|--|--|--|--|--|
| 206. | Cebolha palha                       |  |  |  |  |  |
| 207. | Alface                              |  |  |  |  |  |
| 212. | Abóbora                             |  |  |  |  |  |
| 233. | Couve                               |  |  |  |  |  |
| 237. | Cheiro Verde                        |  |  |  |  |  |
|      |                                     |  |  |  |  |  |
|      | <u>FRUTOS</u><br><u>(plantados)</u> |  |  |  |  |  |
| 300. | Manga                               |  |  |  |  |  |
| 301. | Mamão                               |  |  |  |  |  |
| 302. | Goiaba                              |  |  |  |  |  |
| 303. | Laranja                             |  |  |  |  |  |
| 304. | Banana                              |  |  |  |  |  |
| 306. | Coco                                |  |  |  |  |  |
| 308. | Maracujá                            |  |  |  |  |  |
| 309. | Abacaxi                             |  |  |  |  |  |
| 310. | Açaí                                |  |  |  |  |  |
| 311. | Jaca                                |  |  |  |  |  |
| 313. | Carambola                           |  |  |  |  |  |
| 321. | Abacate                             |  |  |  |  |  |
| 324. | Graviola                            |  |  |  |  |  |
| 325. | Limão                               |  |  |  |  |  |
| 326. | Tamarindo                           |  |  |  |  |  |
|      | OUTROS                              |  |  |  |  |  |

| #   | 1. Item                                                               | 2. Quantidade | 3. Unidades | 4. Preço unitário |
|-----|-----------------------------------------------------------------------|---------------|-------------|-------------------|
| 1.  | Sementes ou mudas, materias para plantação<br>_____<br>_____<br>_____ |               |             |                   |
| 2.  | Fertilizantes/Adubo Químico                                           |               |             |                   |
| 3.  | Esterco                                                               |               |             |                   |
| 4.  | Inseticidas/Herbicidas/Fungicidas                                     |               |             |                   |
| 5.  | Tração animal                                                         |               |             |                   |
| 6.  | Trabalho contratado                                                   |               |             |                   |
| 7.  | Máquinas contratadas                                                  |               |             |                   |
| 8.  | Transporte/(venda)                                                    |               |             |                   |
| 9.  | Pagamento de renda da terra                                           |               |             |                   |
| 10. | Outro: especificar (ex. combustível)                                  |               |             |                   |

1. Materiais para plantação: Ancinho, Draga, Enxada, Esmeril, Foice, Lima, Machado, Terçado  
5. Aluguel boi de arrasto/boi manso;6. Diária Terçado; 7.Diária Trator, Motosserra  
10. Diesel, Óleo Queimado, Óleo 2 Tempos, Gasolina

**RAISSA ELP**

ELP 1 – Quais destes procedimentos o Sr.(a) adota no preparo e cultivo das áreas de plantio e de pasto (anotar o mês da atividade e o número de dias gasto na atividades por hectare ou tarefa) ?

| FASES    | ROÇA(ultima vez) |     |             |                | PASTO |     |             |                |
|----------|------------------|-----|-------------|----------------|-------|-----|-------------|----------------|
|          | Área             | Mês | Dias gastos | N.º de pessoas | Área  | Mês | Dias gastos | N.º de pessoas |
| Broca    |                  |     |             |                | -     | -   | -           | -              |
| Derruba  |                  |     |             |                | -     | -   | -           | -              |
| Aceiro   |                  |     |             |                |       |     |             |                |
| Queima   |                  |     |             |                |       |     |             |                |
| Coivara  |                  |     |             |                | -     | -   | -           | -              |
| Plantio  |                  |     |             |                | -     | -   | -           | -              |
| Capina 1 |                  |     |             |                | -     | -   | -           | -              |
| Capina 2 |                  |     |             |                | -     | -   | -           | -              |
| Colheita |                  |     |             |                | -     | -   | -           | -              |

ELP 2. No último ano, o Sr.(a) contratou diaristas para realizar as atividades listadas (especificar número de diaristas, remuneração e dias gastos)?

| Diaristas                | Quantidade de diaristas | Valor da diária | Total de dias gastos |
|--------------------------|-------------------------|-----------------|----------------------|
| Derrubada                |                         |                 |                      |
| Aceiro                   |                         |                 |                      |
| Plantio                  |                         |                 |                      |
| Colheita                 |                         |                 |                      |
| Limpeza de roça          |                         |                 |                      |
| Limpeza de pasto         |                         |                 |                      |
| Cuidar do gado(vaqueiro) |                         |                 |                      |

**TABELA 3E. POSSE DE GADO E OUTROS ANIMAIS E RENDA DE VENDAS NOS ÚLTIMOS 12 MESES**

Nós gostaríamos de perguntar questões sobre a posse, consumo e vendas de gado e outros animais nos últimos 12 meses.

*Para os tipos de animais marcados com um asterisco (\*) pergunte apenas sobre os animais adultos.*

| 1.<br>Animal               | 2.<br>Inicial | 3.<br>Vendidos | 4.<br>Abatidos<br>ou<br>perdidos | 6.<br>Comprados<br>ou nascidos | 8.<br>Final | 9.<br>Preço | 10. Valor<br>Total |
|----------------------------|---------------|----------------|----------------------------------|--------------------------------|-------------|-------------|--------------------|
| 1. Vaca                    |               |                |                                  |                                |             |             |                    |
| 2. Touro                   |               |                |                                  |                                |             |             |                    |
| 3. Boi                     |               |                |                                  |                                |             |             |                    |
| 4. Novilho/<br>novilha     |               |                |                                  |                                |             |             |                    |
| 5. Bezerro                 |               |                |                                  |                                |             |             |                    |
| 6. Búfalo –<br>adulto      |               |                |                                  |                                |             |             |                    |
| 7. Búfalo –<br>jovem       |               |                |                                  |                                |             |             |                    |
| 8. Cavalo<br>– adulto      |               |                |                                  |                                |             |             |                    |
| 9. Cavalo<br>– jovem       |               |                |                                  |                                |             |             |                    |
| 10. Burro /<br>mula *      |               |                |                                  |                                |             |             |                    |
| 11. Cabra *                |               |                |                                  |                                |             |             |                    |
| 12. Ovelha<br>*            |               |                |                                  |                                |             |             |                    |
| 13. Porco *                |               |                |                                  |                                |             |             |                    |
| 14. Perus *                |               |                |                                  |                                |             |             |                    |
| 15. Patos *                |               |                |                                  |                                |             |             |                    |
| 16. Galinha<br>s *         |               |                |                                  |                                |             |             |                    |
| 17. Galos *                |               |                |                                  |                                |             |             |                    |
| 18. Galinha<br>Angola<br>* |               |                |                                  |                                |             |             |                    |
| 19. Gansos<br>*            |               |                |                                  |                                |             |             |                    |
| 20. Coelhos<br>*           |               |                |                                  |                                |             |             |                    |
| 21. Peixes                 |               |                |                                  |                                |             |             |                    |
| 22. Enxame<br>s<br>Abelhas |               |                |                                  |                                |             |             |                    |
| 23. Outro:                 |               |                |                                  |                                |             |             |                    |

**Tabela 3F. RENDA DA VENDA DE PRODUTOS DE ORIGEM ANIMAL NOS ÚLTIMOS 12 MESES**

Quais foram as quantidades e valores de produtos e serviços de origem animal que você produziu durante os últimos 12 meses?

| 1. Produto            | 2. Unidade | 3. Unidades produzidas (4+5) | 4. Uso próprio | 5. Vendas | 6. Preço unitário | 7. Valor total (3X6) |
|-----------------------|------------|------------------------------|----------------|-----------|-------------------|----------------------|
| TOTAL DE LEITE        |            |                              |                |           |                   |                      |
| 1. Leite <sup>1</sup> |            |                              |                |           |                   |                      |
| 2. Manteiga           |            |                              |                |           |                   |                      |
| 3. Queijo             |            |                              |                |           |                   |                      |
| 4. Ovos               |            |                              |                |           |                   |                      |
| 5. Couros/ peles      |            |                              |                |           |                   |                      |
| 6. Lã                 |            |                              |                |           |                   |                      |
| 7. Esterco            |            |                              |                |           |                   |                      |
| 8. Tração animal      |            |                              |                |           |                   |                      |
| 9. Mel                |            |                              |                |           |                   |                      |
| 10. Sabão             |            |                              |                |           |                   |                      |
| 11. Outro (Favo)      |            |                              |                |           |                   |                      |
| 12. Outro (Propólis)  |            |                              |                |           |                   |                      |
| 13. Outro             |            |                              |                |           |                   |                      |

**RAISSA ELP**

ELP 3 - O Sr(a). sempre trabalhou com agricultura( )/pecuária( ) ? ( )sim ( )não

ELP 4 - Caso não, o que o Sr.(a) fazia antes?

ELP 5 - Caso não, há quantos anos trabalha com agricultura \_\_\_\_\_ / pecuária \_\_\_\_\_ ?

ELP 6 – O Sr(a) já recebeu algum tipo de financiamento? ( )sim ( )não

ELP 7 – Caso sim, qual o tipo de financiamento (especificar a linha de crédito, o ano em que recebeu ou vai receber, se foi para a agricultura, pecuária ou os dois, já recebeu as parcelas e se esta endividado ou não?)

| Tipo de financiamento | Ano em que: |             | Finalidade(*) |          |
|-----------------------|-------------|-------------|---------------|----------|
|                       | Recebeu     | Vai receber | Agricultura   | Pecuária |
|                       |             |             |               |          |

(\*)Marcar com um X, e no caso de ter recebido para os dois, marcar ambos.

<sup>1</sup> Certifique-se de que não há duplicação no cálculo das quantidades de laticínios (leite, manteiga, queijo). Uma parte do leite produzido pode ter sido usada para fazer manteiga ou queijo. Anote na linha correspondente ao 'leite' apenas a quantidade de leite não transformado consumido ou vendido pela unidade familiar.



**Tabela 3J. MUDANÇAS NA COBERTURA E NA RENDA FLORESTAL NOS ÚLTIMOS DOIS ANOS**

Nós queremos conhecer como a sua renda florestal tem mudado nos últimos **DOIS** anos (24 meses) e as razões para estas mudanças.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                              |           |           |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|
| 1. A sua família derrubou alguma área de floresta durante os dois últimos anos?<br><i>1 = sim; 0 = não.</i><br><i>Se 'sim,' vá para 2. Se 'não,' vá para 9.</i>                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                              |           |           |           |
| <b>Se SIM:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2. Qual foi a área total de floresta derrubada nos dois últimos anos?<br><i>Indicar a área total derrubada em hectares, para um total de até 3 parcelas.</i> | Parcela 1 | Parcela 2 | Parcela 3 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3. Qual foi o principal propósito da derrubada?<br><i>Códigos: 1=roça; 2= plantação de árvores; 3=pastagem; 4=uso não-agrícola</i>                           |           |           |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4. [Se foi usada para lavouras, roçado (código '1' na questão acima)] Qual foi o principal produto plantado? <i>Veja o caderno de códigos: Produto</i>       |           |           |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 5. Que tipo de floresta você derrubou?<br><i>Veja o caderno de códigos: tipo de floresta</i>                                                                 |           |           |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 6. Se foi capoeira secundária, qual era a idade da floresta?<br><i>Indicar a idade da floresta em anos.</i>                                                  |           |           |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 7. Qual era a situação de posse da floresta derrubada? <i>Use o código de 3 dígitos. Veja o caderno de códigos: posse.</i>                                   |           |           |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 8. Quanto longe de casa estava localizada a floresta derrubada?<br><i>Indicar a distância em quilômetros.</i>                                                |           |           |           |
| 9. Quanto de terra usada pela família foi deixada em pousio ou abandonada nos últimos 2 anos (deixada para regeneração natural)?<br><i>Indicar a área em hectares.</i>                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                              |           |           |           |
| 10. Nos últimos dois anos, as oportunidades para derrubar uma área com florestas têm aumentado, permanecido a mesma, ou diminuído?<br><i>1 = aumentado</i><br><i>2 = permanecido a mesma</i><br><i>3 = diminuído</i><br><i>-8 = não se aplica (os moradores da localidade não derrubam floresta)</i><br><i>-9 = respondente não sabe</i><br><i>Se Q10=3 ir para Q.12</i>                                                                                                                   |                                                                                                                                                              |           |           |           |
| 11. Se as oportunidades para derrubar uma área com florestas têm aumentado, quais são as principais razões? (selecione todas que se aplicam) (pq tá mais fácil derrubar mata hoje comparado com 2 anos atrás?)<br><i>1 = menor competição de moradores da comunidade</i><br><i>2 = menor competição de pessoas de comunidades vizinhas</i><br><i>3 = redução de restrições do governo</i><br><i>4 = redução da interferência de companhias reivindicando posse sobre terras florestais</i> |                                                                                                                                                              |           |           |           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>5 = outra (especificar) _____</p> <p>6 = outra (especificar) _____</p> <p>-9 = respondente não sabe</p>                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
| <p>12. Se as oportunidades para derrubar uma área com florestas têm diminuído, quais são as principais razões? (selecione todas que se aplicam)</p> <p>1= competição de moradores da comunidade</p> <p>2= competição de pessoas de comunidades vizinhas</p> <p>3= restrições do governo</p> <p>4= interferência de companhias reivindicando posse sobre terras florestais</p> <p>5= novas regras impostas pelo projeto de REDD</p> <p>6=outra (especificar) _____</p> <p>8=outra (especificar) _____</p> <p>-9=respondente não sabe</p> |  |
| <p>13. Nos últimos dois anos, o consumo de produtos florestais pela sua família tem aumentado, permanecido o mesmo, ou diminuído? <b>T. 3H</b></p> <p>1 = aumentado</p> <p>2 = permanecido a mesma</p> <p>3 = diminuído</p> <p>4 = varia por produto</p> <p>-8 = não se aplica (sem consumo de produto florestal)</p> <p>-9 = respondente não sabe</p>                                                                                                                                                                                  |  |
| <p>14. Se a resposta para a questão 13 é 1(=aumentado) ou 3 (=diminuído) pergunte:<br/>Por que o consumo de produtos florestais pela sua família aumentou / diminuiu nos últimos dois anos? Liste até 3 razões. Use -8 (não se aplica) se a resposta para a questão 13 é 2, 4, -8 ou -9</p> <p>1. _____</p> <p>2. _____</p> <p>3. _____</p>                                                                                                                                                                                             |  |
| <p>15. Nos últimos dois anos, a renda florestal em dinheiro (i.e. <u>para venda, não para consumo doméstico</u>) da sua família tem aumentado, permanecido a mesma, ou diminuído? <b>T. 3H</b></p> <p>1 = aumentado</p> <p>2 = permanecido a mesma</p> <p>3 = diminuído</p> <p>4 = varia por produto</p> <p>-8 = não se aplica (sem renda de produto florestal)</p> <p>-9 = respondente não sabe</p>                                                                                                                                    |  |
| <p>16. Se a resposta para a questão 15 é 1(=aumentado) ou 3 (=diminuído) pergunte:<br/>Por que a renda florestal em dinheiro de sua família aumentou / diminuiu nos últimos dois anos? Liste até 3 razões. Use -8 (não se aplica) se a resposta para a questão 15 é 2, 4, -8 ou -9</p> <p>1. _____</p> <p>2. _____</p> <p>3. _____</p> <p>4. _____</p>                                                                                                                                                                                  |  |

**Tabela 3K. RENDA DE NEGÓCIOS DA FAMÍLIA NOS ÚLTIMOS 12 MESES**

Alguém na família tem um negócio – mesmo que seja pequeno – e se sim, qual foi a renda bruta e quais os custos durante os últimos 12 meses?

*Verifique na Tabela 1A se algum membro da família tem uma empresa familiar. Certifique-se de olhar em ambas as colunas de ocupação primária e secundária. Certifique-se de perguntar se há algum pequeno negócio que não foi revelado por esses códigos. Pode haver também uma terceira ocupação. Se isto for mais fácil, você pode obter dados sobre as vendas, custos e renda anuais perguntando sobre os valores mensais e os meses em que o negócio esteve em operação, e então agregando os resultados.*

|                                                                                                                                                                             | Negócio 1 | Negócio 2 | Negócio 3 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|
| 1. Qual é o seu tipo de negócio?<br><br><i>Veja o caderno de códigos: Ocupação</i>                                                                                          |           |           |           |
| 2. Qual é o seu tipo de negócio?<br><br><i>Se o código usado acima não fornece informação suficiente para descrever o negócio, registre informações adicionais aqui.</i>    |           |           |           |
| 3. Qual foi a valor de vendas brutas (receita) no últimos 12 meses?                                                                                                         |           |           |           |
| 4. Quais foram os custos de seu negócio durante os últimos 12 meses (inclua insumos comprados, trabalho contratado, aluguéis, etc.)?                                        |           |           |           |
| 5. Qual foi a renda líquida de seu negócio durante os últimos 12 meses?<br><i>Renda líquida é definida como o valor das vendas brutas menos custos (3 menos 4).</i>         |           |           |           |
| 6. Qual é o valor do seu negócio se fosse vender hoje? <i>Valor do negócio é definido como maquinário, estoque de insumos, etc., se vendido hoje, nas condições atuais.</i> |           |           |           |

**Tabela 3L. RENDA DE SALÁRIO OU REMUNERAÇÃO NOS ÚLTIMOS 12 MESES**

Algum membro da sua família recebeu salário ou diária (em dinheiro ou em outros benefícios) para trabalhar nos últimos 12 meses?

*Uma pessoa pode ser listada mais de uma vez porque esta pessoa pode ter realizado diferentes trabalhos remunerados. Verifique na Tabela 1A as indicações para esse tipo de trabalho, mas tenha cuidado pois alguns trabalhos podem ser considerados de menor importância e, em consequência, eles não aparecem na Tabela 1A.*

| 1. Membro da unidade familiar<br>(Número da Tabela 1A, coluna 1) | 2. Tipo de trabalho<br><br><i>Veja o caderno de códigos: meios de subsistência</i> | 3. Códigos para unidades:<br><i>1 = atividade (sem unidade de tempo)<br/>2 = hora;<br/>3 = dia;<br/>4 = semana;<br/>5 = mês;<br/>6 = ano;<br/>9 = outro (especifique)</i> | 4. Unidades de trabalho nos últimos 12 meses<br><br><i>Use -8 se no. 3 = 1 (pagamento por atividade e não por unidade de tempo)</i> | 5. Relação de pagamento por unidade de tempo<br><br><i>Use -8 se no. 3 = 1 (pagamento por atividade e não por unidade de tempo)</i> | 6. Total renda<br>(4 X 5) |
|------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
|                                                                  |                                                                                    |                                                                                                                                                                           |                                                                                                                                     |                                                                                                                                     |                           |
|                                                                  |                                                                                    |                                                                                                                                                                           |                                                                                                                                     |                                                                                                                                     |                           |
|                                                                  |                                                                                    |                                                                                                                                                                           |                                                                                                                                     |                                                                                                                                     |                           |
|                                                                  |                                                                                    |                                                                                                                                                                           |                                                                                                                                     |                                                                                                                                     |                           |
|                                                                  |                                                                                    |                                                                                                                                                                           |                                                                                                                                     |                                                                                                                                     |                           |

**Tabela 3M. RENDA MISTA NOS ÚLTIMOS 12 MESES**

Por favor, conte-nos sobre qualquer outro tipo de renda que a sua família tem e que ainda não foi mencionada (em dinheiro ou em outros benefícios). Eu vou ler uma lista em voz alta. Por favor, diga-nos se você recebeu renda em alguma dessas categorias, e em caso positivo, quanto recebeu nos últimos 12 meses.

| No. | 1. Tipo de renda                               | 3. Quantia total recebida nos últimos 12 meses |
|-----|------------------------------------------------|------------------------------------------------|
| 1.  | Arrendamento de terra                          |                                                |
| 2.  | Dinheiro mandado de fora                       |                                                |
| 3.  | Presentes, apoio da família ou amigos          |                                                |
| 4.  | Herança                                        |                                                |
| 5.  | Aposentadoria                                  |                                                |
| 6.  | Apoio do governo                               |                                                |
|     | <i>FUNRURAL (16)</i>                           |                                                |
|     | <i>Salário de maternidade (18)</i>             |                                                |
|     | <i>Auxílio da saúde (19)</i>                   |                                                |
|     | <i>Auxílio escolar (20)</i>                    |                                                |
|     | <i>Bolsa família (21)</i>                      |                                                |
| 7.  | Renda de políticos locais                      |                                                |
| 8.  | Apoio de ONG                                   |                                                |
| 9.  | Indenização                                    |                                                |
| 10. | Dividendos de empresa comunitária/ cooperative |                                                |
| 11. | Pagamentos por serviços ambientais             |                                                |
| 12. | Outro (especifique)                            |                                                |

**Raissa**

Algumas questões abertas para o final da entrevista

R5.O sr. acha que alguma atitude deveria ser tomada para combater o desmatamento na sua região? O que, por exemplo?

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R6. O que seria necessário para a sua família continuar ganhando o mesmo valor com sua produção sem ter que abrir novas áreas para produzir?

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R7. O que o sr. faria se sua renda mensal fosse aumentada em:

R\$

200,00? \_\_\_\_\_

\_\_\_\_\_

R\$

500,00? \_\_\_\_\_

\_\_\_\_\_

R\$

1500,00? \_\_\_\_\_

\_\_\_\_\_

**6. AVALIAÇÃO PELO ENUMERADOR**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1. Quanto tempo durou a entrevista?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>min</i> |
| 2. Em termos gerais, quão confiável é a informação fornecida por essa unidade familiar?<br><i>Códigos: 1=pouco; 2=razoavelmente confiável; 3=muito confiável</i>                                                                                                                                                                                                                                                                                                                                                       |            |
| 3. Se a confiabilidade da informação é pequena, qual é a razão?<br><i>Códigos. Liste todas as razões que se aplicam:</i><br><i>1=Nessa comunidade as pessoas não podem falar francamente sobre certas práticas florestais;</i><br><i>2=o respondente não estava entusiasmado para participar da entrevista;</i><br><i>3=o tempo disponível para a entrevista foi muito curto;</i><br><i>4=o foco de atenção do respondente desapareceu porque a entrevista foi muito longa;</i><br><i>5= outra razão (especifique)</i> |            |

APPENDIX B  
CALENDAR OF ACTIVITIES OF RURAL SETTLERS OF COTRIGUAÇU

|                 | JAN                          | FEB | MAR                                                                                             | APR | MAY | JUN | JUL                                                                                 | AUG                                                                                                   | SEP | OCT | NOV | DEC                                                                                             |
|-----------------|------------------------------|-----|-------------------------------------------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----|-----|-----|-------------------------------------------------------------------------------------------------|
| Crops           |                              |     | Harvesting<br> |     |     |     |                                                                                     | Slash and Burn<br> |     |     |     | Planting<br> |
| Cattle          | Treating, giving medicaments |     |                                                                                                 |     |     |     |    |                                                                                                       |     |     |     |                                                                                                 |
| Milk            | Milking                      |     |                                                                                                 |     |     |     |    |                                                                                                       |     |     |     |                                                                                                 |
| Small animals   | Feeding                      |     |                                                                                                 |     |     |     |    |                                                                                                       |     |     |     |                                                                                                 |
| Salaries        | Mostly in close large farms  |     |                                                                                                 |     |     |     |   |                                                                                                       |     |     |     |                                                                                                 |
| Social Programs | Bolsa Floresta Program       |     |                                                                                                 |     |     |     |  |                                                                                                       |     |     |     |                                                                                                 |

Figure B-1. Annual activities of rural settlers of Cotriguaçu

APPENDIX C  
SCHEMATIC OF THE LIVELIHOOD SYSTEMS IN RURAL SETTLERS IN  
COTRIGUAÇU

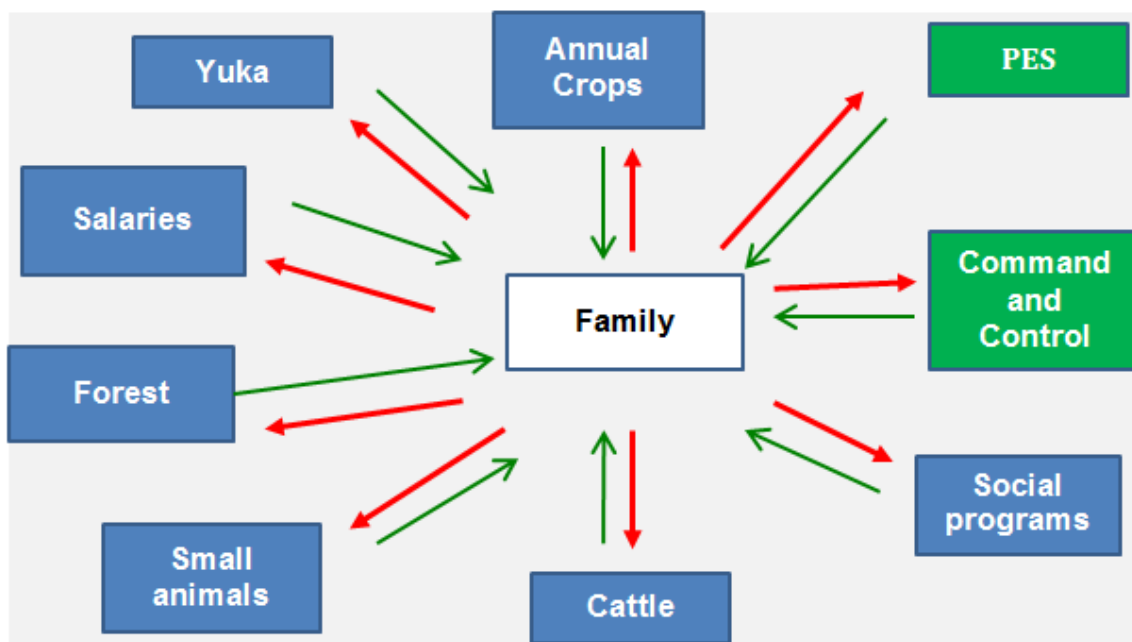


Figure C-1. Description of the livelihood systems in rural settlers in Cotriguaçu

# APPENDIX D LAND DISTRIBUTION IN THE EIGHT HOUSEHOLDS STUDIED FOR SCENARIOS

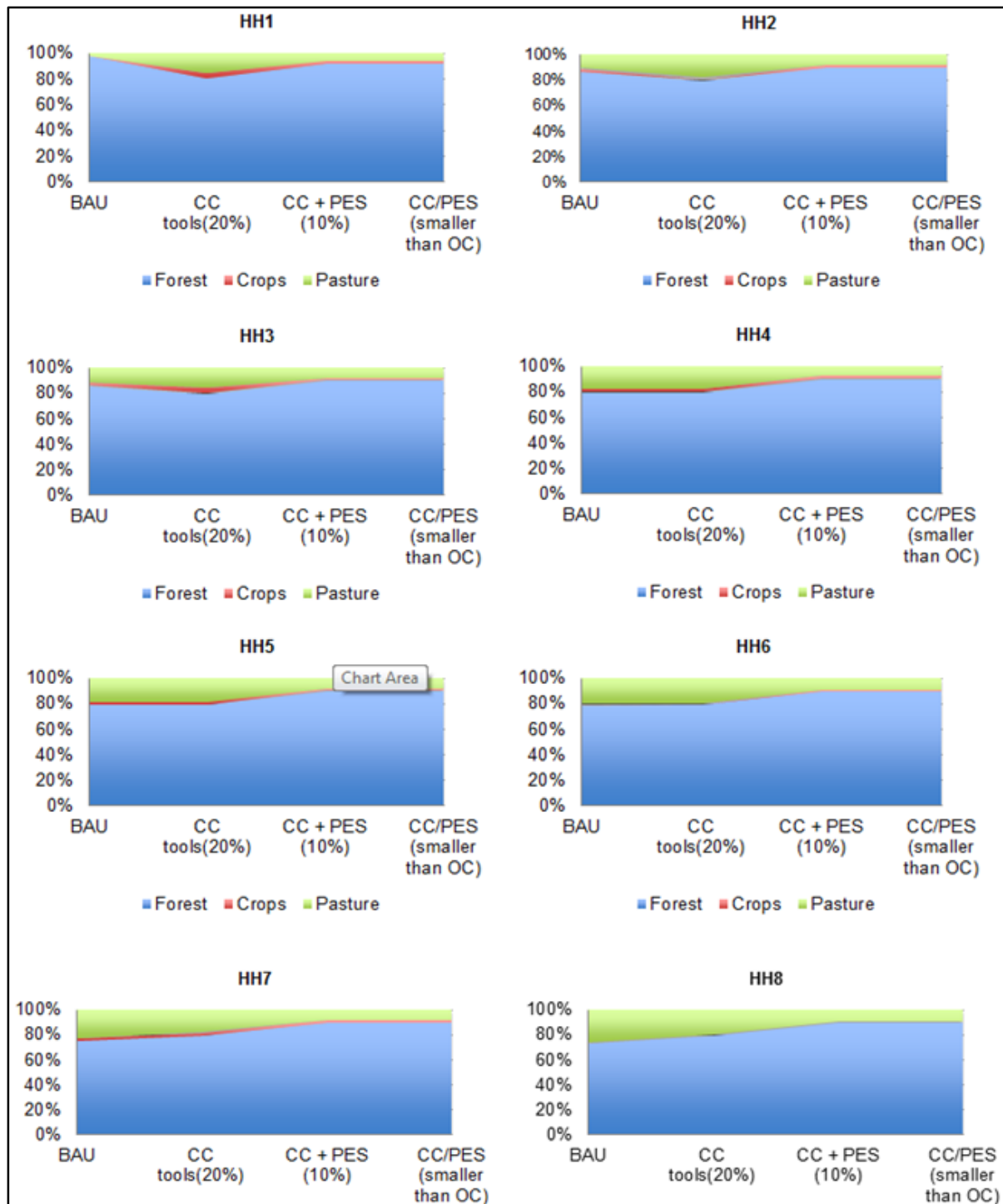


Figure D-1. Land distribution per household for four ELP scenarios



APPENDIX E  
SCENARIOS EMPLOYED TO RUN THE ELP MODEL

Table E-1. Household (HH) scenarios for the ELP model

| HH  | Activity |          | BAU  | CC tools<br>(20%) | CC + PES<br>(10%) | CC/PES<br>(smaller than OC) |
|-----|----------|----------|------|-------------------|-------------------|-----------------------------|
| HH1 | Forest   |          | 44.4 | 36.3              | 40.7              | 40.9                        |
|     | Crops    | Ha       | 0.0  | 2.0               | 1.0               | 1.0                         |
|     | Pasture  |          | 1.0  | 7.1               | 2.5               | 2.5                         |
|     | PES      | R\$/year | 0.0  | 0.0               | 11,413.6          | 11,299.9                    |
| HH2 | Forest   |          | 88.5 | 81.6              | 91.8              | 91.8                        |
|     | Crops    | Ha       | 2.5  | 2.0               | 2.0               | 2.0                         |
|     | Pasture  |          | 11.0 | 18.4              | 8.2               | 8.2                         |
|     | PES      | R\$/year | 0.0  | 0.0               | 9,537.0           | 8,583.3                     |
| HH3 | Forest   |          | 86.4 | 80.6              | 90.7              | 90.7                        |
|     | Crops    | Ha       | 2.4  | 4.0               | 1.4               | 1.4                         |
|     | Pasture  |          | 12.0 | 16.2              | 8.7               | 8.7                         |
|     | PES      | R\$/year | 0.0  | 0.0               | 9,424.8           | 8,482.3                     |
| HH4 | Forest   |          | 33.6 | 33.6              | 37.8              | 37.8                        |
|     | Crops    | Ha       | 1.0  | 1.0               | 1.0               | 1.0                         |
|     | Pasture  |          | 7.4  | 7.4               | 3.2               | 3.2                         |
|     | PES      | R\$/year | 0.0  | 0.0               | 7,774.2           | 6,996.8                     |
| HH5 | Forest   |          | 77.7 | 78.4              | 88.2              | 88.2                        |
|     | Crops    | Ha       | 2.0  | 2.0               | 1.5               | 1.5                         |
|     | Pasture  |          | 18.3 | 17.6              | 8.3               | 8.3                         |
|     | PES      | R\$/year | 0.0  | 0.0               | 9,163.0           | 8,246.7                     |
| HH6 | Forest   |          | 75.7 | 76.8              | 86.4              | 86.4                        |
|     | Crops    | Ha       | 1.0  | 1.0               | 1.0               | 1.0                         |
|     | Pasture  |          | 19.3 | 18.2              | 8.6               | 8.6                         |
|     | PES      | R\$/year | 0.0  | 0.0               | 8,976.0           | 8,078.4                     |
| HH7 | Forest   |          | 75.5 | 80.0              | 90.0              | 90.0                        |
|     | Crops    | ha       | 2.0  | 2.0               | 2.0               | 2.0                         |
|     | Pasture  |          | 22.5 | 18.0              | 8.0               | 8.0                         |
|     | PES      | R\$/year | 0.0  | 0.0               | 9,350.0           | 8,415.0                     |
| HH8 | Forest   |          | 74.4 | 80.6              | 90.7              | 90.7                        |
|     | Crops    | ha       | 0.5  | 0.5               | 0.5               | 0.5                         |
|     | Pasture  |          | 25.9 | 19.7              | 9.6               | 9.6                         |
|     | PES      | R\$/year | 0.0  | 0.0               | 9,424.8           | 8,482.3                     |

## APPENDIX F ELP MODEL RESULTS

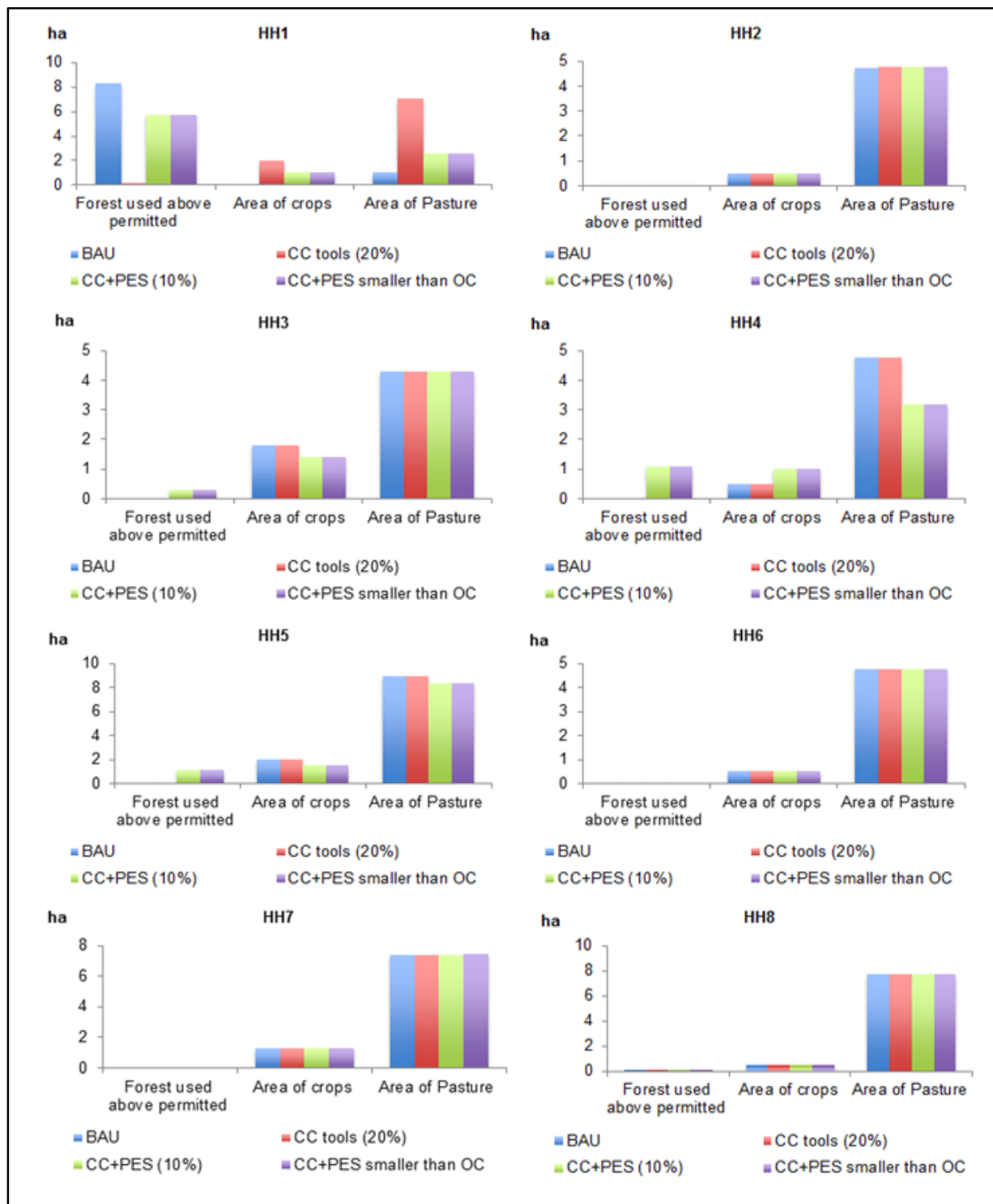


Figure F-1. Area (ha) allocated for different activities, scenarios and households

APPENDIX G  
INCOME VARIATION IN ALL SCENARIOS

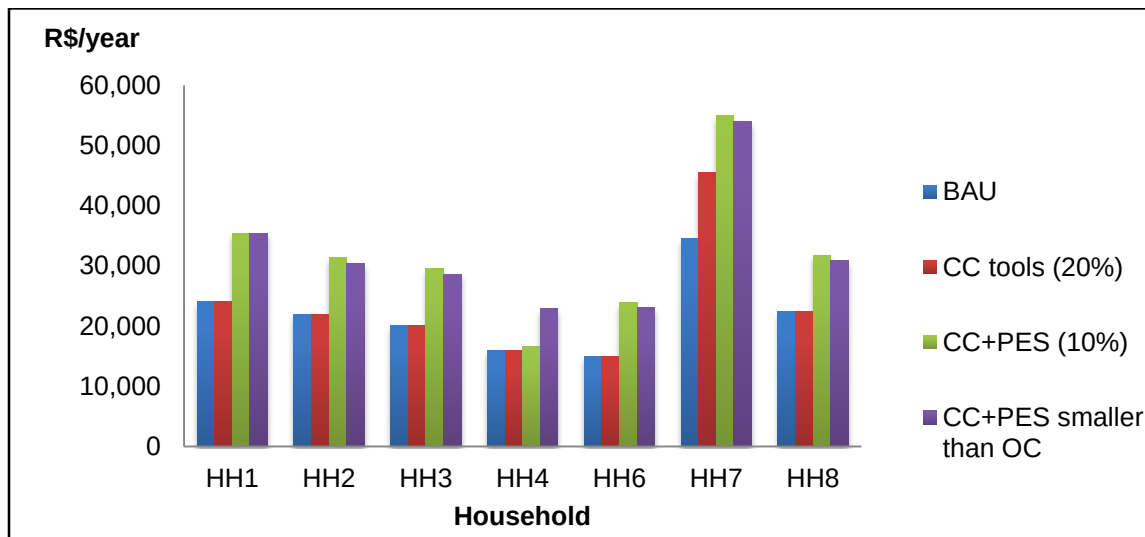


Figure G-1. Income variation (in R\$/year) in all scenarios for all households except for household 5

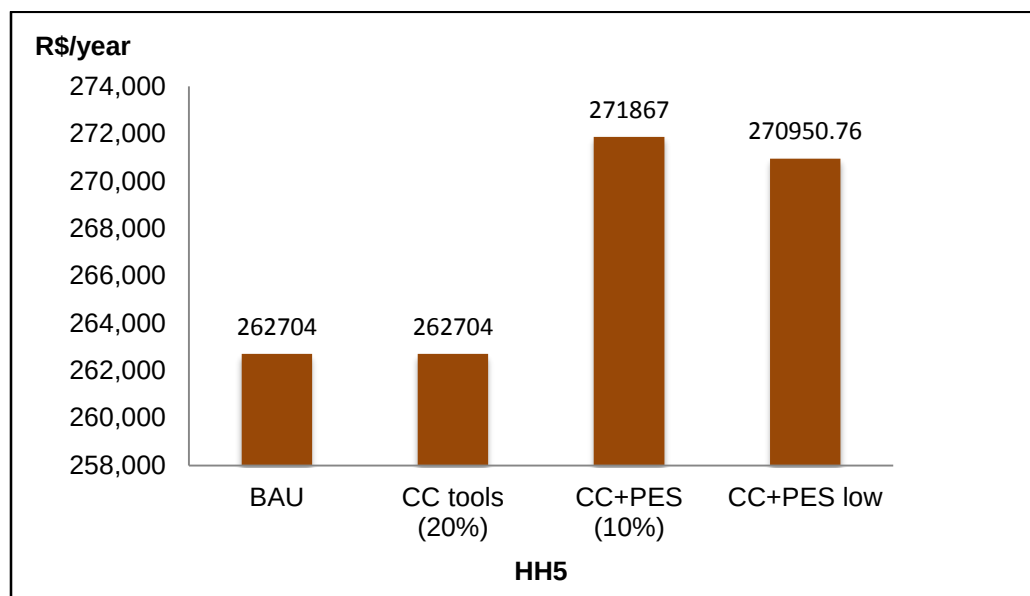


Figure G-2. Income variation (in R\$/year) in all scenarios for household 5

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## BIOGRAPHICAL SKETCH

Raissa Guerra was born in Brasília-DF, Brazil, a city located many kilometers away from the Amazon region. She received her bachelor's degree in Biological Sciences in 1995 from the University of Brasilia. She also obtained her Master's degree in Environmental Science in 2002 from the University of Brasilia. She has dedicated all her professional life to study impacts of public policies in the Brazilian Amazon. She worked for almost ten years at the Ministry of Environment of Brazil, as part of the Pilot Program to Protect the Rainforest (PPG7 program) in the Environmental Ministry. She came to the University of Florida where she was part of the Amazon Conservation Leadership Initiative.

She started her Ph.D. program at the University of Florida in the fall of 2007, under the supervision of Dr. Marianne Schmink. Meanwhile, Manuela Aitana arrived to complete happiness to her life, and to somewhat happily postpone her doctoral studies. She is currently working at Amazon Institute for Environmental Research (IPAM). After the completion of her Ph.D. program, she intends to keep engaged in understanding the impacts of policies over the protection of the Amazon forest, as well as to contribute to build a better world for all of us.